

Part II: Foundations

IN THIS CHAPTER

3. Functions	60
Appendix. Ordered Pairs	75
4. Graphs	77
Appendix 1. Vectors	96
Appendix 2. The Conic Sections	101
Appendix 3. Polar Coordinates	105
5. Limits	111
6. Continuous Functions	136
7. Three Hard Theorems	143
8. Least Upper Bounds	154
Appendix. Uniform Continuity	165

PART
FOUNDATIONS

*The statement is so frequently made
that the differential calculus deals with
continuous magnitude, and yet
an explanation of this continuity is
nowhere given;
even the most rigorous expositions
of the differential calculus do not base
their proofs upon continuity but,
with more or less consciousness of the fact,
they either appeal to geometric notions
or those suggested by geometry,
or depend upon theorems which are never
established in a purely arithmetic manner.
Among these, for example,
belongs the above-mentioned theorem,
and a more careful investigation
convinced me that this theorem, or
any one equivalent to it, can be regarded
in some way as a sufficient basis
for infinitesimal analysis.
It then only remained to discover its true
origin in the elements of arithmetic
and thus at the same time
to secure a real definition of
the essence of continuity.
I succeeded Nov. 24, 1858, and
a few days afterward I communicated
the results
of my meditations to my dear friend
Durège with whom I had a long
and lively discussion.*

RICHARD DEDEKIND

CHAPTER 3 FUNCTIONS

Undoubtedly the most important concept in all of mathematics is that of a function—in almost every branch of modern mathematics functions turn out to be the central objects of investigation. It will therefore probably not surprise you to learn that the concept of a function is one of great generality. Perhaps it will be a relief to learn that, for the present, we will be able to restrict our attention to functions of a very special kind; even this small class of functions will exhibit sufficient variety to engage our attention for quite some time. We will not even begin with a proper definition. For the moment a provisional definition will enable us to discuss functions at length, and will illustrate the intuitive notion of functions, as understood by mathematicians. Later, we will consider and discuss the advantages of the modern mathematical definition. Let us therefore begin with the following:

PROVISIONAL DEFINITION

A function is a rule which assigns, to each of certain real numbers, some other real number.

The following examples of functions are meant to illustrate and amplify this definition, which, admittedly, requires some such clarification.

Example 1 The rule which assigns to each number the square of that number.

Example 2 The rule which assigns to each number y the number

$$\frac{y^3 + 3y + 5}{y^2 + 1}.$$

Example 3 The rule which assigns to each number $c \neq 1, -1$ the number

$$\frac{c^3 + 3c + 5}{c^2 - 1}.$$

Example 4 The rule which assigns to each number x satisfying $-17 \leq x \leq \pi/3$ the number x^2 .

Example 5 The rule which assigns to each number a the number 0 if a is irrational, and the number 1 if a is rational.

Example 6 The rule which assigns

to 2 the number 5,
to 17 the number $\frac{36}{\pi}$,

to $\frac{\pi^2}{17}$ the number 28,

to $\frac{36}{\pi}$ the number 28,

and to any $y \neq 2, 17, \pi^2/17$, or $36/\pi$, the number 16 if y is of the form $a + b\sqrt{2}$ for a, b in $\mathbf{Q}$.

Example 7 The rule which assigns to each number t the number $t^3 + x$. (This rule depends, of course, on what the number x is, so we are really describing infinitely many different functions, one for each number x .)

Example 8 The rule which assigns to each number z the number of 7's in the decimal expansion of z , if this number is finite, and $-\pi$ if there are infinitely many 7's in the decimal expansion of z .

One thing should be abundantly clear from these examples—a function is *any* rule that assigns numbers to certain other numbers, not just a rule which can be expressed by an algebraic formula, or even by one uniform condition which applies to every number; nor is it necessarily a rule which you, or anybody else, can actually apply in practice (no one knows, for example, what rule 8 associates to π). Moreover, the rule may neglect some numbers and it may not even be clear to which numbers the function applies (try to determine, for example, whether the function in Example 6 applies to π). The set of numbers to which a function *does* apply is called the *domain* of the function.

Before saying anything else about functions we badly need some notation. Since throughout this book we shall frequently be talking about functions (indeed we shall hardly ever talk about anything else) we need a convenient way of naming functions, and of referring to functions in general. The standard practice is to denote a function by a letter. For obvious reasons the letter “ f ” is a favorite, thereby making “ g ” and “ h ” other obvious candidates, but any letter (or any reasonable symbol, for that matter) will do, not excluding “ x ” and “ y ”, although these letters are usually reserved for indicating numbers. If f is a function, then the number which f associates to a number x is denoted by $f(x)$ —this symbol is read “ f of x ” and is often called the **value of f at x** . Naturally, if we denote a function by x , some other letter must be chosen to denote the number (a perfectly legitimate, though perverse, choice would be “ f ,” leading to the symbol $x(f)$). Note that the symbol $f(x)$ makes sense only for x in the domain of f ; for other x the symbol $f(x)$ is not defined.

If the functions defined in Examples 1–8 are denoted by $f, g, h, r, s, \theta, \alpha_x$, and y , then we can rewrite their definitions as follows:

$$(1) \quad f(x) = x^2 \quad \text{for all } x.$$

$$(2) \quad g(y) = \frac{y^3 + 3y + 5}{y^2 + 1} \quad \text{for all } y.$$

$$(3) \quad h(c) = \frac{c^3 + 3c + 5}{c^2 - 1} \quad \text{for all } c \neq 1, -1.$$

$$(4) \quad r(x) = x^2 \quad \text{for all } x \text{ such that } -17 \leq x \leq \pi/3.$$

$$(5) \quad s(x) = \begin{cases} 0, & x \text{ irrational} \\ 1, & x \text{ rational.} \end{cases}$$

$$(6) \quad \theta(x) = \begin{cases} 5, & x = 2 \\ \frac{36}{\pi}, & x = 17 \\ 28, & x = \frac{\pi^2}{17} \\ 28, & x = \frac{36}{\pi} \\ 16, & x \neq 2, 17, \frac{\pi^2}{17}, \text{ or } \frac{36}{\pi}, \text{ and } x = a + b\sqrt{2} \text{ for } a, b \text{ in } \mathbf{Q}. \end{cases}$$

$$(7) \quad \alpha_x(t) = t^3 + x \quad \text{for all numbers } t.$$

$$(8) \quad y(x) = \begin{cases} n, & \text{exactly } n \text{ 7's appear in the decimal expansion of } x \\ -\pi, & \text{infinitely many 7's appear in the decimal expansion of } x. \end{cases}$$

These definitions illustrate the common procedure adopted for defining a function f —indicating what $f(x)$ is for every number x in the domain of f . (Notice that this is exactly the same as indicating $f(a)$ for every number a , or $f(b)$ for every number b , etc.) In practice, certain abbreviations are tolerated. Definition (1) could be written simple

$$(1) \quad f(x) = x^2$$

the qualifying phrase “for all x ” being understood. Of course, for definition (4) the only possible abbreviation is

$$(4) \quad r(x) = x^2, \quad -17 \leq x \leq \pi/3.$$

It is usually understood that a definition such as

$$k(x) = \frac{1}{x} + \frac{1}{x-1}, \quad x \neq 0, 1$$

can be shortened to

$$k(x) = \frac{1}{x} + \frac{1}{x-1};$$

in other words, *unless the domain is explicitly restricted further, it is understood to consist of all numbers for which the definition makes any sense at all.*

You should have little difficulty checking the following assertions about the functions defined above:

$$f(x+1) = f(x) + 2x + 1;$$

$$g(x) = h(x) \text{ if } x^3 + 3x + 5 = 0;$$

$$r(x+1) = r(x) + 2x + 1 \text{ if } -17 \leq x \leq \frac{\pi}{3} - 1;$$

$$s(x + y) = s(x) \text{ if } y \text{ is rational;}$$

$$\theta\left(\frac{\pi^2}{17}\right) = \theta\left(\frac{36}{\pi}\right);$$

$$\alpha_x(x) = x \cdot [f(x) + 1];$$

$$y\left(\frac{1}{3}\right) = 0, \quad y\left(\frac{7}{9}\right) = -\pi.$$

If the expression $f(s(a))$ looks unreasonable to you, then you are forgetting that $s(a)$ is a number like any other number, so that $f(s(a))$ makes sense. As a matter of fact, $f(s(a)) = s(a)$ for all a . Why? Even more complicated expressions than $f(s(a))$ are, after a first exposure, no more difficult to unravel. The expression

$$f(r(s(\theta(\alpha_3(y(\frac{1}{3})))))),$$

formidable as it appears, may be evaluated quite easily with a little patience:

$$\begin{aligned} & f(r(s(\theta(\alpha_3(y(\frac{1}{3})))))) \\ &= f(r(s(\theta(\alpha_3(0))))) \\ &= f(r(s(\theta(3)))) \\ &= f(r(s(16))) \\ &= f(r(1)) \\ &= f(1) \\ &= 1. \end{aligned}$$

The first few problems at the end of this chapter give further practice manipulating this symbolism.

The function defined in (1) is a rather special example of an extremely important class of functions, the polynomial functions. A function f is a **polynomial function** if there are real numbers $a_0, \dots, a_n$ such that

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_2 x^2 + a_1 x + a_0, \quad \text{for all } x$$

(when $f(x)$ is written in this form it is usually tacitly assumed that $a_n \neq 0$). The highest power of x with a nonzero coefficient is called the **degree** of f ; for example, the polynomial function f defined by $f(x) = 5x^6 + 137x^4 - \pi$ has degree 6.

The functions defined in (2) and (3) belong to a somewhat larger class of functions, the **rational functions**; these are the functions of the form p/q where p and q are polynomial functions (and q is not the function which is always 0). The rational functions are themselves quite special examples of an even larger class of functions, very thoroughly studied in calculus, which are simpler than many of the functions first mentioned in this chapter. The following are examples of this kind of function:

$$(9) \quad f(x) = \frac{x + x^2 + x \sin^2 x}{x \sin x + x \sin^2 x}$$

$$(10) \quad f(x) = \sin(x^2).$$

$$(11) \quad f(x) = \sin(\sin(x^2)).$$

$$(12) \quad f(x) = \sin^2(\sin(\sin^2(x \sin^2 x^2))) \cdot \sin\left(\frac{x + \sin(x \sin x)}{x + \sin x}\right).$$

By what criterion, you may feel impelled to ask, can such functions, especially a monstrosity like (12), be considered simple? The answer is that they can be built up from a few simple functions using a few simple means of combining functions. In order to construct the functions (9)–(12) we need to start with the “identity function” I , for which $I(x) = x$, and the “sine function” $\sin$, whose value $\sin(x)$ at x is often written simply $\sin x$. The following are some of the important ways in which functions may be combined to produce new functions.

If f and g are any two functions, we can define a new function $f + g$, called the **sum** of f and g , by the equation

$$(f + g)(x) = f(x) + g(x).$$

Note that according to the conventions we have adopted, the domain of $f + g$ consists of all x for which “ $f(x) + g(x)$ ” makes sense, i.e., the set of all x in both domain f and domain g . If A and B are any two sets, then $A \cap B$ (read “ A intersect B ” or “the intersection of A and B ”) denotes the set of x in both A and B ; this notation allows us to write $\text{domain}(f + g) = \text{domain } f \cap \text{domain } g$.

In a similar vein, we define the **product** $f \cdot g$ and the **quotient** $\frac{f}{g}$ (or f/g) of f and g by

$$(f \cdot g)(x) = f(x) \cdot g(x)$$

and

$$\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}.$$

Moreover, if g is a function and c is a number, we define a new function $c \cdot g$ by

$$(c \cdot g)(x) = c \cdot g(x).$$

This becomes a special case of the notation $f \cdot g$ if we agree that the symbol c should also represent the function f defined by $f(x) = c$; such a function, which has the same value for all numbers x , is called a **constant function**.

The domain of $f \cdot g$ is $\text{domain } f \cap \text{domain } g$, and the domain of $c \cdot g$ is simply the domain of g . On the other hand, the domain of f/g is rather complicated—it may be written $\text{domain } f \cap \text{domain } g \cap \{x : g(x) \neq 0\}$, the symbol $\{x : g(x) \neq 0\}$ denoting the set of numbers x such that $g(x) \neq 0$. In general, $\{x : \dots\}$ denotes the set of all x such that “ $\dots$ ” is true. Thus $\{x : x^3 + 3 < 11\}$ denotes the set of all numbers x such that $x^3 + 3 < 11$, and consequently $\{x : x^3 + 3 < 11\} = \{x : x < 2\}$. Either of these symbols could just as well have been written using y everywhere instead of x . Variations of this notation are common, but hardly require any discussion. Any one can guess that $\{x > 0 : x^3 < 8\}$ denotes the set of positive numbers whose cube is less than 8; it could be expressed more formally as $\{x : x > 0 \text{ and } x^3 < 8\}$. Incidentally, this set is equal to the set $\{x : 0 < x < 2\}$. One

variation is slightly less transparent, but very standard. The set $\{1, 3, 2, 4\}$, for example, contains just the four numbers 1, 2, 3, and 4; it can also be denoted by $\{x : x = 1 \text{ or } x = 3 \text{ or } x = 2 \text{ or } x = 4\}$.

Certain facts about the sum, product, and quotient of functions are obvious consequences of facts about sums, products, and quotients of numbers. For example, it is very easy to prove that

$$(f + g) + h = f + (g + h).$$

The proof is characteristic of almost every proof which demonstrates that two functions are equal—the two functions must be shown to have the same domain, and the same value at any number in the domain. For example, to prove that $(f + g) + h = f + (g + h)$, note that unraveling the definition of the two sides gives

$$\begin{aligned} [(f + g) + h](x) &= (f + g)(x) + h(x) \\ &= [f(x) + g(x)] + h(x) \end{aligned}$$

and

$$\begin{aligned} [f + (g + h)](x) &= f(x) + (g + h)(x) \\ &= f(x) + [g(x) + h(x)], \end{aligned}$$

and the equality of $[f(x) + g(x)] + h(x)$ and $f(x) + [g(x) + h(x)]$ is a fact about numbers. In this proof the equality of the two domains was not explicitly mentioned because this is obvious, as soon as we begin to write down these equations; the domain of $(f + g) + h$ and of $f + (g + h)$ is clearly $\text{domain } f \cap \text{domain } g \cap \text{domain } h$. We naturally write $f + g + h$ for $(f + g) + h = f + (g + h)$, precisely as we did for numbers.

It is just as easy to prove that $(f \cdot g) \cdot h = f \cdot (g \cdot h)$, and this function is denoted by $f \cdot g \cdot h$. The equations $f + g = g + f$ and $f \cdot g = g \cdot f$ should also present no difficulty.

Using the operations $+$, $\cdot$, $/$ we can now express the function f defined in (9) by

$$f = \frac{I + I \cdot I + I \cdot \sin \cdot \sin}{I \cdot \sin + I \cdot \sin \cdot \sin}.$$

It should be clear, however, that we cannot express function (10) this way. We require yet another way of combining functions. This combination, the composition of two functions, is by far the most important.

If f and g are any two functions, we define a new function $f \circ g$, the **composition** of f and g , by

$$(f \circ g)(x) = f(g(x));$$

the domain of $f \circ g$ is $\{x : x \text{ is in domain } g \text{ and } g(x) \text{ is in domain } f\}$. The symbol “ $f \circ g$ ” is often read “ f circle g .” Compared to the phrase “the composition of f and g ” this has the advantage of brevity, of course, but there is another advantage of far greater import: there is much less chance of confusing $f \circ g$ with $g \circ f$, and

these must *not* be confused, since they are not usually equal; in fact, almost any f and g chosen at random will illustrate this point (try $f = I \cdot I$ and $g = \sin$, for example). Lest you become too apprehensive about the operation of composition, let us hasten to point out that composition *is* associative:

$$(f \circ g) \circ h = f \circ (g \circ h)$$

(and the proof is a triviality); this function is denoted by $f \circ g \circ h$. We can now write the functions (10), (11), (12) as

$$(10) \quad f = \sin \circ (I \cdot I),$$

$$(11) \quad f = \sin \circ \sin \circ (I \cdot I),$$

$$(12) \quad f = (\sin \cdot \sin) \circ \sin \circ (\sin \cdot \sin) \circ (I \cdot [(\sin \cdot \sin) \circ (I \cdot I)]) \cdot$$

$$\sin \circ \left(\frac{I + \sin \circ (I \cdot \sin)}{I + \sin} \right).$$

One fact has probably already become clear. Although this method of writing functions reveals their “structure” very clearly, it is hardly short or convenient. The shortest name for the function f such that $f(x) = \sin(x^2)$ for all x unfortunately seems to be “the function f such that $f(x) = \sin(x^2)$ for all x .” The need for abbreviating this clumsy description has been clear for two hundred years, but no reasonable abbreviation has received universal acclaim. At present the strongest contender for this honor is something like

$$x \rightarrow \sin(x^2)$$

(read “ x goes to $\sin(x^2)$ ” or just “ x arrow $\sin(x^2)$ ”), but it is hardly popular among writers of calculus textbooks. In this book we will tolerate a certain amount of ellipsis, and speak of “the function $f(x) = \sin(x^2)$.” Even more popular is the quite drastic abbreviation: “the function $\sin(x^2)$.” For the sake of precision we will never use this description, which, strictly speaking, confuses a number and a function, but it is so convenient that you will probably end up adopting it for personal use. As with any convention, utility is the motivating factor, and this criterion is reasonable so long as the slight logical deficiencies cause no confusion. On occasion, confusion *will* arise unless a more precise description is used. For example, “the function $x + t^3$ ” is an ambiguous phrase; it could mean either

$$x \rightarrow x + t^3, \text{ i.e., the function } f \text{ such that } f(x) = x + t^3 \text{ for all } x$$

or

$$t \rightarrow x + t^3, \text{ i.e., the function } f \text{ such that } f(t) = x + t^3 \text{ for all } t.$$

As we shall see, however, for many important concepts associated with functions, calculus has a notation which contains the “ $x \rightarrow$ ” built in.

By now we have made a sufficiently extensive investigation of functions to warrant reconsidering our definition. We have defined a function as a “rule,” but it is hardly clear what this means. If we ask “What happens if you break this rule?” it is not easy to say whether this question is merely facetious or actually profound.

A more substantial objection to the use of the word “rule” is that

$$f(x) = x^2$$

and

$$f(x) = x^2 + 3x + 3 - 3(x + 1)$$

are certainly *different* rules, if by a rule we mean the actual instructions given for determining $f(x)$; nevertheless, we want

$$f(x) = x^2$$

and

$$f(x) = x^2 + 3x + 3 - c(x + 1)$$

to define the same function. For this reason, a function is sometimes defined as an “association” between numbers; unfortunately the word “association” escapes the objections raised against “rule” only because it is even more vague.

There is, of course, a satisfactory way of defining functions, or we should never have gone to the trouble of criticizing our original definition. But a satisfactory definition can never be constructed by finding synonyms for English words which are troublesome. The definition which mathematicians have finally accepted for “function” is a beautiful example of the means by which intuitive ideas have been incorporated into rigorous mathematics. The correct question to ask about a function is not “What is a rule?” or “What is an association?” but “What does one have to know about a function in order to know all about it?” The answer to the last question is easy—for each number x one needs to know the number $f(x)$; we can imagine a table which would display all the information one could desire about the function $f(x) = x^2$:

x	$f(x)$
1	1
−1	1
2	4
−2	4
$\sqrt{2}$	2
$-\sqrt{2}$	2
π	π^2
$-\pi$	π^2

It is not even necessary to arrange the numbers in a table (which would actually be impossible if we wanted to list all of them). Instead of a two column array we can consider various pairs of numbers

$$(1, 1), (-1, 1), (2, 4), (-2, 4), (\pi, \pi^2), (\sqrt{2}, 2), \dots$$

simply collected together into a set.* To find $f(1)$ we simply take the second number of the pair whose first member is 1; to find $f(\pi)$ we take the second number of the pair whose first member is π . We seem to be saying that a function might as well be defined as a collection of pairs of numbers. For example, if we were given the following collection (which contains just 5 pairs):

$$f = \{(1, 7), (3, 7), (5, 3), (4, 8), (8, 4)\},$$

then $f(1) = 7$, $f(3) = 7$, $f(5) = 3$, $f(4) = 8$, $f(8) = 4$ and 1, 3, 4, 5, 8 are the only numbers in the domain of f . If we consider the collection

$$f = \{(1, 7), (3, 7), (2, 5), (1, 8), (8, 4)\},$$

then $f(3) = 7$, $f(2) = 5$, $f(8) = 4$; but it is impossible to decide whether $f(1) = 7$ or $f(1) = 8$. In other words, a function cannot be defined to be any old collection of pairs of numbers; we must rule out the possibility which arose in this case. We are therefore led to the following definition.

DEFINITION

A **function** is a collection of pairs of numbers with the following property: if (a, b) and (a, c) are both in the collection, then $b = c$; in other words, the collection must not contain two different pairs with the same first element.

This is our first full-fledged definition, and illustrates the format we shall always use to define significant new concepts. These definitions are so important (at least as important as theorems) that it is essential to know when one is actually at hand, and to distinguish them from comments, motivating remarks, and casual explanations. They will be preceded by the word **DEFINITION**, contain the term being defined in boldface letters, and constitute a paragraph unto themselves.

There is one more definition (actually defining two things at once) which can now be made rigorously:

DEFINITION

If f is a function, the **domain** of f is the set of all a for which there is some b such that (a, b) is in f . If a is in the domain of f , it follows from the definition of a function that there is, in fact, a *unique* number b such that (a, b) is in f . This unique b is denoted by $f(a)$.

With this definition we have reached our goal: the important thing about a function f is that a number $f(x)$ is determined for each number x in its domain. You may feel that we have also reached the point where an intuitive definition has been replaced by an abstraction with which the mind can hardly grapple. Two consolations may be offered. First, although a function has been defined as a

*The pairs occurring here are often called “ordered pairs,” to emphasize that, for example, $(2, 4)$ is not the same pair as $(4, 2)$. It is only fair to warn that we are going to define functions in terms of ordered pairs, another undefined term. Ordered pairs *can* be defined, however, and an appendix to this chapter has been provided for skeptics.

collection of pairs, there is nothing to stop you from *thinking* of a function as a rule. Second, neither the intuitive nor the formal definition indicates the best way of thinking about functions. The best way is to draw pictures; but this requires a chapter all by itself.

PROBLEMS

1. Let $f(x) = 1/(1+x)$. What is

- (i) $f(f(x))$ (for which x does this make sense?).
- (ii) $f\left(\frac{1}{x}\right)$.
- (iii) $f(cx)$.
- (iv) $f(x+y)$.
- (v) $f(x) + f(y)$.
- (vi) For which numbers c is there a number x such that $f(cx) = f(x)$.
Hint: There are a lot more than you might think at first glance.
- (vii) For which numbers c is it true that $f(cx) = f(x)$ for two different numbers x ?

2. Let $g(x) = x^2$, and let

$$h(x) = \begin{cases} 0, & x \text{ rational} \\ 1, & x \text{ irrational.} \end{cases}$$

- (i) For which y is $h(y) \leq y$?
- (ii) For which y is $h(y) \leq g(y)$?
- (iii) What is $g(h(z)) - h(z)$?
- (iv) For which w is $g(w) \leq w$?
- (v) For which ε is $g(g(\varepsilon)) = g(\varepsilon)$?

3. Find the domain of the functions defined by the following formulas.

- (i) $f(x) = \sqrt{1-x^2}$.
- (ii) $f(x) = \sqrt{1-\sqrt{1-x^2}}$.
- (iii) $f(x) = \frac{1}{x-1} + \frac{1}{x-2}$.
- (iv) $f(x) = \sqrt{1-x^2} + \sqrt{x^2-1}$.
- (v) $f(x) = \sqrt{1-x} + \sqrt{x-2}$.

4. Let $S(x) = x^2$, let $P(x) = 2^x$, and let $s(x) = \sin x$. Find each of the following. In each case your answer should be a *number*.

- (i) $(S \circ P)(y)$.
- (ii) $(S \circ s)(y)$.
- (iii) $(S \circ P \circ s)(t) + (s \circ P)(t)$.
- (iv) $s(t^3)$.

5. Express each of the following functions in terms of S , P , s , using only $+$, $\cdot$, and $\circ$ (for example, the answer to (i) is $P \circ s$). In each case your

answer should be a *function*.

- (i) $f(x) = 2^{\sin x}$.
- (ii) $f(x) = \sin 2^x$.
- (iii) $f(x) = \sin x^2$.
- (iv) $f(x) = \sin^2 x$ (remember that $\sin^2 x$ is an abbreviation for $(\sin x)^2$).
- (v) $f(t) = 2^{2^t}$. (Note: a^{b^c} *always* means $a^{(b^c)}$; this convention is adopted because $(a^b)^c$ can be written more simply as a^{bc} .)
- (vi) $f(u) = \sin(2^u + 2^{u^2})$.
- (vii) $f(y) = \sin(\sin(\sin(2^{2^{\sin y}})))$.
- (viii) $f(a) = 2^{\sin^2 a} + \sin(a^2) + 2^{\sin(a^2 + \sin a)}$.

Polynomial functions, because they are simple, yet flexible, occupy a favored role in most investigations of functions. The following two problems illustrate their flexibility, and guide you through a derivation of their most important elementary properties.

6. (a) If $x_1, \dots, x_n$ are distinct numbers, find a polynomial function f_i of degree $n - 1$ which is 1 at x_i and 0 at x_j for $j \neq i$. Hint: the product of all $(x - x_j)$ for $j \neq i$, is 0 at x_j if $j \neq i$. (This product is usually denoted by

$$\prod_{\substack{j=1 \\ j \neq i}}^n (x - x_j),$$

the symbol Π (capital pi) playing the same role for products that Σ plays for sums.)

- (b) Now find a polynomial function f of degree $n - 1$ such that $f(x_i) = a_i$, where $a_1, \dots, a_n$ are given numbers. (You should use the functions f_i from part (a). The formula you will obtain is called the “Lagrange interpolation formula.”)
7. (a) Prove that for any polynomial function f , and any number a , there is a polynomial function g , and a number b , such that $f(x) = (x - a)g(x) + b$ for all x . (The idea is simply to divide $(x - a)$ into $f(x)$ by long division, until a constant remainder is left. For example, the calculation

$$\begin{array}{r}
 x^2 \quad +x \quad -2 \\
 x-1 \overline{) x^3 } \\
 \underline{x^3 } \\
 -x^2 \\
 \underline{-x^2 } \\
 -3x \\
 \underline{-3x } \\
 -x \\
 \underline{-x } \\
 -2x + 1 \\
 \underline{-2x + 2} \\
 -1
 \end{array}$$

shows that $x^3 - 3x + 1 = (x - 1)(x^2 + x - 2) - 1$. A formal proof is possible by induction on the degree of f .)

- (b) Prove that if $f(a) = 0$, then $f(x) = (x - a)g(x)$ for some polynomial function g . (The converse is obvious.)
- (c) Prove that if f is a polynomial function of degree n , then f has at most n roots, i.e., there are at most n numbers a with $f(a) = 0$.
- (d) Show that for each n there is a polynomial function of degree n with n roots. If n is even find a polynomial function of degree n with no roots, and if n is odd find one with only one root.

8. For which numbers a, b, c , and d will the function

$$f(x) = \frac{ax + b}{cx + d}$$

satisfy $f(f(x)) = x$ for all x (for which this equation makes sense)?

9. (a) If A is any set of real numbers, define a function C_A as follows:

$$C_A(x) = \begin{cases} 1, & x \text{ in } A \\ 0, & x \text{ not in } A. \end{cases}$$

Find expressions for $C_{A \cap B}$ and $C_{A \cup B}$ and $C_{\mathbf{R} - A}$, in terms of C_A and C_B . (The symbol $A \cap B$ was defined in this chapter, but the other two may be new to you. They can be defined as follows:

$$\begin{aligned} A \cup B &= \{x : x \text{ is in } A \text{ or } x \text{ is in } B\}, \\ \mathbf{R} - A &= \{x : x \text{ is in } \mathbf{R} \text{ but } x \text{ is not in } A\}. \end{aligned}$$

- (b) Suppose f is a function such that $f(x) = 0$ or 1 for each x . Prove that there is a set A such that $f = C_A$.
 - (c) Show that $f = f^2$ if and only if $f = C_A$ for some set A .
10. (a) For which functions f is there a function g such that $f = g^2$? Hint: You can certainly answer this question if “function” is replaced by “number.”
- (b) For which functions f is there a function g such that $f = 1/g$?
 - * (c) For which functions b and c can we find a function x such that

$$(x(t))^2 + b(t)x(t) + c(t) = 0$$

for all numbers t ?

- * (d) What conditions must the functions a and b satisfy if there is to be a function x such that

$$a(t)x(t) + b(t) = 0$$

for all numbers t ? How many such functions x will there be?

11. (a) Suppose that H is a function and y is a number such that $H(H(y)) = y$. What is

$$\underbrace{H(H(H(\cdots(H(y)\cdots)))}_{80 \text{ times}}?$$

- (b) Same question if 80 is replaced by 81.
 - (c) Same question if $H(H(y)) = H(y)$.
 - * (d) Find a function H such that $H(H(x)) = H(x)$ for all numbers x , and such that $H(1) = 36$, $H(2) = \pi/3$, $H(13) = 47$, $H(36) = 36$, $H(\pi/3) = \pi/3$, $H(47) = 47$. (Don't try to "solve" for $H(x)$; there are many functions H with $H(H(x)) = H(x)$. The extra conditions on H are supposed to suggest a way of finding a suitable H .)
 - * (e) Find a function H such that $H(H(x)) = H(x)$ for all x , and such that $H(1) = 7$, $H(17) = 18$.
12. A function f is **even** if $f(x) = f(-x)$ and **odd** if $f(x) = -f(-x)$. For example, f is even if $f(x) = x^2$ or $f(x) = |x|$ or $f(x) = \cos x$, while f is odd if $f(x) = x$ or $f(x) = \sin x$.
- (a) Determine whether $f + g$ is even, odd, or not necessarily either, in the four cases obtained by choosing f even or odd, and g even or odd. (Your answers can most conveniently be displayed in a 2×2 table.)
 - (b) Do the same for $f \cdot g$.
 - (c) Do the same for $f \circ g$.
 - (d) Prove that every even function f can be written $f(x) = g(|x|)$, for infinitely many functions g .
- *13. (a) Prove that any function f with domain $\mathbf{R}$ can be written $f = E + O$, where E is even and O is odd.
- (b) Prove that this way of writing f is unique. (If you try to do part (b) first, by "solving" for E and O you will probably find the solution to part (a).)
14. If f is any function, define a new function $|f|$ by $|f|(x) = |f(x)|$. If f and g are functions, define two new functions, $\max(f, g)$ and $\min(f, g)$, by
- $$\begin{aligned}\max(f, g)(x) &= \max(f(x), g(x)), \\ \min(f, g)(x) &= \min(f(x), g(x)).\end{aligned}$$
- Find an expression for $\max(f, g)$ and $\min(f, g)$ in terms of $| \cdot |$.
15. (a) Show that $f = \max(f, 0) + \min(f, 0)$. This particular way of writing f is fairly useful; the functions $\max(f, 0)$ and $\min(f, 0)$ are called the **positive** and **negative parts** of f .
- (b) A function f is called **nonnegative** if $f(x) \geq 0$ for all x . Prove that any function f can be written $f = g - h$, where g and h are nonnegative, in infinitely many ways. (The "standard way" is $g = \max(f, 0)$ and $h = -\min(f, 0)$.) Hint: Any *number* can certainly be written as the difference of two nonnegative *numbers* in infinitely many ways.
- *16. Suppose f satisfies $f(x + y) = f(x) + f(y)$ for all x and y .
- (a) Prove that $f(x_1 + \cdots + x_n) = f(x_1) + \cdots + f(x_n)$.
 - (b) Prove that there is some number c such that $f(x) = cx$ for all *rational* numbers x (at this point we're not trying to say anything about $f(x)$ for irrational x). Hint: First figure out what c must be. Now prove that

$f(x) = cx$, first when x is a natural number, then when x is an integer, then when x is the reciprocal of an integer and, finally, for all rational x .

- *17.** If $f(x) = 0$ for all x , then f satisfies $f(x + y) = f(x) + f(y)$ for all x and y , and also $f(x \cdot y) = f(x) \cdot f(y)$ for all x and y . Now suppose that f satisfies these two properties, but that $f(x)$ is not always 0. Prove that $f(x) = x$ for all x , as follows:

- Prove that $f(1) = 1$.
- Prove that $f(x) = x$ if x is rational.
- Prove that $f(x) > 0$ if $x > 0$. (This part is tricky, but if you have been paying attention to the philosophical remarks accompanying the problems in the last two chapters, you will know what to do.)
- Prove that $f(x) > f(y)$ if $x > y$.
- Prove that $f(x) = x$ for all x . Hint: Use the fact that between any two numbers there is a rational number.

- *18.** Precisely what conditions must f , g , h , and k satisfy in order that $f(x)g(y) = h(x)k(y)$ for all x and y ?

- *19.** (a) Prove that there do *not* exist functions f and g with either of the following properties:

- $f(x) + g(y) = xy$ for all x and y .
- $f(x) \cdot g(y) = x + y$ for all x and y .

Hint: Try to get some information about f or g by choosing particular values of x and y .

- (b) Find functions f and g such that $f(x + y) = g(xy)$ for all x and y .

- *20.** (a) Find a function f , other than a constant function, such that $|f(y) - f(x)| \leq |y - x|$.
- (b) Suppose that $f(y) - f(x) \leq (y - x)^2$ for all x and y . (Why does this imply that $|f(y) - f(x)| \leq (y - x)^2$?) Prove that f is a constant function. Hint: Divide the interval from x to y into n equal pieces.

- 21.** Prove or give a counterexample for each of the following assertions:

- $f \circ (g + h) = f \circ g + f \circ h$.
- $(g + h) \circ f = g \circ f + h \circ f$.
- $\frac{1}{f \circ g} = \frac{1}{f} \circ g$.
- $\frac{1}{f \circ g} = f \circ \left(\frac{1}{g}\right)$.

- 22.** (a) Suppose $g = h \circ f$. Prove that if $f(x) = f(y)$, then $g(x) = g(y)$.
- (b) Conversely, suppose that f and g are two functions such that $g(x) = g(y)$ whenever $f(x) = f(y)$. Prove that $g = h \circ f$ for some function h . Hint: Just try to define $h(z)$ when z is of the form $z = f(x)$ (these are the only z that matter) and use the hypotheses to show that your definition will not run into trouble.

23. Suppose that $f \circ g = I$, where $I(x) = x$. Prove that
- (a) if $x \neq y$, then $g(x) \neq g(y)$;
 - (b) every number b can be written $b = f(a)$ for some number a .
- *24. (a) Suppose g is a function with the property that $g(x) \neq g(y)$ if $x \neq y$. Prove that there is a function f such that $f \circ g = I$.
 (b) Suppose that f is a function such that every number b can be written $b = f(a)$ for some number a . Prove that there is a function g such that $f \circ g = I$.
- *25. Find a function f such that $g \circ f = I$ for some g , but such that there is no function h with $f \circ h = I$.
- *26. Suppose $f \circ g = I$ and $h \circ f = I$. Prove that $g = h$. Hint: Use the fact that composition is associative.
27. (a) Suppose $f(x) = x + 1$. Are there any functions g such that $f \circ g = g \circ f$?
 (b) Suppose f is a constant function. For which functions g does $f \circ g = g \circ f$?
 (c) Suppose that $f \circ g = g \circ f$ for *all* functions g . Show that f is the identity function, $f(x) = x$.
28. (a) Let F be the set of all functions whose domain is $\mathbf{R}$. Prove that, using $+$ and $\cdot$ as defined in this chapter, all of properties P1–P9 except P7 hold for F , provided 0 and 1 are interpreted as constant functions.
 (b) Show that P7 does not hold.
 *(c) Show that P10–P12 cannot hold. In other words, show that there is no collection P of functions in F , such that P10–P12 hold for P . (It is sufficient, and will simplify things, to consider only functions which are 0 except at two points x_0 and x_1 .)
 (d) Suppose we define $f < g$ to mean that $f(x) < g(x)$ for all x . Which of P'10–P'13 (in Problem 1-8) now hold?
 (e) If $f < g$, is $h \circ f < h \circ g$? Is $f \circ h < g \circ h$?

APPENDIX. ORDERED PAIRS

Not only in the definition of a function, but in other parts of the book as well, it is necessary to use the notion of an ordered pair of objects. A definition has not yet been given, and we have never even stated explicitly what properties an ordered pair is supposed to have. The one property which we will require states formally that the ordered pair (a, b) should be determined by a and b , and the order in which they are given:

$$\text{if } (a, b) = (c, d), \text{ then } a = c \text{ and } b = d.$$

Ordered pairs may be treated most conveniently by simply introducing (a, b) as an undefined term and adopting the basic property as an axiom—since this property is the only significant fact about ordered pairs, there is not much point worrying about what an ordered pair “really” is. Those who find this treatment satisfactory need read no further.

The rest of this short appendix is for the benefit of those readers who will feel uncomfortable unless ordered pairs are somehow defined so that this basic property becomes a theorem. There is no point in restricting our attention to ordered pairs of numbers; it is just as reasonable, and just as important, to have available the notion of an ordered pair of any two mathematical objects. This means that our definition ought to involve only concepts common to all branches of mathematics. The one common concept which pervades all areas of mathematics is that of a set, and ordered pairs (like everything else in mathematics) can be defined in this context; an ordered pair will turn out to be a set of a rather special sort.

The set $\{a, b\}$, containing the two elements a and b , is an obvious first choice, but will not do as a definition for (a, b) , because there is no way of determining from $\{a, b\}$ which of a or b is meant to be the first element. A more promising candidate is the rather startling set:

$$\{\{a\}, \{a, b\}\}.$$

This set has two members, both of which are *themselves* sets; one member is the set $\{a\}$, containing the single member a , the other is the set $\{a, b\}$. Shocking as it may seem, we are going to define (a, b) to be this set. The justification for this choice is given by the theorem immediately following the definition—the definition works, and there really isn’t anything else worth saying.

DEFINITION

$$(a, b) = \{\{a\}, \{a, b\}\}.$$

THEOREM 1

If $(a, b) = (c, d)$, then $a = c$ and $b = d$.

PROOF

The hypothesis means that

$$\{\{a\}, \{a, b\}\} = \{\{c\}, \{c, d\}\}.$$

Now $\{\{a\}, \{a, b\}\}$ contains just two members, $\{a\}$ and $\{a, b\}$; and a is the only common element of these two members of $\{\{a\}, \{a, b\}\}$. Similarly, c is the unique

common member of both members of $\{\{c\}, \{c, d\}\}$. Therefore $a = c$. We therefore have

$$\{\{a\}, \{a, b\}\} = \{\{a\}, \{a, d\}\},$$

and only the proof that $b = d$ remains. It is convenient to distinguish 2 cases.

Case 1. $b = a$. In this case, $\{a, b\} = \{a\}$, so the set $\{\{a\}, \{a, b\}\}$ really has only one member, namely, $\{a\}$. The same must be true of $\{\{a\}, \{a, d\}\}$, so $\{a, d\} = \{a\}$, which implies that $d = a = b$.

Case 2. $b \neq a$. In this case, b is in one member of $\{\{a\}, \{a, b\}\}$ but not in the other. It must therefore be true that b is in one member of $\{\{a\}, \{a, d\}\}$ but not in the other. This can happen only if b is in $\{a, d\}$, but b is not in $\{a\}$; thus $b = a$ or $b = d$, but $b \neq a$; so $b = d$. ■

4

CHAPTER GRAPHS

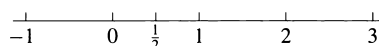


FIGURE 1

Mention the real numbers to a mathematician and the image of a straight line will probably form in her mind, quite involuntarily. And most likely she will neither banish nor too eagerly embrace this mental picture of the real numbers. “Geometric intuition” will allow her to interpret statements about numbers in terms of this picture, and may even suggest methods of proving them. Although the properties of the real numbers which were studied in Part I are not greatly illuminated by a geometric picture, such an interpretation will be a great aid in Part II.

You are probably already familiar with the conventional method of considering the straight line as a picture of the real numbers, i.e., of associating to each real number a point on a line. To do this (Figure 1) we pick, arbitrarily, a point which we label 0, and a point to the right, which we label 1. The point twice as far to the right is labeled 2, the point the same distance from 0 to 1, but to the left of 0, is labeled -1 , etc. With this arrangement, if $a < b$, then the point corresponding to a lies to the left of the point corresponding to b . We can also draw rational numbers, such as $\frac{1}{2}$, in the obvious way. It is usually taken for granted that the irrational numbers also somehow fit into this scheme, so that every real number can be drawn as a point on the line. We will not make too much fuss about justifying this assumption, since this method of “drawing” numbers is intended solely as a method of picturing certain abstract ideas, and our proofs will never rely on these pictures (although we will frequently use a picture to suggest or help explain a proof). Because this geometric picture plays such a prominent, albeit inessential role, geometric terminology is frequently employed when speaking of numbers—thus a number is sometimes called a *point*, and $\mathbf{R}$ is often called the *real line*.

The number $|a - b|$ has a simple interpretation in terms of this geometric picture: it is the distance between a and b , the length of the line segment which has a as one end point and b as the other. This means, to choose an example whose frequent occurrence justifies special consideration, that the set of numbers x which satisfy $|x - a| < \varepsilon$ may be pictured as the collection of points whose distance from a is less than ε . This set of points is the “interval” from $a - \varepsilon$ to $a + \varepsilon$, which may also be described as the points corresponding to numbers x with $a - \varepsilon < x < a + \varepsilon$ (Figure 2).



FIGURE 2

Sets of numbers which correspond to intervals arise so frequently that it is desirable to have special names for them. The set $\{x : a < x < b\}$ is denoted by (a, b) and called the **open interval** from a to b . This notation naturally creates some ambiguity, since (a, b) is also used to denote a pair of numbers, but in context it is always clear (or can easily be made clear) whether one is talking about a pair or an interval. Note that if $a \geq b$, then $(a, b) = \emptyset$, the set with no elements; in prac-

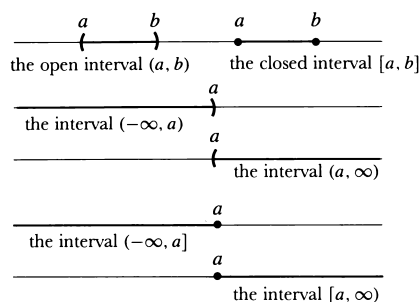


FIGURE 3

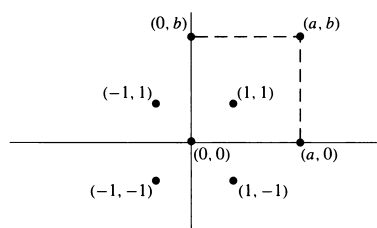


FIGURE 4

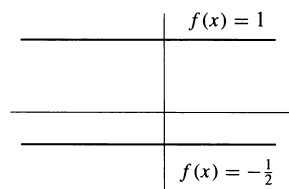


FIGURE 5

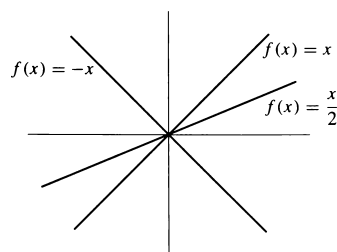


FIGURE 6

tice, however, it is almost always assumed (explicitly if one has been careful, and implicitly otherwise), that whenever an interval (a, b) is mentioned, the number a is less than b .

The set $\{x : a \leq x \leq b\}$ is denoted by $[a, b]$ and is called the **closed interval** from a to b . This symbol is usually reserved for the case $a < b$, but it is sometimes used for $a = b$, also. The usual pictures for the intervals (a, b) and $[a, b]$ are shown in Figure 3; since no reasonably accurate picture could ever indicate the difference between the two intervals, various conventions have been adopted. Figure 3 also shows certain “infinite” intervals. The set $\{x : x > a\}$ is denoted by (a, ∞) , while the set $\{x : x \geq a\}$ is denoted by $[a, \infty)$; the sets $(-\infty, a)$ and $(-\infty, a]$ are defined similarly. At this point a standard warning must be issued: the symbols ∞ and $-\infty$, though usually read “infinity” and “minus infinity,” are *purely* suggestive; there is no number “ ∞ ” which satisfies $\infty \geq a$ for all numbers a . While the symbols ∞ and $-\infty$ will appear in many contexts, it is always necessary to define these uses in ways that refer only to numbers. The set $\mathbf{R}$ of all real numbers is also considered to be an “interval,” and is sometimes denoted by $(-\infty, \infty)$.

Of even greater interest to us than the method of drawing numbers is a method of drawing pairs of numbers. This procedure, probably also familiar to you, requires a “coordinate system,” two straight lines intersecting at right angles. To distinguish these straight lines, we call one the *horizontal axis*, and one the *vertical axis*. (More prosaic terminology, such as the “first” and “second” axes, is probably preferable from a logical point of view, but most people hold their books, or at least their blackboards, in the same way, so that “horizontal” and “vertical” are more descriptive.) Each of the two axes could be labeled with real numbers, but we can also label points on the horizontal axis with pairs $(a, 0)$ and points on the vertical axis with pairs $(0, b)$, so that the intersection of the two axes, the “origin” of the coordinate system, is labeled $(0, 0)$. Any pair (a, b) can now be drawn as in Figure 4, lying at the vertex of the rectangle whose other three vertices are labeled $(0, 0)$, $(a, 0)$, and $(0, b)$. The numbers a and b are called the *first* and *second coordinates*, respectively, of the point determined in this way.

Our real concern, let us recall, is a method of drawing functions. Since a function is just a collection of pairs of numbers, we can draw a function by drawing each of the pairs in the function. The drawing obtained in this way is called the **graph** of the function. In other words, the graph of f contains all the points corresponding to pairs $(x, f(x))$. Since most functions contain infinitely many pairs, drawing the graph promises to be a laborious undertaking, but, in fact, many functions have graphs which are quite easy to draw.

Not surprisingly, the simplest functions of all, the constant functions $f(x) = c$, have the simplest graphs. It is easy to see that the graph of the function $f(x) = c$ is a straight line parallel to the horizontal axis, at distance c from it (Figure 5).

The functions $f(x) = cx$ also have particularly simple graphs—straight lines through $(0, 0)$, as in Figure 6. A proof of this fact is indicated in Figure 7:

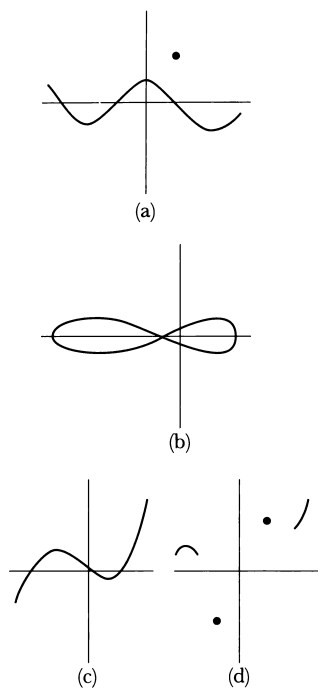


FIGURE 10

f is to be of the form $f(x) = \alpha x + \beta$, then we must have

$$\alpha a + \beta = b,$$

$$\alpha c + \beta = d;$$

therefore $\alpha = (d - b)/(c - a)$ and $\beta = b - [(d - b)/(c - a)]a$, so

$$f(x) = \frac{d - b}{c - a}x + b - \frac{d - b}{c - a}a = \frac{d - b}{c - a}(x - a) + b,$$

a formula most easily remembered by using the “point-slope form” (see Problem 6).

Of course, this solution is possible only if $a \neq c$; the graphs of linear functions account only for the straight lines which are not parallel to the vertical axis. The vertical straight lines are not the graph of *any* function at all; in fact, the graph of a function can never contain even two distinct points on the same vertical line. This conclusion is immediate from the definition of a function—two points on the same vertical line correspond to pairs of the form (a, b) and (a, c) and, by definition, a function cannot contain (a, b) and (a, c) if $b \neq c$. Conversely, if a set of points in the plane has the property that no two points lie on the same vertical line, then it is surely the graph of a function. Thus, the first two sets in Figure 10 are not graphs of functions and the last two are; notice that the fourth is the graph of a function whose domain is not all of $\mathbf{R}$, since some vertical lines have no points on them at all.

After the linear functions the simplest is perhaps the function $f(x) = x^2$. If we draw some of the pairs in f , i.e., some of the pairs of the form (x, x^2) , we obtain a picture like Figure 11.

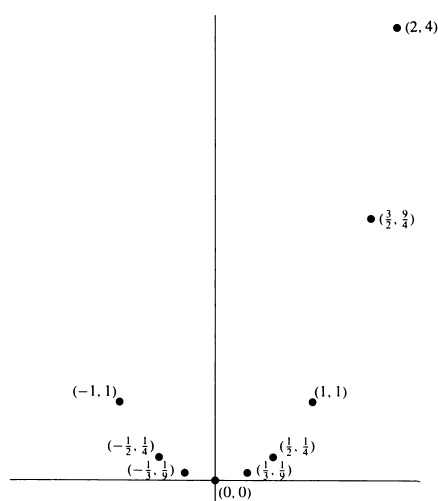


FIGURE 11

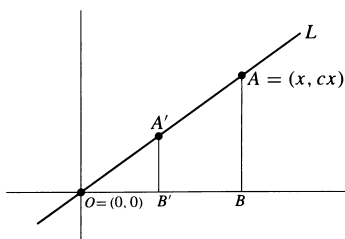


FIGURE 7

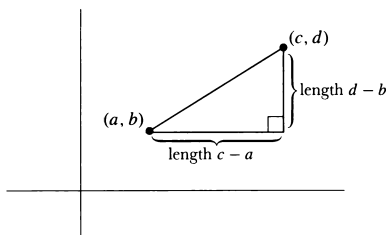


FIGURE 8

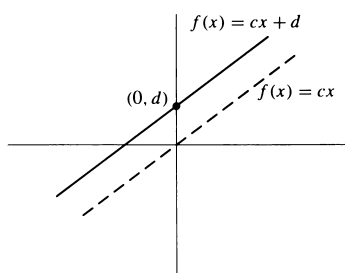


FIGURE 9

Let x be some number not equal to 0, and let L be the straight line which passes through the origin O , corresponding to $(0, 0)$, and through the point A , corresponding to (x, cx) . A point A' , with first coordinate y , will lie on L when the triangle $A'B'O$ is similar to the triangle ABO , thus when

$$\frac{A'B'}{OB'} = \frac{AB}{OB} = c;$$

this is precisely the condition that A' corresponds to the pair (y, cy) , i.e., that A' lies on the graph of f . The argument has implicitly assumed that $c > 0$, but the other cases are treated easily enough. The number c , which measures the ratio of the sides of the triangles appearing in the proof, is called the *slope* of the straight line, and a line parallel to this line is also said to have slope c .

This demonstration has neither been labeled nor treated as a formal proof. Indeed, a rigorous demonstration would necessitate a digression which we are not at all prepared to follow. The rigorous proof of *any* statement connecting geometric and algebraic concepts would first require a real proof (or a precisely stated assumption) that the points on a straight line correspond in an exact way to the real numbers. Aside from this, it would be necessary to develop plane geometry as precisely as we intend to develop the properties of real numbers. Now the detailed development of plane geometry is a beautiful subject, but it is by no means a prerequisite for the study of calculus. We shall use geometric pictures only as an aid to intuition; for our purposes (and for most of mathematics) it is perfectly satisfactory to *define* the plane to be the set of all pairs of real numbers, and to *define* straight lines as certain collections of pairs, including, among others, the collections $\{(x, cx) : x \text{ a real number}\}$. To provide this artificially constructed geometry with all the structure of geometry studied in high school, one more definition is required. If (a, b) and (c, d) are two points in the plane, i.e., pairs of real numbers, we *define* the **distance** between (a, b) and (c, d) to be

$$\sqrt{(a - c)^2 + (b - d)^2}.$$

If the motivation for this definition is not clear, Figure 8 should serve as adequate explanation—with this definition the Pythagorean theorem has been built into our geometry.*

Reverting once more to our informal geometric picture, it is not hard to see (Figure 9) that the graph of the function $f(x) = cx + d$ is a straight line with slope c , passing through the point $(0, d)$. For this reason, the functions $f(x) = cx + d$ are called **linear functions**. Simple as they are, linear functions occur frequently, and you should feel comfortable working with them. The following is a typical problem whose solution should not cause any trouble. Given two distinct points (a, b) and (c, d) , find the linear function f whose graph goes through (a, b) and (c, d) . This amounts to saying that $f(a) = b$ and $f(c) = d$. If

*The fastidious reader might object to this definition on the grounds that nonnegative numbers are not yet known to have square roots. This objection is really unanswerable at the moment—the definition will just have to be accepted with reservations, until this little point is settled.

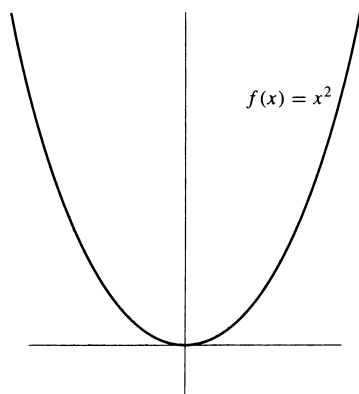
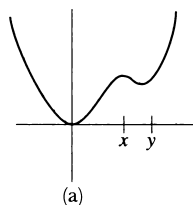
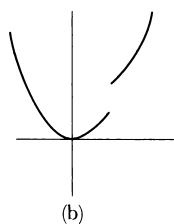


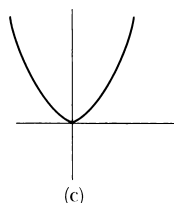
FIGURE 12



(a)



(b)



(c)

FIGURE 13

It is not hard to convince yourself that all the pairs (x, x^2) lie along a curve like the one shown in Figure 12; this curve is known as a **parabola**.

Since a graph is just a drawing on paper, made (in this case) with printer's ink, the question "Is this what the graph really looks like?" is hard to phrase in any sensible manner. No drawing is ever *really* correct since the line has thickness. Nevertheless, there are some questions which one *can* ask: for example, how can you be sure that the graph does not look like one of the drawings in Figure 13? It is easy to see, and even to prove, that the graph cannot look like (a); for if $0 < x < y$, then $x^2 < y^2$, so the graph should be higher at y than at x , which is not the case in (a). It is also easy to see, simply by drawing a very accurate graph, first plotting many pairs (x, x^2) , that the graph cannot have a large "jump" as in (b) or a "corner" as in (c). In order to prove these assertions, however, we first need to say, in a mathematical way, what it means for a function not to have a "jump" or "corner"; these ideas already involve some of the fundamental concepts of calculus. Eventually we will be able to define them rigorously, but meanwhile you may amuse yourself by attempting to define these concepts, and then examining your definitions critically. Later these definitions may be compared with the ones mathematicians have agreed upon. If they compare favorably, you are certainly to be congratulated!

The functions $f(x) = x^n$, for various natural numbers n , are sometimes called **power functions**. Their graphs are most easily compared as in Figure 14, by drawing several at once.

The power functions are only special cases of polynomial functions, introduced in the previous chapter. Two particular polynomial functions are graphed in

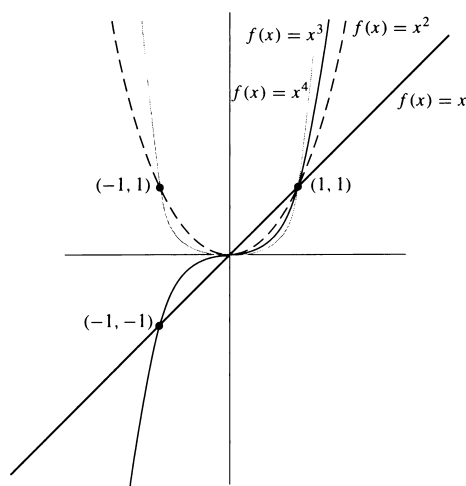


FIGURE 14

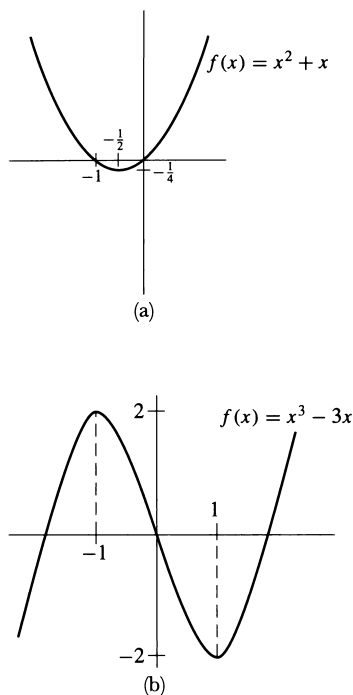


FIGURE 15

Figure 15, while Figure 16 is meant to give a general idea of the graph of the polynomial function

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_0,$$

in the case $a_n > 0$.

In general, the graph of f will have at most $n - 1$ “peaks” or “valleys” (a “peak” is a point like $(x, f(x))$ in Figure 16, while a “valley” is a point like $(y, f(y))$). The number of peaks and valleys may actually be much smaller (the power functions, for example, have at most one valley). Although these assertions are easy to make, we will not even contemplate giving proofs until Part III (once the powerful methods of Part III are available, the proofs will be very easy).

Figure 17 illustrates the graphs of several rational functions. The rational functions exhibit even greater variety than the polynomial functions, but their behavior will also be easy to analyze once we can use the derivative, the basic tool of Part III.

Many interesting graphs can be constructed by “piecing together” the graphs of functions already studied. The graph in Figure 18 is made up entirely of straight lines. The function f with this graph satisfies

$$f\left(\frac{1}{n}\right) = (-1)^{n+1},$$

$$f\left(\frac{-1}{n}\right) = (-1)^{n+1},$$

$$f(x) = 1, \quad |x| \geq 1,$$

and is a linear function on each interval $[1/(n+1), 1/n]$ and $[-1/n, -1/(n+1)]$. (The number 0 is not in the domain of f .) Of course, one can write out an explicit formula for $f(x)$, when x is in $[1/(n+1), 1/n]$; this is a good exercise in the use of linear functions, and will also convince you that a picture is worth a thousand words.

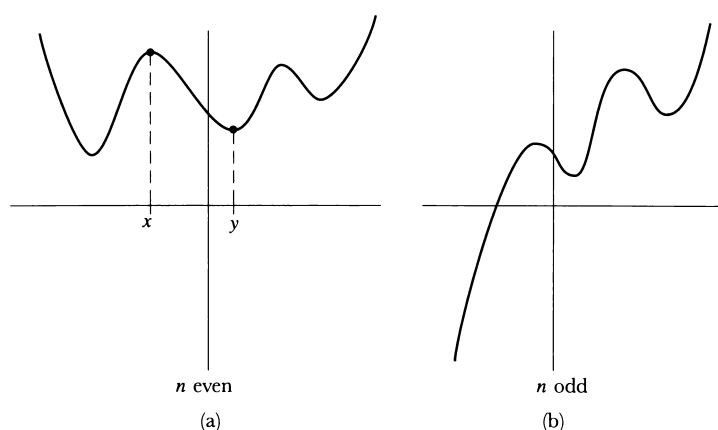


FIGURE 16

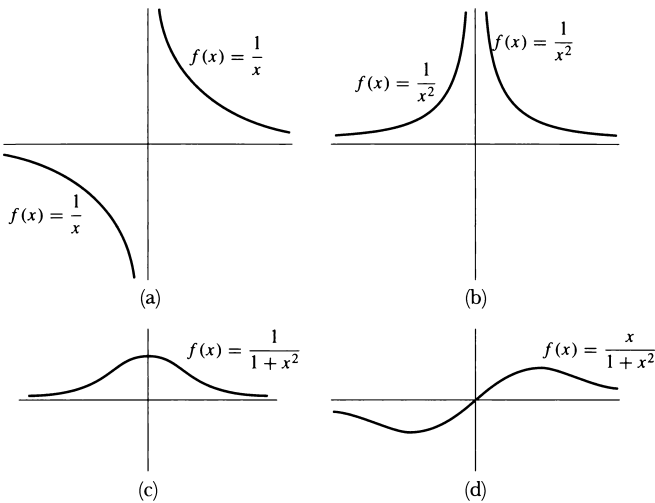


FIGURE 17

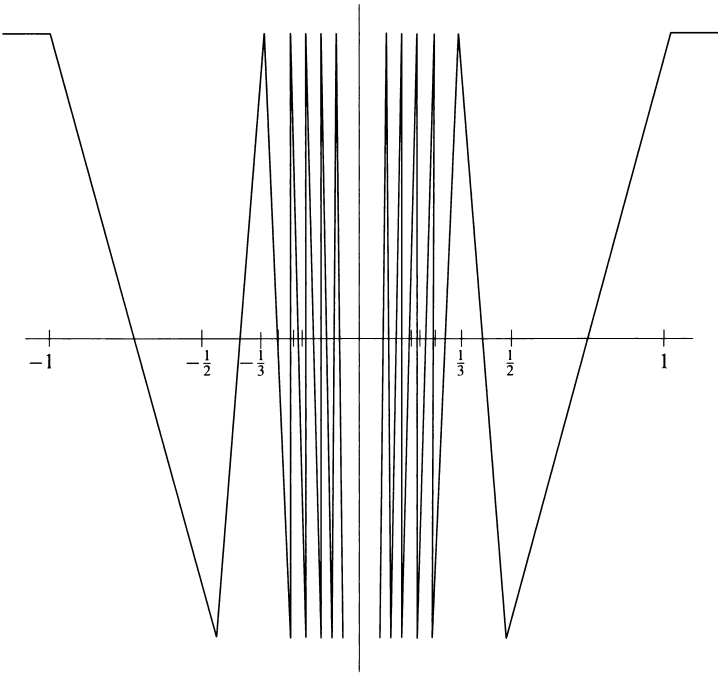


FIGURE 18

It is actually possible to define, in a much simpler way, a function which exhibits this same property of oscillating infinitely often near 0, by using the sine function, which we will discuss in detail in Chapter 15. As usual, we are using radian measure, so an angle of 2π means an angle “all the way around” a circle, an angle of π an angle half way around (or 180° in layman’s terms), an angle of $\pi/2$ a right angle, etc.

The graph of the sine function is shown in Figure 19.

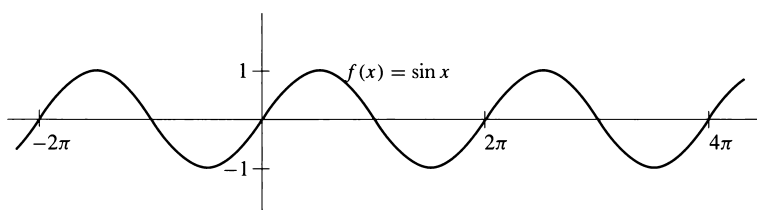


FIGURE 19

Now consider the function $f(x) = \sin 1/x$. The graph of f is shown in Figure 20. To draw this graph it helps to first observe that

$$f(x) = 0 \quad \text{for } x = \frac{1}{\pi}, \frac{1}{2\pi}, \frac{1}{3\pi}, \dots,$$

$$f(x) = 1 \quad \text{for } x = \frac{1}{\frac{1}{2}\pi}, \frac{1}{\frac{1}{2}\pi + 2\pi}, \frac{1}{\frac{1}{2}\pi + 4\pi}, \dots,$$

$$f(x) = -1 \quad \text{for } x = \frac{1}{\frac{3}{2}\pi}, \frac{1}{\frac{3}{2}\pi + 2\pi}, \frac{1}{\frac{3}{2}\pi + 4\pi}, \dots$$

Notice that when x is large, so that $1/x$ is small, $f(x)$ is also small; when x is

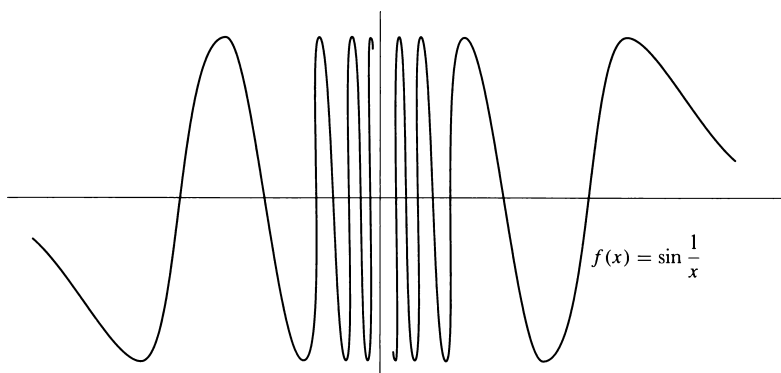


FIGURE 20

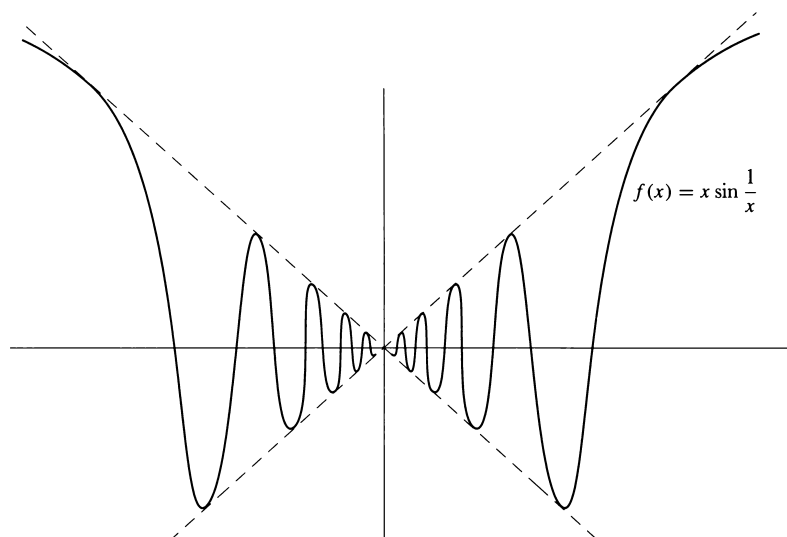
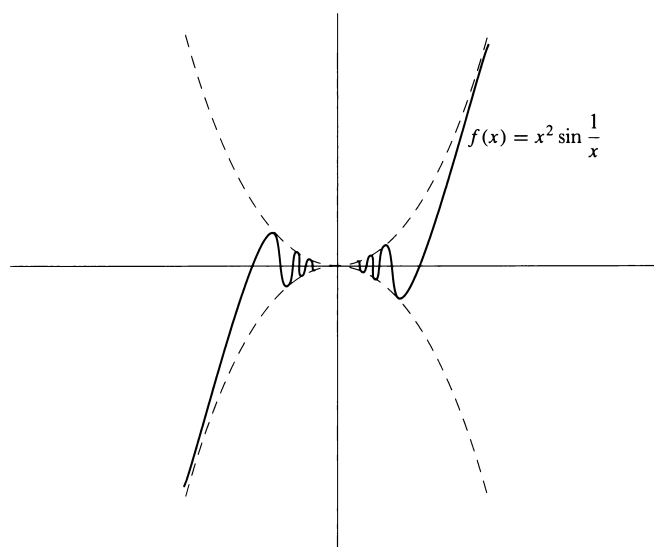


FIGURE 21

“large negative,” that is, when $|x|$ is large for negative x , again $f(x)$ is close to 0, although $f(x) < 0$.

An interesting modification of this function is $f(x) = x^2 \sin 1/x$. The graph of this function is sketched in Figure 21. Since $\sin 1/x$ oscillates infinitely often near 0 between 1 and -1 , the function $f(x) = x^2 \sin 1/x$ oscillates infinitely often between x and $-x$. The behavior of the graph for x large or large negative is harder to



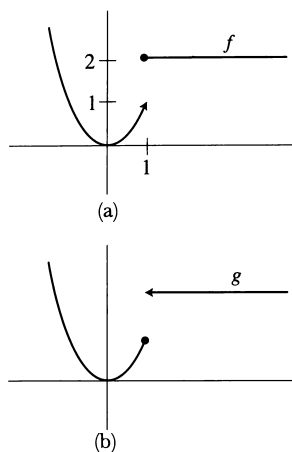


FIGURE 23

analyze. Since $\sin 1/x$ is getting close to 0, while x is getting larger and larger, there seems to be no telling what the product will do. It is possible to decide, but this is another question that is best deferred to Part III. The graph of $f(x) = x^2 \sin 1/x$ has also been illustrated (Figure 22).

For these infinitely oscillating functions, it is clear that the graph cannot hope to be really “accurate.” The best we can do is to show part of it, and leave out the part near 0 (which is the interesting part). Actually, it is easy to find much simpler functions whose graphs cannot be “accurately” drawn. The graphs of

$$f(x) = \begin{cases} x^2, & x < 1 \\ 2, & x \geq 1 \end{cases} \quad \text{and} \quad g(x) = \begin{cases} x^2, & x \leq 1 \\ 2, & x > 1 \end{cases}$$

can only be distinguished by some convention similar to that used for open and closed intervals (Figure 23).

Our last example is a function whose graph is spectacularly nondrawable:

$$f(x) = \begin{cases} 0, & x \text{ irrational} \\ 1, & x \text{ rational} \end{cases}$$

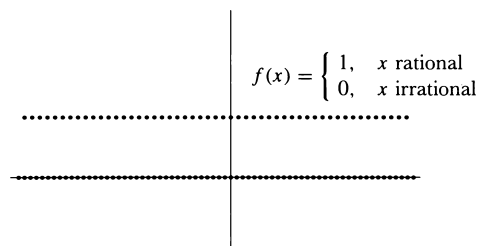


FIGURE 24

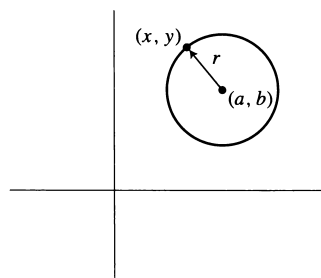


FIGURE 25

The graph of f must contain infinitely many points on the horizontal axis and also infinitely many points on a line parallel to the horizontal axis, but it must not contain either of these lines entirely. Figure 24 shows the usual textbook picture of the graph. To distinguish the two parts of the graph, the dots are placed closer together on the line corresponding to irrational x . (There is actually a mathematical reason behind this convention, but it depends on some sophisticated ideas, introduced in Problems 21-5 and 21-6.)

The peculiarities exhibited by some functions are so engrossing that it is easy to forget some of the simplest, and most important, subsets of the plane, which are not the graphs of functions. The most important example of all is the **circle**. A circle with center (a, b) and radius $r > 0$ contains, by definition, all the points (x, y) whose distance from (a, b) is equal to r . The circle thus consists (Figure 25) of all points (x, y) with

$$\sqrt{(x-a)^2 + (y-b)^2} = r$$

or

$$(x-a)^2 + (y-b)^2 = r^2.$$

The circle with center $(0,0)$ and radius 1, often regarded as a sort of standard copy, is called the *unit circle*.

A close relative of the circle is the **ellipse**. This is defined as the set of points, the *sum* of whose distances from two “focus” points is a constant. (When the two foci are the same, we obtain a circle.) If, for convenience, the focus points are taken to be $(-c, 0)$ and $(c, 0)$, and the sum of the distances is taken to be $2a$ (the factor 2 simplifies some algebra), then (x, y) is on the ellipse if and only if

$$\sqrt{(x - (-c))^2 + y^2} + \sqrt{(x - c)^2 + y^2} = 2a$$

or

$$\sqrt{(x + c)^2 + y^2} = 2a - \sqrt{(x - c)^2 + y^2}$$

or

$$x^2 + 2cx + c^2 + y^2 = 4a^2 - 4a\sqrt{(x - c)^2 + y^2} + x^2 - 2cx + c^2 + y^2$$

or

$$4(cx - a^2) = -4a\sqrt{(x - c)^2 + y^2}$$

or

$$c^2x^2 - 2cxa^2 + a^4 = a^2(x^2 - 2cx + c^2 + y^2)$$

or

$$(c^2 - a^2)x^2 - a^2y^2 = a^2(c^2 - a^2)$$

or

$$\frac{x^2}{a^2} + \frac{y^2}{a^2 - c^2} = 1.$$

This is usually written simply

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1,$$

where $b = \sqrt{a^2 - c^2}$ (since we must clearly choose $a > c$, it follows that $a^2 - c^2 > 0$). A picture of an ellipse is shown in Figure 26. The ellipse intersects the horizontal axis when $y = 0$, so that

$$\frac{x^2}{a^2} = 1, \quad x = \pm a,$$

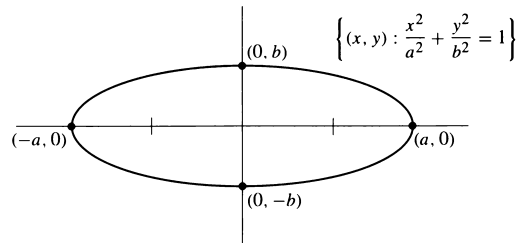


FIGURE 26

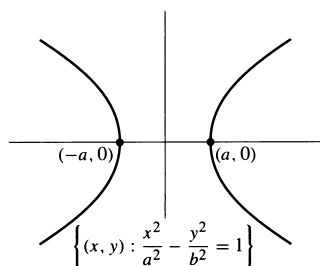


FIGURE 27

and it intersects the vertical axis when $x = 0$, so that

$$\frac{y^2}{b^2} = 1, \quad y = \pm b.$$

The **hyperbola** is defined analogously, except that we require the *difference* of the two distances to be constant. Choosing the points $(-c, 0)$ and $(c, 0)$ once again, and the constant difference as $2a$, we obtain, as the condition that (x, y) be on the hyperbola,

$$\sqrt{(x+c)^2 + y^2} - \sqrt{(x-c)^2 + y^2} = \pm 2a,$$

which may be simplified to

$$\frac{x^2}{a^2} + \frac{y^2}{a^2 - c^2} = 1.$$

In this case, however, we must clearly choose $c > a$, so that $a^2 - c^2 < 0$. If $b = \sqrt{c^2 - a^2}$, then (x, y) is on the hyperbola if and only if

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1.$$

The picture is shown in Figure 27. It contains two pieces, because the difference between the distances of (x, y) from $(-c, 0)$ and $(c, 0)$ may be taken in two different orders. The hyperbola intersects the horizontal axis when $y = 0$, so that $x = \pm a$, but it never intersects the vertical axis.

It is interesting to compare (Figure 28) the hyperbola with $a = b = \sqrt{2}$ and the graph of the function $f(x) = 1/x$. The drawings look quite similar, and the two sets are actually identical, except for a rotation through an angle of $\pi/4$ (Problem 23).

Clearly no rotation of the plane will change circles or ellipses into the graphs of functions. Nevertheless, the study of these important geometric figures can often be reduced to the study of functions. Ellipses, for example, are made up of the

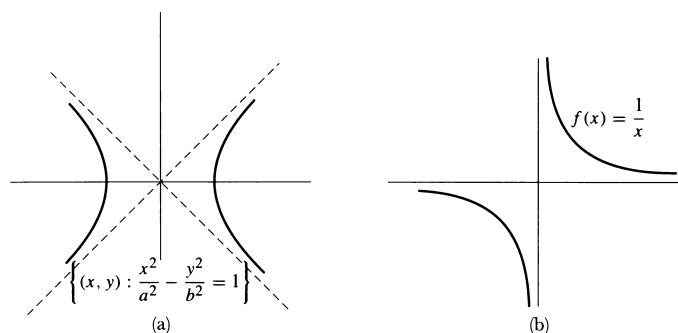


FIGURE 28

graphs of two functions,

$$f(x) = b\sqrt{1 - (x^2/a^2)}, \quad -a \leq x \leq a$$

and

$$g(x) = -b\sqrt{1 - (x^2/a^2)}, \quad -a \leq x \leq a.$$

Of course, there are many other pairs of functions with this same property. For example, we can take

$$f(x) = \begin{cases} b\sqrt{1 - (x^2/a^2)}, & 0 < x \leq a \\ -b\sqrt{1 - (x^2/a^2)}, & -a \leq x \leq 0 \end{cases}$$

and

$$g(x) = \begin{cases} -b\sqrt{1 - (x^2/a^2)}, & 0 < x \leq a \\ b\sqrt{1 - (x^2/a^2)}, & -a \leq x \leq 0. \end{cases}$$

We could also choose

$$f(x) = \begin{cases} b\sqrt{1 - (x^2/a^2)}, & x \text{ rational, } -a \leq x \leq a \\ -b\sqrt{1 - (x^2/a^2)}, & x \text{ irrational, } -a \leq x \leq a \end{cases}$$

and

$$g(x) = \begin{cases} -b\sqrt{1 - (x^2/a^2)}, & x \text{ rational, } -a \leq x \leq a \\ b\sqrt{1 - (x^2/a^2)}, & x \text{ irrational, } -a \leq x \leq a. \end{cases}$$

But all these other pairs necessarily involve unreasonable functions which jump around. A proof, or even a precise statement of this fact, is too difficult at present. Although you have probably already begun to make a distinction between those functions with reasonable graphs, and those with unreasonable graphs, you may find it very difficult to state a reasonable definition of reasonable functions. A mathematical definition of this concept is by no means easy, and a great deal of this book may be viewed as successive attempts to impose more and more conditions that a “reasonable” function must satisfy. As we define some of these conditions, we will take time out to ask if we have really succeeded in isolating the functions which deserve to be called reasonable. The answer, unfortunately, will always be “no,” or at best, a qualified “yes.”

PROBLEMS

1. Indicate on a straight line the set of all x satisfying the following conditions. Also name each set, using the notation for intervals (in some cases you will also need the $\cup$ sign).
 - (i) $|x - 3| < 1$.
 - (ii) $|x - 3| \leq 1$.
 - (iii) $|x - a| < \varepsilon$.
 - (iv) $|x^2 - 1| < \frac{1}{2}$.

- (v) $\frac{1}{1+x^2} \geq \frac{1}{5}$.
- (vi) $\frac{1}{1+x^2} \leq a$ (give an answer in terms of a , distinguishing various cases).
- (vii) $x^2 + 1 \geq 2$.
- (viii) $(x+1)(x-1)(x-2) > 0$.

2. There is a very useful way of describing the points of the closed interval $[a, b]$ (where we assume, as usual, that $a < b$).

- (a) First consider the interval $[0, b]$, for $b > 0$. Prove that if x is in $[0, b]$, then $x = tb$ for some t with $0 \leq t \leq 1$. What is the significance of the number t ? What is the mid-point of the interval $[0, b]$?
- (b) Now prove that if x is in $[a, b]$, then $x = (1-t)a + tb$ for some t with $0 \leq t \leq 1$. Hint: This expression can also be written as $a + t(b-a)$. What is the midpoint of the interval $[a, b]$? What is the point $1/3$ of the way from a to b ?
- (c) Prove, conversely, that if $0 \leq t \leq 1$, then $(1-t)a + tb$ is in $[a, b]$.
- (d) The points of the *open* interval (a, b) are those of the form $(1-t)a + tb$ for $0 < t < 1$.

3. Draw the set of all points (x, y) satisfying the following conditions. (In most cases your picture will be a sizable portion of a plane, not just a line or curve.)

- (i) $x > y$.
- (ii) $x + a > y + b$.
- (iii) $y < x^2$.
- (iv) $y \leq x^2$.
- (v) $|x - y| < 1$.
- (vi) $|x + y| < 1$.
- (vii) $x + y$ is an integer.
- (viii) $\frac{1}{x+y}$ is an integer.
- (ix) $(x-1)^2 + (y-2)^2 < 1$.
- (x) $x^2 < y < x^4$.

4. Draw the set of all points (x, y) satisfying the following conditions:

- (i) $|x| + |y| = 1$.
- (ii) $|x| - |y| = 1$.
- (iii) $|x-1| = |y-1|$.
- (iv) $|1-x| = |y-1|$.
- (v) $x^2 + y^2 = 0$.
- (vi) $xy = 0$.
- (vii) $x^2 - 2x + y^2 = 4$.
- (viii) $x^2 = y^2$.

5. Draw the set of all points (x, y) satisfying the following conditions:

- (i) $x = y^2$.
- (ii) $\frac{y^2}{a^2} - \frac{x^2}{b^2} = 1$.
- (iii) $x = |y|$.
- (iv) $x = \sin y$.

Hint: You already know the answers when x and y are interchanged.

6. (a) Show that the straight line through (a, b) with slope m is the graph of the function $f(x) = m(x - a) + b$. This formula, known as the “point-slope form” is far more convenient than the equivalent expression $f(x) = mx + (b - ma)$; it is immediately clear from the point-slope form that the slope is m , and that the value of f at a is b .
- (b) For $a \neq c$, show that the straight line through (a, b) and (c, d) is the graph of the function

$$f(x) = \frac{d - b}{c - a}(x - a) + b.$$

- (c) When are the graphs of $f(x) = mx + b$ and $g(x) = m'x + b'$ parallel straight lines?
7. (a) For any numbers A, B , and C , with A and B not both 0, show that the set of all (x, y) satisfying $Ax + By + C = 0$ is a straight line (possibly a vertical one). Hint: First decide when a vertical straight line is described.
- (b) Show conversely that every straight line, including vertical ones, can be described as the set of all (x, y) satisfying $Ax + By + C = 0$.

8. (a) Prove that the graphs of the functions

$$\begin{aligned} f(x) &= mx + b, \\ g(x) &= nx + c, \end{aligned}$$

are perpendicular if $mn = -1$, by computing the squares of the lengths of the sides of the triangle in Figure 29. (Why is this special case, where the lines intersect at the origin, as good as the general case?)

- (b) Prove that the two straight lines consisting of all (x, y) satisfying the conditions

$$\begin{aligned} Ax + By + C &= 0, \\ A'x + B'y + C' &= 0, \end{aligned}$$

are perpendicular if and only if $AA' + BB' = 0$.

9. (a) Prove, using Problem 1-19, that

$$\sqrt{(x_1 + y_1)^2 + (x_2 + y_2)^2} \leq \sqrt{x_1^2 + x_2^2} + \sqrt{y_1^2 + y_2^2}.$$

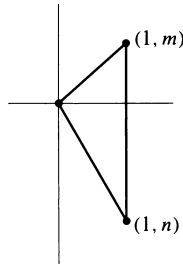


FIGURE 29

(b) Prove that

$$\sqrt{(x_3 - x_1)^2 + (y_3 - y_1)^2} \leq \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} + \sqrt{(x_3 - x_2)^2 + (y_3 - y_2)^2}.$$

Interpret this inequality geometrically (it is called the “triangle inequality”). When does strict inequality hold?

10. Sketch the graphs of the following functions, plotting enough points to get a good idea of the general appearance. (Part of the problem is to make a reasonable decision how many is “enough”; the queries posed below are meant to show that a little thought will often be more valuable than hundreds of individual points.)

(i) $f(x) = x + \frac{1}{x}$. (What happens for x near 0, and for large x ? Where does the graph lie in relation to the graph of the identity function? Why does it suffice to consider only positive x at first?)

(ii) $f(x) = x - \frac{1}{x}$.

(iii) $f(x) = x^2 + \frac{1}{x^2}$.

(iv) $f(x) = x^2 - \frac{1}{x^2}$.

11. Describe the general features of the graph of f if

(i) f is even.

(ii) f is odd.

(iii) f is nonnegative.

(iv) $f(x) = f(x + a)$ for all x (a function with this property is called **periodic**, with **period** a).

12. Graph the functions $f(x) = \sqrt[m]{x}$ for $m = 1, 2, 3, 4$. (There is an easy way to do this, using Figure 14. Be sure to remember, however, that $\sqrt[m]{x}$ means the *positive* m th root of x when m is even; you should also note that there will be an important difference between the graphs when m is even and when m is odd.)

13. (a) Graph $f(x) = |x|$ and $f(x) = x^2$.

(b) Graph $f(x) = |\sin x|$ and $f(x) = \sin^2 x$. (There is an important difference between the graphs, which we cannot yet even describe rigorously. See if you can discover what it is; part (a) is meant to be a clue.)

14. Describe the graph of g in terms of the graph of f if

- (i) $g(x) = f(x) + c$.
- (ii) $g(x) = f(x + c)$. (It is easy to make a mistake here.)
- (iii) $g(x) = cf(x)$.
- (iv) $g(x) = f(cx)$. (Distinguish the cases $c = 0$, $c > 0$, $c < 0$.)
- (v) $g(x) = f(1/x)$.
- (vi) $g(x) = f(|x|)$.
- (vii) $g(x) = |f(x)|$.
- (viii) $g(x) = \max(f, 0)$.
- (ix) $g(x) = \min(f, 0)$.
- (x) $g(x) = \max(f, 1)$.

15. Draw the graph of $f(x) = ax^2 + bx + c$. Hint: Use the methods of Problem 1-18.

16. Suppose that A and C are not both 0. Show that the set of all (x, y) satisfying

$$Ax^2 + Bx + Cy^2 + Dy + E = 0$$

is either a parabola, an ellipse, or an hyperbola (or a “degenerate case”: two lines [either intersecting or parallel], one line, a point, or $\emptyset$). Hint: The case $C = 0$ is essentially Problem 15, and the case $A = 0$ is just a minor variant. Now consider separately the cases where A and B are both positive or negative, and where one is positive while the other is negative. When do we have a circle?

17. The symbol $[x]$ denotes the largest integer which is $\leq x$. Thus, $[2.1] = [2] = 2$ and $[-0.9] = [-1] = -1$. Draw the graph of the following functions (they are all quite interesting, and several will reappear frequently in other problems).

- (i) $f(x) = [x]$.
- (ii) $f(x) = x - [x]$.
- (iii) $f(x) = \sqrt{x - [x]}$.
- (iv) $f(x) = [x] + \sqrt{x - [x]}$.
- (v) $f(x) = \left[\frac{1}{x} \right]$.
- (vi) $f(x) = \frac{1}{\left[\frac{1}{x} \right]}$.

18. Graph the following functions.

- (i) $f(x) = \{x\}$, where $\{x\}$ is defined to be the distance from x to the nearest integer.
- (ii) $f(x) = \{2x\}$.
- (iii) $f(x) = \{x\} + \frac{1}{2}\{2x\}$.
- (iv) $f(x) = \{4x\}$.
- (v) $f(x) = \{x\} + \frac{1}{2}\{2x\} + \frac{1}{4}\{4x\}$.

Many functions may be described in terms of the decimal expansion of a number. Although we will not be in a position to describe infinite decimals rigorously until Chapter 23, your intuitive notion of infinite decimals should suffice to carry you through the following problem, and others which occur before Chapter 23. There is one ambiguity about infinite decimals which must be eliminated: Every decimal ending in a string of 9's is equal to another ending in a string of 0's (e.g., $1.23999\ldots = 1.24000\ldots$). We will always use the one ending in 9's.

***19.** Describe as best you can the graphs of the following functions (a complete picture is usually out of the question).

- (i) $f(x)$ = the 1st number in the decimal expansion of x .
- (ii) $f(x)$ = the 2nd number in the decimal expansion of x .
- (iii) $f(x)$ = the number of 7's in the decimal expansion of x if this number is finite, and 0 otherwise.
- (iv) $f(x) = 0$ if the number of 7's in the decimal expansion of x is finite, and 1 otherwise.
- (v) $f(x)$ = the number obtained by replacing all digits in the decimal expansion of x which come after the first 7 (if any) by 0.
- (vi) $f(x) = 0$ if 1 never appears in the decimal expansion of x , and n if 1 first appears in the n th place.

***20.** Let

$$f(x) = \begin{cases} 0, & x \text{ irrational} \\ \frac{1}{q}, & x = \frac{p}{q} \text{ rational in lowest terms.} \end{cases}$$

(A number p/q is in **lowest terms** if p and q are integers with no common factor, and $q > 0$). Draw the graph of f as well as you can (don't sprinkle points randomly on the paper; consider first the rational numbers with $q = 2$, then those with $q = 3$, etc.).

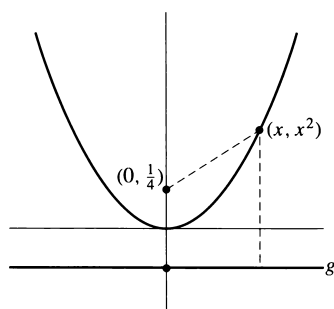


FIGURE 30

- 21.** (a) The points on the graph of $f(x) = x^2$ are the ones of the form (x, x^2) . Prove that each such point is equidistant from the point $(0, \frac{1}{4})$ and the graph of $g(x) = -\frac{1}{4}$. (See Figure 30.)
- (b) Given a horizontal line L , the graph of $g(x) = \gamma$, and a point $P = (\alpha, \beta)$ not on L , so that $\gamma \neq \beta$, show that the set of all points (x, y) equidistant from P and L is the graph of a function of the form $f(x) = ax^2 + bx + c$. What is this set if $\gamma = \beta$?

***22.** (a) Show that the square of the distance from (c, d) to (x, mx) is

$$x^2(m^2 + 1) + x(-2md - 2c) + d^2 + c^2.$$

Using Problem 1-18 to find the minimum of these numbers, show that the distance from (c, d) to the graph of $f(x) = mx$ is

$$|cm - d|/\sqrt{m^2 + 1}.$$

- (b) Find the distance from (c, d) to the graph of $f(x) = mx + b$. (Reduce this case to part (a).)

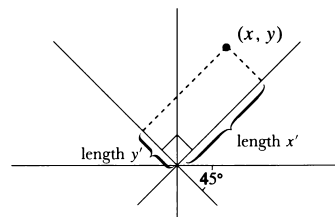


FIGURE 31

- *23. (a) Using Problem 22, show that the numbers x' and y' indicated in Figure 31 are given by

$$x' = \frac{1}{\sqrt{2}}x + \frac{1}{\sqrt{2}}y,$$

$$y' = -\frac{1}{\sqrt{2}}x + \frac{1}{\sqrt{2}}y.$$

- (b) Show that the set of all (x, y) with $(x'/\sqrt{2})^2 - (y'/\sqrt{2})^2 = 1$ is the same as the set of all (x, y) with $xy = 1$.

APPENDIX 1. VECTORS

Suppose that v is a point in the plane; in other words, v is a pair of numbers

$$v = (v_1, v_2).$$

For convenience, we will use this convention that subscripts indicate the first and second pairs of a point that has been described by a single letter. Thus, if we mention the points w and z , it will be understood that w is the pair (w_1, w_2) , while z is the pair (z_1, z_2) .

Instead of the actual pair of numbers (v_1, v_2) , we often picture v as an arrow from the origin O to this point (Figure 1), and we refer to these arrows as *vectors* in the plane. Of course, we've haven't really said anything new yet, we've simply introduced an alternate term for a point of the plane, and another mental picture. The real point of the new terminology is to emphasize that we are going to do some new things with points in the plane.

For example, suppose that we have two vectors (i.e., points) in the plane,

$$v = (v_1, v_2), \quad w = (w_1, w_2).$$

Then we can define a new vector (a new point of the plane) $v + w$ by the equation

$$(1) \quad v + w = (v_1 + w_1, v_2 + w_2).$$

Notice that all the letters on the right side of this equation are numbers, and the $+$ sign is just our usual addition of numbers. On the other hand, the $+$ sign on the left side is new: previously, the sum of two points in the plane wasn't defined, and we've simply used equation (1) as a *definition*.

A very fussy mathematician might want to use some new symbol for this newly defined operation, like

$$v \oplus w, \quad \text{or perhaps} \quad v \oplus w,$$

but there's really no need to insist on this; since $v + w$ hasn't been defined before, there's no possibility of confusion, so we might as well keep the notation simple.

Of course, any one can make new notation; for example, since it's our definition, we could just as well have defined $v + w$ as $(v_1 + w_1 \cdot w_2, v_2 + w_1^2)$, or by some other equally weird formula. The real question is, does our new construction have any particular significance?

Figure 2 shows two vectors v and w , as well as the point

$$(v_1 + w_1, v_2 + w_2),$$

which, for the moment, we have simply indicated in the usual way, without drawing an arrow. Note that it is easy to compute the slope of the line L between v and our new point: as indicated in Figure 2, this slope is just

$$\frac{(v_2 + w_2) - v_2}{(v_1 + w_1) - v_1} = \frac{w_2}{w_1},$$

and this, of course, is the slope of our vector w , from the origin O to (w_1, w_2) . In other words, the line L is parallel to w .

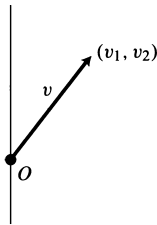


FIGURE 1

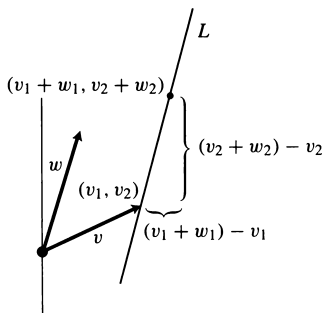


FIGURE 2

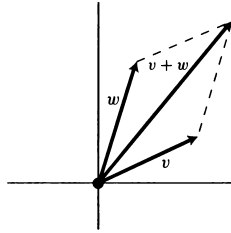


FIGURE 3

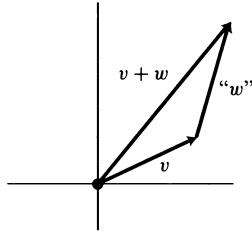


FIGURE 4

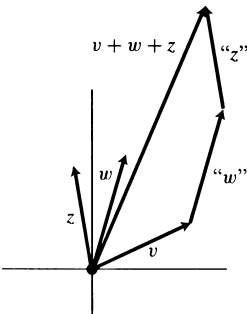


FIGURE 5

Similarly, the slope of the line M between (w_1, w_2) and our new point is

$$\frac{(v_2 + w_2) - w_2}{(v_1 + w_1) - w_1} = \frac{v_2}{v_1},$$

which is the slope of the vector v ; so M is parallel to v . In short, the new point $v + w$ lies on the parallelogram having v and w as sides. When we draw $v + w$ as an arrow (Figure 3), it points along the diagonal of this parallelogram. In physics, vectors are used to symbolize forces, and the sum of two vectors represents the resultant force when two different forces are applied simultaneously to the same object.

Figure 4 shows another way of visualizing the sum $v + w$. If we use “ w ” to denote an arrow parallel to w , and having the same length, but starting at v instead of at the origin, then $v + w$ is the vector from O to the final endpoint; thus we get to $v + w$ by first following v , and then following w .

Many of the properties of $+$ for ordinary numbers also hold for this new $+$ for vectors. For example, the “commutative law”

$$v + w = w + v,$$

is obvious from the geometric picture, since the parallelogram spanned by v and w is the same as the parallelogram spanned by w and v . It is also easily checked analytically, since it states that

$$(v_1 + w_1, v_2 + w_2) = (w_1 + v_1, w_2 + v_2),$$

and thus simply depends on the commutative law for numbers:

$$\begin{aligned} v_1 + w_1 &= w_1 + v_1, \\ v_2 + w_2 &= w_2 + v_2. \end{aligned}$$

Similarly, unraveling definitions, we find the “associative law”

$$[v + w] + z = v + [w + z].$$

Figure 5 indicates a method of finding $v + w + z$.

The origin $O = (0, 0)$ is an “additive identity,”

$$O + v = v + O = v,$$

and if we define

$$-v = (-v_1, -v_2),$$

then we also have

$$v + (-v) = -v + v = O.$$

Naturally we can also define

$$w - v = w + (-v),$$

exactly as with numbers; equivalently,

$$w - v = (w_1 - v_1, w_2 - v_2).$$

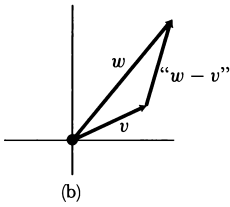
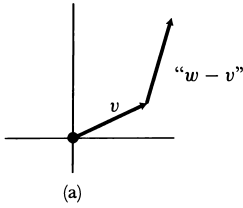


FIGURE 6

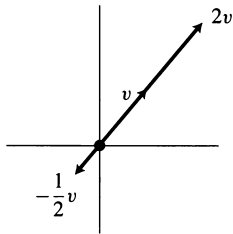


FIGURE 7

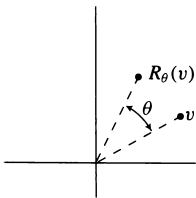


FIGURE 8

Just as with numbers, our definition of $w - v$ simply means that it satisfies

$$v + (w - v) = w.$$

Figure 6(a) shows v and an arrow “ $w - v$ ” that is parallel to $w - v$ but that starts at the endpoint of v . As we established with Figure 4, the vector from the origin to the endpoint of this arrow is just $v + (w - v) = w$ (Figure 6(b)). In other words, we can picture $w - v$ geometrically as the arrow that goes from v to w (except that it must then be moved back to the origin).

There is also a way of multiplying a number by a vector: For a number a and a vector $v = (v_1, v_2)$, we define

$$a \cdot v = (av_1, av_2)$$

(We sometimes simply write av instead of $a \cdot v$; of course, it is then especially important to remember that v denotes a vector, rather than a number.) The vector $a \cdot v$ points in the same direction as v when $a > 0$ and in the opposite direction when $a < 0$ (Figure 7).

You can easily check the following formulas:

$$a \cdot (b \cdot v) = (ab) \cdot v,$$

$$1 \cdot v = v,$$

$$0 \cdot v = O,$$

$$-1 \cdot v = -v.$$

Notice that we have only defined a product of a number and a vector; we have not defined a way of ‘multiplying’ two vectors to get another vector.* However, there are various ways of ‘multiplying’ vectors to get numbers, which are explored in the following problems.

PROBLEMS

- Given a point v of the plane, let $R_\theta(v)$ be the result of rotating v around the origin through an angle of θ (Figure 8). The aim of this problem is to obtain a formula for R_θ , with minimal calculation.

- Show that

$$R_\theta(1, 0) = (\cos \theta, \sin \theta), \quad [\text{we should really write } R_\theta((1, 0)), \text{ etc.}]$$

$$R_\theta(0, 1) = (-\sin \theta, \cos \theta).$$

- Explain why we have

$$R_\theta(v + w) = R_\theta(v) + R_\theta(w),$$

$$R_\theta(a \cdot w) = a \cdot R_\theta(w).$$

- Now show that for any point (x, y) we have

$$R_\theta(x, y) = (x \cos \theta - y \sin \theta, x \sin \theta + y \cos \theta).$$

* If you jump to Chapter 25, you'll find that there is an important way of defining a product, but this is something very special for the plane—it doesn't work for vectors in 3-space, for example, even though the other constructions do.

(d) Use this result to give another solution to Problem 4-23.

2. Given v and w , we define the *number*

$$v \cdot w = v_1 w_1 + v_2 w_2;$$

this is often called the ‘dot product’ or ‘scalar product’ of v and w (‘scalar’ being a rather old-fashioned word for a number, as opposed to a vector).

- (a) Given v , find a vector w such that $v \cdot w = 0$. Now describe the set of all such vectors w .
 (b) Show that

$$\begin{aligned} v \cdot w &= w \cdot v \\ v \cdot (w + z) &= v \cdot w + v \cdot z \end{aligned}$$

and that

$$a \cdot (v \cdot w) = (a \cdot v) \cdot w = v \cdot (a \cdot w).$$

Notice that the last of these equations involves *three* products: the dot product $\cdot$ of two vectors; the product $\cdot$ of a number and a vector; and the ordinary product $\cdot$ of two numbers.

- (c) Show that $v \cdot v \geq 0$, and that $v \cdot v = 0$ only when $v = O$. Hence we can define the *norm* $\|v\|$ as

$$\|v\| = \sqrt{v \cdot v},$$

which will be 0 only for $v = O$. What is the geometric interpretation of the norm?

- (d) Prove that

$$\|v + w\| \leq \|v\| + \|w\|,$$

and that equality holds if and only if $v = 0$ or $w = 0$ or $w = a \cdot v$ for some number $a > 0$.

- (e) Show that

$$v \cdot w = \frac{\|v + w\|^2 - \|v - w\|^2}{4}.$$

3. (a) Let R_θ be rotation by an angle of θ (Problem 1). Show that

$$R_\theta(v) \cdot R_\theta(w) = v \cdot w.$$

- (b) Let $e = (1, 0)$ be the vector of length 1 pointing along the first axis, and let $w = (\cos \theta, \sin \theta)$; this is a vector of length 1 that makes an angle of θ with the first axis (compare Problem 1). Calculate that

$$e \cdot w = \cos \theta.$$

Conclude that in general

$$v \cdot w = \|v\| \cdot \|w\| \cdot \cos \theta,$$

where θ is the angle between v and w .

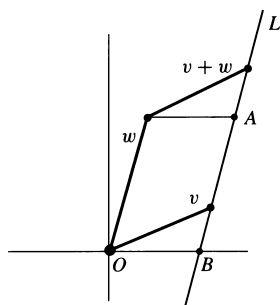


FIGURE 9

4. Given two vectors v and w , we'd expect to have a simple formula, involving the coordinates v_1, v_2, w_1, w_2 , for the area of the parallelogram they span. Figure 9 indicates a strategy for finding such a formula: since the triangle with vertices $w, A, v + w$ is congruent to the triangle OBv , we can reduce the problem to an easier one where one side of the parallelogram lies along the horizontal axis:

- (a) The line L passes through v and is parallel to w , so has slope w_2/w_1 . Conclude that the point B has coordinate

$$\frac{v_1 w_2 - w_1 v_2}{w_2},$$

and that the parallelogram therefore has area

$$\det(v, w) = v_1 w_2 - w_1 v_2.$$

This formula, which defines the *determinant* $\det$, certainly seems to be simple enough, but it can't really be true that $\det(v, w)$ always gives the area. After all, we clearly have

$$\det(w, v) = -\det(v, w),$$

so sometimes $\det$ will be negative! Indeed, it is easy to see that our "derivation" made all sorts of assumptions (that w_2 was positive, that B had a positive coordinate, etc.) Nevertheless, it seems likely that $\det(v, w)$ is $\pm$ the area; the next problem gives an independent proof.

5. (a) If v points along the positive horizontal axis, show that $\det(v, w)$ is the area of the parallelogram spanned by v and w for w above the horizontal axis ($w_2 > 0$), and the negative of the area for w below this axis.
(b) If R_θ is rotation by an angle of θ (Problem 1), show that

$$\det(R_\theta v, R_\theta w) = \det(v, w).$$

Conclude that $\det(v, w)$ is the area of the parallelogram spanned by v and w when the rotation from v to w is counterclockwise, and the negative of the area when it is clockwise.

6. Show that

$$\det(v, w + z) = \det(v, w) + \det(v, z)$$

$$\det(v + w, z) = \det(v, z) + \det(w, z)$$

and that

$$a \det(v, w) = \det(a \cdot v, w) = \det(v, a \cdot w).$$

7. Using the method of Problem 3, show that

$$\det(v, w) = \|v\| \cdot \|w\| \cdot \sin \theta,$$

which is also obvious from the geometric interpretation (Figure 10).

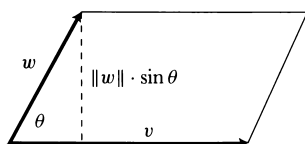


FIGURE 10

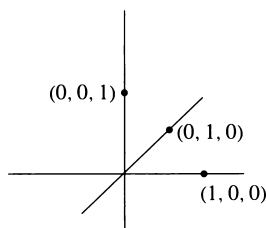


FIGURE 1

APPENDIX 2. THE CONIC SECTIONS

Although we will be concerned almost exclusively with figures in the plane, defined formally as the set of all pairs of real numbers, in this Appendix we want to consider three-dimensional space, which we can describe in terms of triples of real numbers, using a “three-dimensional coordinate system,” consisting of three straight lines intersecting at right angles (Figure 1). Our *horizontal* and *vertical* axes now mutate to two axes in a horizontal plane, with the third axis perpendicular to both.

One of the simplest subsets of this three-dimensional space is the (infinite) *cone* illustrated in Figure 2; this cone may be produced by rotating a “generating line,” of slope C say, around the third axis.

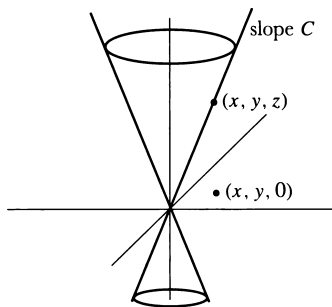


FIGURE 2

For any given first two coordinates x and y , the point $(x, y, 0)$ in the horizontal plane has distance $\sqrt{x^2 + y^2}$ from the origin, and thus

$$(1) \quad (x, y, z) \text{ is on the cone if and only if } z = \pm C\sqrt{x^2 + y^2}.$$

We can descend from these three-dimensional vistas to the more familiar two-dimensional one by asking what happens when we intersect this cone with some plane P (Figure 3).

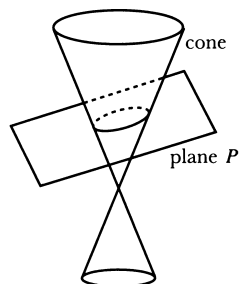


FIGURE 3

If the plane is parallel to the horizontal plane, there's certainly no mystery—the intersection is just a circle. Otherwise, the plane P intersects the horizontal plane

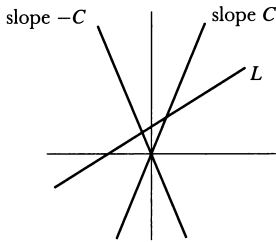


FIGURE 4

in a straight line. We can make things a lot simpler for ourselves if we rotate everything around the vertical axis so that this intersection line points straight out from the plane of the paper, while the first axis is in the usual position that we are familiar with. The plane P is thus viewed “straight on,” so that all we see (Figure 4) is its intersection L with the plane of the first and third axes; from this view-point the cone itself simply appears as two straight lines.

If this line L happens to be vertical, consisting of all points (a, z) for some a , then equation (1) says that the intersection of the cone and the plane consists of all points (a, y, z) with

$$z^2 - C^2 y^2 = C^2 a^2,$$

which is an hyperbola.

Otherwise, in the plane of the first and third axes, the line L can be described as the collection of all points of the form

$$(x, Mx + B),$$

where M is the slope of L . For an arbitrary point (x, y, z) it follows that

$$(2) \quad (x, y, z) \text{ is in the plane } P \text{ if and only if } z = Mx + B.$$

Combining (1) and (2), we see that (x, y, z) is in the intersection of the cone and the plane if and only if

$$(*) \quad Mx + B = \pm C\sqrt{x^2 + y^2}.$$

Now we have to choose coordinate axes in the plane P . We can choose L as the first axis, measuring distances from the intersection Q with the horizontal plane (Figure 5); for the second axis we just choose the line through Q parallel to our original second axis. If the first coordinate of a point in P with respect to these axes is x , then the first coordinate of this point with respect to the original axes can be written in the form

$$\alpha x + \beta$$

for some α and β . On the other hand, if the second coordinate of the point with respect to these axes is y , then y is also the second coordinate with respect to the original axes.

Consequently, $(*)$ says that the point lies on the intersection of the plane and the cone if and only if

$$M(\alpha x + \beta) + B = \pm C\sqrt{(\alpha x + \beta)^2 + y^2}.$$

Although this looks fairly complicated, after squaring we can write this as

$$C^2 y^2 - \alpha^2 (M^2 - C^2) x^2 + Ex + F = 0$$

for some E and F that we won't bother writing out.

Now Problem 4-16 indicates that this is either a parabola, an ellipse, or an hyperbola. In fact, looking a little more closely at the solution, we see that the

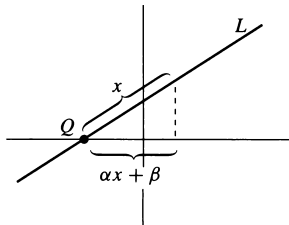


FIGURE 5

values of E and F are irrelevant:

- (1) If $M = \pm C$ we obtain a parabola;
- (2) If $C^2 > M^2$ we obtain an ellipse;
- (3) If $C^2 < M^2$ we obtain an hyperbola.

These analytic conditions are easy to interpret geometrically (Figure 6):

- (1) If our plane is parallel to one of the generating lines of the cone we obtain a parabola;
- (2) If our plane slopes less than the generating line of the cone (so that our intersection omits one half of the cone) we obtain an ellipse;
- (3) If our plane slopes more than the generating line of the cone we obtain an hyperbola.

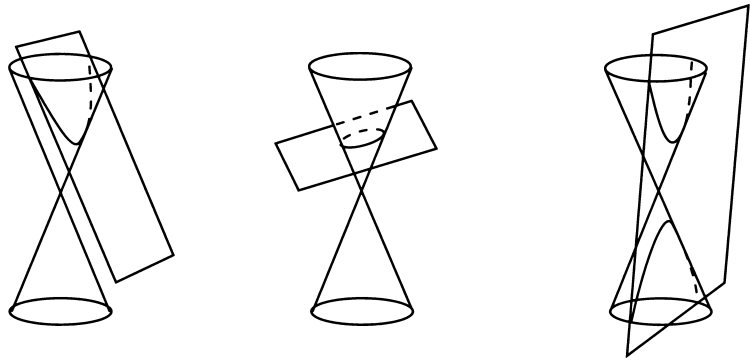


FIGURE 6

In fact, the very names of these “conic sections” are related to this description. The word *parabola* comes from a Greek root meaning ‘alongside,’ the same root that appears in parable, not to mention paradigm, paradox, paragon, paragraph, paralegal, parallax, parallel, and even parachute. *Ellipse* comes from a Greek root meaning ‘defect,’ or omission, as in ellipsis (an omission, . . . or the dots that indicate it). And *hyperbola* comes from a Greek root meaning ‘throwing beyond,’ or excess. With the currency of words like hyperactive, hypersensitive, and hyperventilate, not to mention hype, one can probably say, without risk of hyperbole, that this root is familiar to almost everyone.*

PROBLEMS

1. Consider a cylinder with a generator perpendicular to the horizontal plane (Figure 7); the only requirement for a point (x, y, z) to lie on this cylinder is

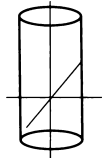


FIGURE 7

* Although the correspondence between these roots and the geometric picture correspond so beautifully, for the sake of dull accuracy it has to be reported that the Greeks originally applied the words to describe features of certain equations involving the conic sections.

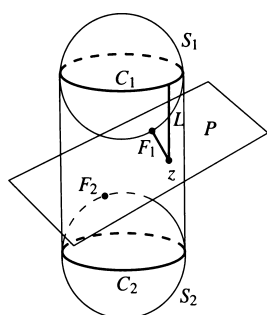


FIGURE 8

that (x, y) lies on a circle:

$$x^2 + y^2 = C^2.$$

Show that the intersection of a plane with this cylinder can be described by an equation of the form

$$(\alpha x + \beta)^2 + y^2 = C^2.$$

What possibilities are there?

2. In Figure 8, the sphere S_1 has the same diameter as the cylinder, so that its equator C_1 lies along the cylinder; it is also tangent to the plane P at F_1 . Similarly, the equator C_2 of S_2 lies along the cylinder, and S_2 is tangent to P at F_2 .
 - (a) Let z be any point on the intersection of P and the cylinder. Explain why the length of the line from z to F_1 is equal to the length of the vertical line L from z to C_1 .
 - (b) By proving a similar fact for the length of the line from z to F_2 , show that the distance from z to F_1 plus the distance from z to F_2 is a constant, so that the intersection is an ellipse, with foci F_1 and F_2 .
3. Similarly, use Figure 9(a) to prove geometrically that the intersection of a plane and a cone is an ellipse when the plane intersects just one half of the cone. Similarly, use (b) to prove that the intersection is a hyperbola when the plane intersects both halves of the cone.

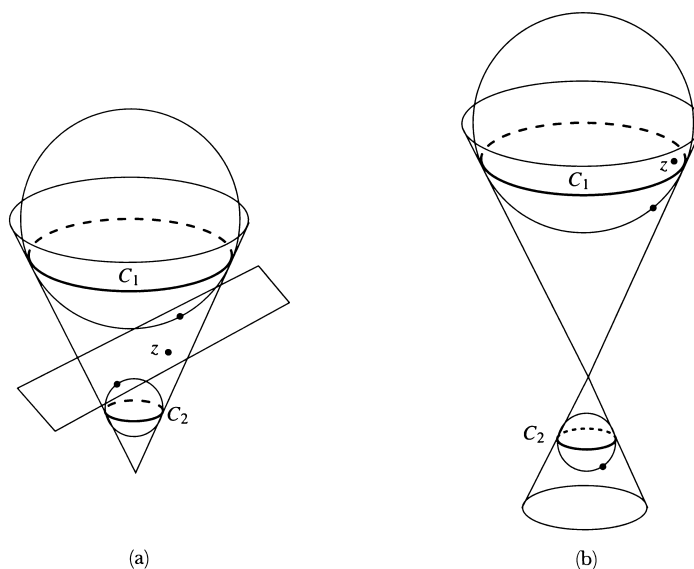


FIGURE 9

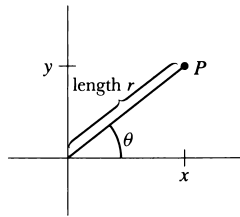
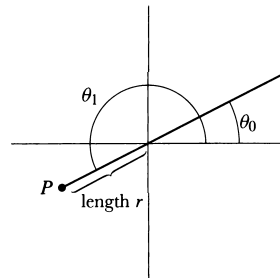


FIGURE 1

APPENDIX 3. POLAR COORDINATES

In this chapter we've been acting all along as if there's only one way to label points in the plane with pairs of numbers. Actually, there are many different ways, each giving rise to a different "coordinate system." The usual coordinates of a point are called its cartesian coordinates, after the French mathematician and philosopher René Descartes (1596–1650), who first introduced the idea of coordinate systems. In many situations it is more convenient to introduce polar coordinates, which are illustrated in Figure 1. To the point P we assign the polar coordinates (r, θ) , where r is the distance from the origin O to P , and θ is the measure, in radians, of the angle between the horizontal axis and the line from O to P . This θ is not determined unambiguously. For example, points on the right side of the horizontal axis could have either $\theta = 0$ or $\theta = 2\pi$; moreover, θ is completely ambiguous at the origin O . So it is necessary to exclude some ray through the origin if we want to assign a unique pair (r, θ) to each point under consideration.

On the other hand, there is no problem associating a unique point to any pair (r, θ) . In fact, it is possible (though not approved of by all) to associate a point to (r, θ) when $r < 0$, according to the scheme indicated in Figure 2. Thus, it always makes sense to talk about "the point with polar coordinates (r, θ) ," (with or without the possibility of $r < 0$), even though there is some ambiguity when we talk about "the polar coordinates" of a given point.



P is the point with polar coordinates (r, θ_1)
and also the point with polar coordinates
 $(-r, \theta_2)$.

FIGURE 2

It is clear from Figure 1 (and Figure 2) that the point with polar coordinates (r, θ) has cartesian coordinates (x, y) given by

$$x = r \cos \theta, \quad y = r \sin \theta.$$

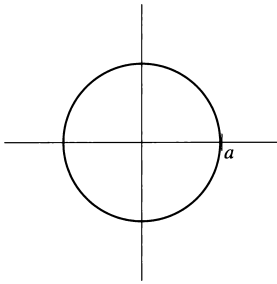


FIGURE 3

Conversely, if a point has cartesian coordinates (x, y) , then (any of) its polar coordinates (r, θ) satisfy

$$r = \pm\sqrt{x^2 + y^2}$$

$$\tan \theta = \frac{y}{x} \quad \text{if } x \neq 0.$$

Now suppose that f is a function. Then by the **graph of f in polar coordinates** we mean the collection of all points P with polar coordinates (r, θ) satisfying $r = f(\theta)$. In other words, the graph of f in polar coordinates is the collection of all points with polar coordinates $(f(\theta), \theta)$. No special significance should be attached to the fact that we are considering pairs $(f(\theta), \theta)$, with $f(\theta)$ first, as opposed to pairs $(x, f(x))$ in the usual graph of f ; it is purely a matter of convention that r is considered the first polar coordinate and θ is considered the second.

The graph of f in polar coordinates is often described as “the graph of the equation $r = f(\theta)$.” For example, suppose that f is a constant function, $f(\theta) = a$ for all θ . The graph of the equation $r = a$ is simply a circle with center O and radius a (Figure 3). This example illustrates, in a rather blatant way, that polar coordinates are likely to make things simpler in situations that involve symmetry with respect to the origin O .

The graph of the equation $r = \theta$ is shown in Figure 4. The solid line corresponds to all values of $\theta \geq 0$, while the dashed line corresponds to values of $\theta \leq 0$.

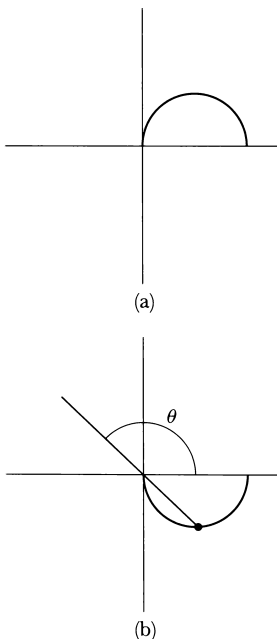


FIGURE 5

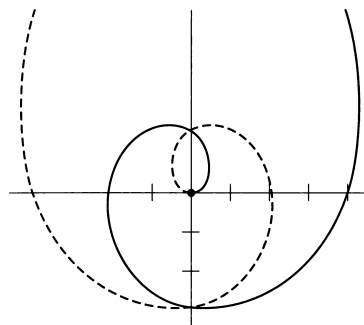


FIGURE 4

Spiral of Archimedes

As another example involving both positive and negative r , consider the graph of the equation $r = \cos \theta$. Figure 5(a) shows the part that corresponds to $0 \leq \theta \leq \pi/2$. Figure 5(b) shows the part corresponding to $\pi/2 \leq \theta \leq \pi$; here $r < 0$. You can check that no new points are added for $\theta > \pi$ or $\theta < 0$. It is easy to describe this same graph in terms of the cartesian coordinates of its points. Since the polar coordinates of any point on the graph satisfy

$$r = \cos \theta,$$

and hence

$$r^2 = r \cos \theta,$$

its cartesian coordinates satisfy the equation

$$x^2 + y^2 = x$$

which describes a circle (Problem 4-16). [Conversely, it is clear that if the cartesian coordinates of a point satisfy $x^2 + y^2 = x$, then it lies on the graph of the equation $r = \cos \theta$.]

Although we've now gotten a circle in two different ways, we might well be hesitant about trying to find the equation of an ellipse in polar coordinates. But it turns out that we can get a very nice equation if we choose one of the *foci* as the origin. Figure 6 shows an ellipse with one focus at O , with the sum of the distances of all points from O and the other focus $\mathbf{f}$ being $2a$. We've chosen $\mathbf{f}$ to the left of O , with coordinates written as

$$(-2\epsilon a, 0).$$

(We have $0 \leq \epsilon < 1$, since we must have $2a > \text{distance from } \mathbf{f} \text{ to } O$).

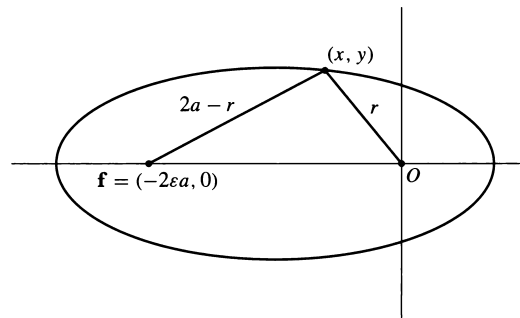


FIGURE 6

The distance r from (x, y) to O is given by

$$(1) \quad r^2 = x^2 + y^2.$$

By assumption, the distance from (x, y) to $\mathbf{f}$ is $2a - r$, hence

$$(2a - r)^2 = (x - [-2\epsilon a])^2 + y^2,$$

or

$$(2) \quad 4a^2 - 4ar + r^2 = x^2 + 4\epsilon ax + 4\epsilon^2 a^2 + y^2.$$

Subtracting (1) from (2), and dividing by $4a$, we get

$$a - r = \epsilon x + \epsilon^2 a,$$

or

$$\begin{aligned} r &= a - \varepsilon x - \varepsilon^2 a \\ &= (1 - \varepsilon^2)a - \varepsilon x, \end{aligned}$$

which we can write as

$$(3) \quad r = \Lambda - \varepsilon x, \quad \text{for } \Lambda = (1 - \varepsilon^2)a.$$

Substituting $r \cos \theta$ for x , we have

$$\begin{aligned} r &= \Lambda - \varepsilon r \cos \theta, \\ r(1 + \varepsilon \cos \theta) &= \Lambda, \end{aligned}$$

and thus

$$(4) \quad r = \frac{\Lambda}{1 + \varepsilon \cos \theta}.$$

In Chapter 4 we found that

$$(5) \quad \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

is the equation in cartesian coordinates for an ellipse with $2a$ as the sum of the distances to the foci, but with the foci at $(-c, 0)$ and $(c, 0)$, where

$$b = \sqrt{a^2 - c^2}.$$

Since the distance between the foci is $2c$, when this ellipse is moved left by c units, so that the focus $(c, 0)$ is now at the origin, we get the ellipse (4) when we take $c = \varepsilon a$ or $\varepsilon = c/a$ (with equation (3) determining Λ). Conversely, given the ellipse described by (4), for the corresponding equation (5) the value of a is determined by (3),

$$a = \frac{\Lambda}{1 - \varepsilon^2},$$

and again using $c = \varepsilon a$, we get

$$b = \sqrt{a^2 - c^2} = \sqrt{a^2 - \varepsilon^2 a^2} = a\sqrt{1 - \varepsilon^2} = \frac{\Lambda}{\sqrt{1 - \varepsilon^2}}.$$

Thus, we can obtain a and b , the lengths of the major and minor axes, immediately from ε and Λ .

The number

$$\varepsilon = \frac{c}{a} = \frac{\sqrt{a^2 - b^2}}{a} = \sqrt{1 - \left(\frac{b}{a}\right)^2},$$

the *eccentricity* of the ellipse, determines the “shape” of the ellipse (the ratio of the major and minor axes), while the number Λ determines its “size,” as shown by (4).

PROBLEMS

1. If two points have polar coordinates (r_1, θ_1) and (r_2, θ_2) , show that the distance d between them is given by

$$d^2 = r_1^2 + r_2^2 - 2r_1r_2 \cos(\theta_1 - \theta_2).$$

What does this say geometrically?

2. Describe the general features of the graph of f in polar coordinates if
- (i) f is even.
 - (ii) f is odd.
 - (iii) $f(\theta) = f(\theta + \pi)$.
3. Sketch the graphs of the following equations.
- (i) $r = a \sin \theta$.
 - (ii) $r = a \sec \theta$. Hint: It is a very simple graph!
 - (iii) $r = \cos 2\theta$. Good luck on this one!
 - (iv) $r = \cos 3\theta$.
 - (v) $r = |\cos 2\theta|$.
 - (vi) $r = |\cos 3\theta|$.
4. Find equations for the cartesian coordinates of points on the graphs (i), (ii) and (iii) in Problem 3.
5. Consider a hyperbola, where the difference of the distance between the two foci is the constant $2a$, and choose one focus at O and the other at $(-2\epsilon a, 0)$. (In this case, we must have $\epsilon > 1$). Show that we obtain the exact same equation in polar coordinates

$$r = \frac{\Lambda}{1 + \epsilon \cos \theta}$$

as we obtained for an ellipse.

6. Consider the set of points (x, y) such that the distance (x, y) to O is equal to the distance from (x, y) to the line $y = a$ (Figure 7). Show that the distance to the line is $a - r \cos \theta$, and conclude that the equation can be written

$$a = r(1 + \cos \theta).$$

Notice that this equation for a parabola is again of the same form as (4).

7. Now, for any Λ and ϵ , consider the graph in polar coordinates of the equation (4), which implies (3). Show that the points satisfying this equation satisfy

$$(1 - \epsilon^2)x^2 + y^2 = \Lambda^2 - 2\Lambda\epsilon x.$$

Using Problem 4-16, show that this is an ellipse for $\epsilon < 1$, a parabola for $\epsilon = 1$, and a hyperbola for $\epsilon > 1$.

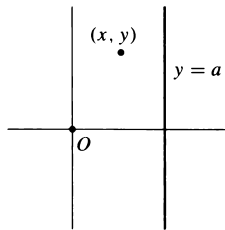


FIGURE 7

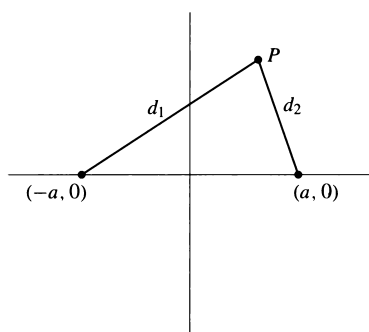


FIGURE 8

8. (a) Sketch the graph of the *cardioid* $r = 1 - \sin \theta$.
 (b) Show that it is also the graph of $r = -1 - \sin \theta$.
 (c) Show that it can be described by the equation

$$x^2 + y^2 = \sqrt{x^2 + y^2} - y,$$

and conclude that it can be described by the equation

$$(x^2 + y^2 + y)^2 = x^2 + y^2$$

9. Sketch the graphs of the following equations.

- (i) $r = 1 - \frac{1}{2} \sin \theta$.
 (ii) $r = 1 - 2 \sin \theta$.
 (iii) $r = 2 + \cos \theta$.

10. (a) Sketch the graph of the *lemniscate*

$$r^2 = 2a^2 \cos 2\theta.$$

- (b) Find an equation for its cartesian coordinates.
 (c) Show that it is the collection of all points P in Figure 8 satisfying $d_1 d_2 = a^2$.
 (d) Make a guess about the shape of the curves formed by the set of all P satisfying $d_1 d_2 = b$, when $b > a^2$ and when $b < a^2$.

CHAPTER 5 LIMITS

The concept of a limit is surely the most important, and probably the most difficult one in all of calculus. The goal of this chapter is the definition of limits, but we are, once more, going to begin with a provisional definition; what we shall define is not the word “limit” but the notion of a function approaching a limit.

PROVISIONAL DEFINITION

The function f approaches the limit l near a , if we can make $f(x)$ as close as we like to l by requiring that x be sufficiently close to, but unequal to, a .

Of the six functions graphed in Figure 1, only the first three approach l at a . Notice that although $g(a)$ is not defined, and $h(a)$ is defined “the wrong way,” it is still true that g and h approach l near a . This is because we explicitly ruled out, in our definition, the necessity of ever considering the value of the function at a —it is only necessary that $f(x)$ should be close to l for x close to a , but *unequal to* a . We are simply not interested in the value of $f(a)$, or even in the question of whether $f(a)$ is defined.

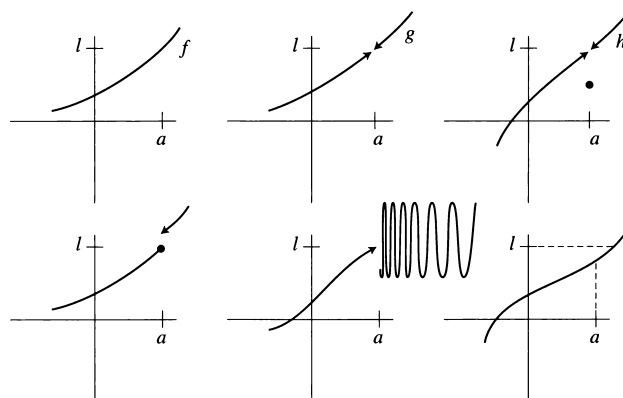


FIGURE 1

One convenient way of picturing the assertion that f approaches l near a is provided by a method of drawing functions that was not mentioned in Chapter 4. In this method, we draw two straight lines, each representing $\mathbf{R}$, and arrows from a point x in one, to $f(x)$ in the other. Figure 2 illustrates such a picture for two different functions.

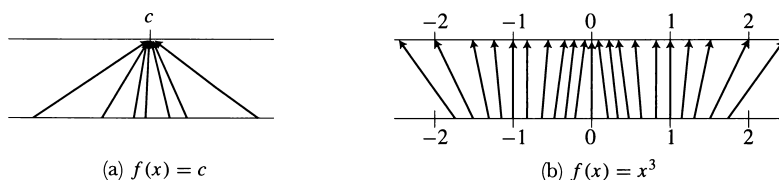


FIGURE 2

Now consider a function f whose drawing looks like Figure 3. Suppose we ask that $f(x)$ be close to l , say within the open interval B which has been drawn in Figure 3. This can be guaranteed if we consider only the numbers x in the interval A of Figure 3. (In this diagram we have chosen the largest interval which will work; any smaller interval containing a could have been chosen instead.) If we

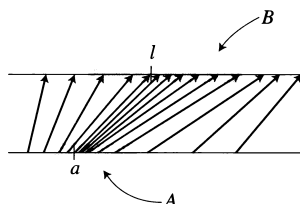


FIGURE 3

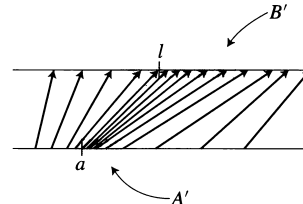


FIGURE 4

choose a smaller interval B' (Figure 4) we will, usually, have to choose a smaller A' , but no matter how small we choose the open interval B , there is always supposed to be some open interval A which works.

A similar pictorial interpretation is possible in terms of the graph of f , but in this case the interval B must be drawn on the vertical axis, and the set A on the horizontal axis. The fact that $f(x)$ is in B when x is in A means that the part of the graph lying over A is contained in the region which is bounded by the horizontal lines through the end points of B ; compare Figure 5(a), where a valid interval A has been chosen, with Figure 5(b), where A is too large.

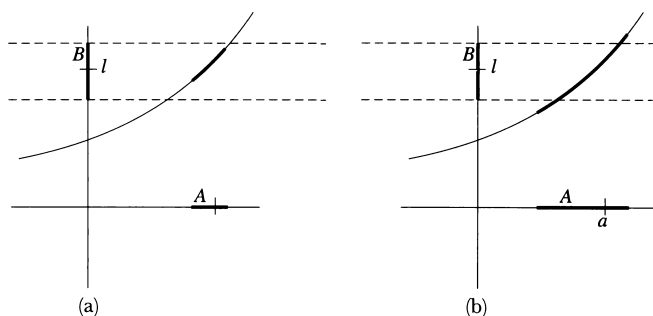


FIGURE 5

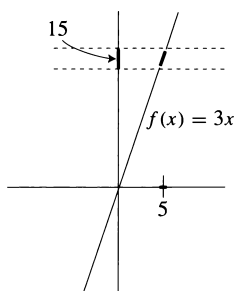


FIGURE 6

To take a specific simple example, let's consider the function $f(x) = 3x$ with $a = 5$ (Figure 6). Presumably f should approach the limit 15 near 5—we ought to be able to get $f(x)$ as close to 15 as we like if we require that x be sufficiently close to 5. To be specific, suppose we want to make sure that $3x$ is within $\frac{1}{10}$ of 15. This means that we want to have

$$15 - \frac{1}{10} < 3x < 15 + \frac{1}{10},$$

which we can also write as

$$-\frac{1}{10} < 3x - 15 < \frac{1}{10}.$$

To do this we just have to require that

$$-\frac{1}{30} < x - 5 < \frac{1}{30},$$

or simply $|x - 5| < \frac{1}{30}$. There is nothing special about the number $\frac{1}{10}$. It is just as easy to guarantee that $|3x - 15| < \frac{1}{100}$; simply require that $|x - 5| < \frac{1}{300}$. In fact, if we take any positive number ε we can make $|3x - 15| < \varepsilon$ simply by requiring that $|x - 5| < \varepsilon/3$.

There's also nothing special about the choice $a = 5$. It's just as easy to see that f approaches the limit $3a$ at a for any a : To ensure that

$$|3x - 3a| < \varepsilon$$

we just have to require that

$$|x - a| < \frac{\varepsilon}{3}.$$

Naturally, the same sort of argument works for the function $f(x) = 3,000,000x$. We just have to be 1,000,000 times as careful, choosing $|x - a| < \varepsilon/3,000,000$ in order to ensure that $|f(x) - a| < \varepsilon$.

The function $f(x) = x^2$ is a little more interesting. Presumably, we should be able to show that $f(x)$ approaches 9 near 3. This means that we need to show how to ensure the inequality

$$|x^2 - 9| < \varepsilon$$

for any given positive number ε by requiring $|x - 3|$ to be small enough. The obvious first step is to write

$$|x^2 - 9| = |x - 3| \cdot |x + 3|,$$

which gives us the useful $|x - 3|$ factor. Unlike the situation with the previous examples, however, the extra factor here is $|x + 3|$, which isn't a convenient constant like 3 or 3,000,000. But the only crucial thing is to make sure that we can say *something* about how big $|x + 3|$ is. So the first thing we'll do is to require that $|x - 3| < 1$. Once we've specified that $|x - 3| < 1$, or $2 < x < 4$, we have $5 < x + 3 < 7$ and we've guaranteed that $|x + 3| < 7$. So we now have

$$|x^2 - 9| = |x - 3| \cdot |x + 3| < 7|x - 3|,$$

which shows that we have $|x^2 - 9| < \varepsilon$ for $|x - 3| < \varepsilon/7$, provided that we've also required that $|x - 3| < 1$. Or, to make it look more official: we require that $|x - 3| < \min(\varepsilon/7, 1)$.

The initial specification $|x - 3| < 1$ was simply made for convenience. We could just as well have specified that $|x - 3| < \frac{1}{10}$ or $|x - 3| < 10$ or any other convenient number. To make sure you understand the reasoning in the previous paragraph, it is a good exercise to figure out how the argument would read if we chose $|x - 3| < 10$.

Our argument to show that f approaches 9 near 3 will basically work to show that f approaches a^2 near a for any a , except that we need to worry a bit more about getting the proper inequality for $|x + a|$. We first require that $|x - a| < 1$, again with the expectation that this will ensure that $|x + a|$ is not too large. In fact, Problem 1-12 shows that

$$|x| - |a| \leq |x - a| < 1,$$

so

$$|x| < 1 + |a|,$$

and consequently

$$|x + a| \leq |x| + |a| < 2|a| + 1,$$

so that we then have

$$\begin{aligned} |x^2 - a^2| &= |x - a| \cdot |x + a| \\ &< |x - a| \cdot (2|a| + 1), \end{aligned}$$

which shows that we have $|x^2 - a^2| < \varepsilon$ for $|x - a| < \varepsilon/(2|a| + 1)$, provided that we also have $|x - a| < 1$. Officially: we require that $|x - a| < \min(\varepsilon/(2|a| + 1), 1)$.

In contrast to this example, we'll now consider the function $f(x) = 1/x$ (for $x \neq 0$), and try to show that f approaches $1/3$ near 3. This means that we need to show how to guarantee the inequality

$$\left| \frac{1}{x} - \frac{1}{3} \right| < \varepsilon$$

for any given positive number ε by requiring $|x - 3|$ to be small enough. We begin by writing

$$\left| \frac{1}{x} - \frac{1}{3} \right| = \left| \frac{3 - x}{3x} \right| = \frac{1}{3} \cdot \frac{1}{|x|} \cdot |x - 3|,$$

giving us the nice factor $|x - 3|$, and even an extra $\frac{1}{3}$ for good measure, along with the problem factor $1/|x|$. In this case, we first need to make sure that $|x|$ isn't too *small*, so that $1/|x|$ won't be too large.

We can first require that $|x - 3| < 1$, because this gives $2 < x < 4$, so that

$$\frac{1}{4} < \frac{1}{x} < \frac{1}{2},$$

which not only tells us that $\frac{1}{x} < \frac{1}{2}$, but also that $x > 0$, which is important in order to conclude that $\frac{1}{|x|} < \frac{1}{2}$. We now have

$$\left| \frac{1}{x} - \frac{1}{3} \right| = \frac{1}{3} \cdot \frac{1}{|x|} \cdot |x - 3| < \frac{1}{6} |x - 3|,$$

which shows that we have $|1/x - 1/3| < \varepsilon$ for $|x - 3| < 6\varepsilon$, provided that we've also required that $|x - 3| < 1$. Or, to make it look official again: we require that $|x - 3| < \min(6\varepsilon, 1)$.

If we instead wanted to show that f approaches $-1/3$ near -3 , we would begin by stipulating that $|x - (-3)| < 1$, giving $-4 < x < -2$, once again implying that $|1/x| < 1/2$, so that everything works as before.

To show in general that f approaches $1/a$ near a for any a we proceed in basically the same way, except that, again, we have to be a little more careful in formulating our initial stipulation. It's not good enough simply to require that $|x - a|$ should be less than 1, or any other particular number, because if a is close to 0 this would allow values of x that are negative (not to mention the embarrassing possibility that $x = 0$, so that $f(x)$ isn't even defined!).

The trick in this case is to first require that

$$|x - a| < \frac{|a|}{2};$$

in other words, we require that x be less than half as far from 0 as a (Figure 7). You should be able to check first that $x \neq 0$ and that $1/|x| < 2/|a|$, and then work out the rest of the argument.

With all the work required for these simple examples, you may have begun to quail at the prospect of tackling even more complicated functions. But that won't really be necessary, since we will eventually have some basic theorems that we can rely on. Instead of worrying about the unpleasant algebra that might be involved in functions like $f(x) = x^3$ or $f(x) = 1/x^3$, we'll turn our attention to some examples that might appear to be even more frightening.

Consider first the function $f(x) = x \sin 1/x$ (Figure 8). Despite the erratic behavior of this function near 0 it is clear, at least intuitively, that f approaches $l = 0$ near $a = 0$ (remember that our provisional definition specifically exempts $x = a$ from consideration, so it doesn't matter that this function isn't even defined at 0). We want to show that we can get $f(x) = x \sin 1/x$ as close to 0 as desired if we require that x be sufficiently close to 0, but $\neq 0$. In other words, for any number $\varepsilon > 0$, we want to show that we can ensure that

$$|f(x) - 0| = \left| x \sin \frac{1}{x} \right| < \varepsilon$$

by requiring that $|x| = |x - 0|$ is sufficiently small (but $\neq 0$). But this is easy. Since

$$\left| \sin \frac{1}{x} \right| \leq 1, \quad \text{for all } x \neq 0,$$

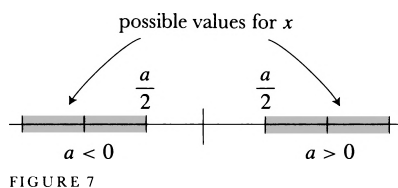


FIGURE 7

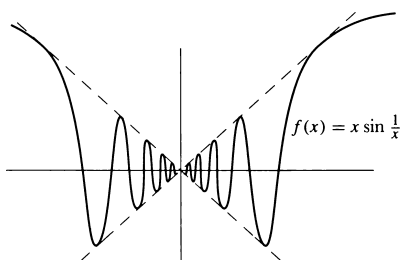


FIGURE 8

we have

$$\left| x \sin \frac{1}{x} \right| \leq |x|, \quad \text{for all } x \neq 0,$$

so we can make $|x \sin 1/x| < \varepsilon$ simply by requiring that $|x| < \varepsilon$ and $x \neq 0$.

For the function $f(x) = x^2 \sin 1/x$ (Figure 9) it seems even clearer that f approaches 0 near 0. If, for example, we want

$$\left| x^2 \sin \frac{1}{x} \right| < \frac{1}{10},$$

then we certainly need only require that $|x| < \frac{1}{10}$ and $x \neq 0$, since this implies that $|x^2| < \frac{1}{100}$ and consequently

$$\left| x^2 \sin \frac{1}{x} \right| \leq |x^2| < \frac{1}{100} < \frac{1}{10}.$$

(We could do even better, and allow $|x| < 1/\sqrt{10}$ and $x \neq 0$, but there is no particular virtue in being as economical as possible.) In general, if $\varepsilon > 0$, to ensure that

$$\left| x^2 \sin \frac{1}{x} \right| < \varepsilon,$$

we need only require that

$$|x| < \varepsilon \quad \text{and} \quad x \neq 0,$$

provided that $\varepsilon \leq 1$. If we are given an ε which is greater than 1 (it might be, even though it is “small” ε ’s which are of interest), then it does not suffice to require that $|x| < \varepsilon$, but it certainly suffices to require that $|x| < 1$ and $x \neq 0$.

As a third example, consider the function $f(x) = \sqrt{|x|} \sin 1/x$ (Figure 10). In order to make $|\sqrt{|x|} \sin 1/x| < \varepsilon$ we can require that

$$|x| < \varepsilon^2 \quad \text{and} \quad x \neq 0$$

(the algebra is left to you).

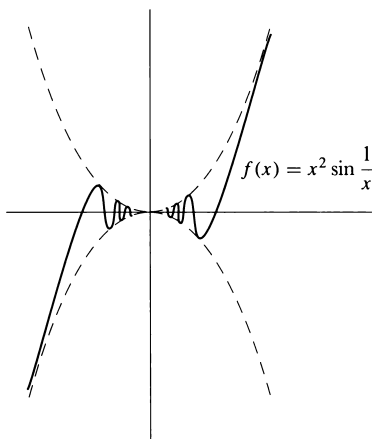


FIGURE 9

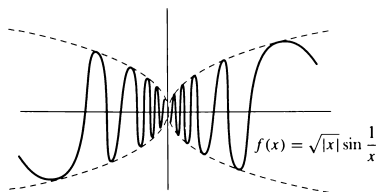


FIGURE 10

Finally, let us consider the function $f(x) = \sin 1/x$ (Figure 11). For this function it is *false* that f approaches 0 near 0. This amounts to saying that it is not true for every number $\varepsilon > 0$ that we can get $|f(x) - 0| < \varepsilon$ by choosing x sufficiently small, and $\neq 0$. To show this we simply have to find *one* $\varepsilon > 0$ for which the condition $|f(x) - 0| < \varepsilon$ cannot be guaranteed, no matter how small we require $|x|$ to be. In fact, $\varepsilon = \frac{1}{2}$ will do: it is impossible to ensure that $|f(x)| < \frac{1}{2}$ no matter how small we require $|x|$ to be; for if A is any interval containing 0, there is some number $x = 1/(\frac{1}{2}\pi + 2n\pi)$ which is in this interval, and for this x we have $f(x) = 1$.

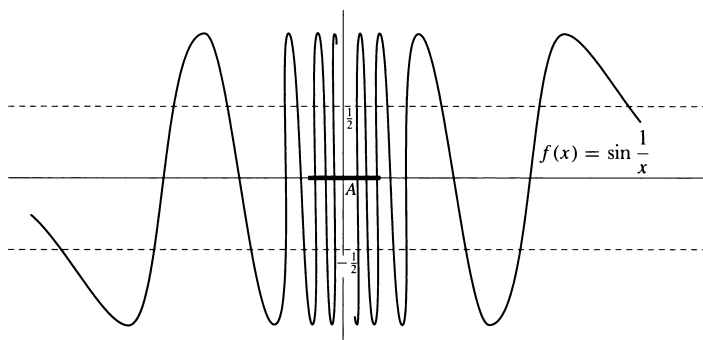


FIGURE 11

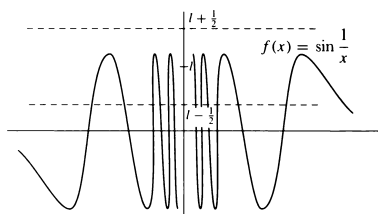


FIGURE 12

This same argument can be used (Figure 12) to show that f does not approach *any* number near 0. To show this we must again find, for any particular number l , some number $\varepsilon > 0$ so that $|f(x) - l| < \varepsilon$ is *not* true, no matter how small x is required to be. The choice $\varepsilon = \frac{1}{2}$ works for any number l ; that is, no matter how small we require $|x|$ to be, we cannot ensure that $|f(x) - l| < \frac{1}{2}$. The reason is, that for any interval A containing 0 we can find both x_1 and x_2 in this interval with

$$f(x_1) = 1 \quad \text{and} \quad f(x_2) = -1,$$

namely

$$x_1 = \frac{1}{\frac{1}{2}\pi + 2n\pi} \quad \text{and} \quad x_2 = \frac{1}{\frac{3}{2}\pi + 2m\pi}$$

for large enough n and m . But the interval from $l - \frac{1}{2}$ to $l + \frac{1}{2}$ cannot contain both -1 and 1 , since its total length is only 1; so we cannot have

$$|1 - l| < \frac{1}{2} \quad \text{and also} \quad |-1 - l| < \frac{1}{2},$$

no matter what l is.

The phenomenon exhibited by $f(x) = \sin 1/x$ near 0 can occur in many ways. If we consider the function

$$f(x) = \begin{cases} 0, & x \text{ irrational} \\ 1, & x \text{ rational,} \end{cases}$$

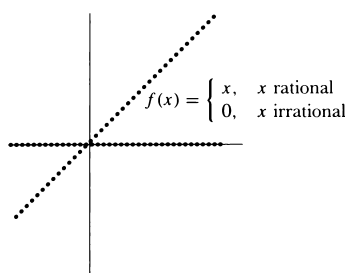


FIGURE 13

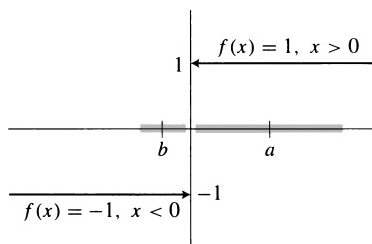


FIGURE 14

then, no matter what a is, f does not approach any number l near a . In fact, we cannot make $|f(x) - l| < \frac{1}{4}$ no matter how close we bring x to a , because in any interval around a there are numbers x with $f(x) = 0$, and also numbers x with $f(x) = 1$, so that we would need $|0 - l| < \frac{1}{4}$ and also $|1 - l| < \frac{1}{4}$.

An amusing variation on this behavior is presented by the function shown in Figure 13:

$$f(x) = \begin{cases} x, & x \text{ rational} \\ 0, & x \text{ irrational} \end{cases}$$

The behavior of this function is “opposite” to that of $g(x) = \sin 1/x$; it approaches 0 at 0, but does not approach any number at a , if $a \neq 0$. By now you should have no difficulty convincing yourself that this is true.

We conclude with a very simple example (Figure 14):

$$f(x) = \begin{cases} -1, & x < 0 \\ 1, & x > 0 \end{cases}$$

If $a > 0$, then f approaches 1 near a : indeed, to ensure that $|f(x) - 1| < \varepsilon$ it certainly suffices to require that $|x - a| < a$, since this implies

$$\begin{aligned} -a &< x - a \\ \text{or } 0 &< x \end{aligned}$$

so that $f(x) = 1$. Similarly, if $b < 0$, then f approaches -1 near b : to ensure that $|f(x) - (-1)| < \varepsilon$ it suffices to require that $|x - b| < -b$. Finally, as you may easily check, f does not approach any number near 0.

The time has now come to point out that of the many demonstrations about limits which we have given, not one has been a real proof. The fault lies not with our reasoning, but with our definition. If our provisional definition of a function was open to criticism, our provisional definition of approaching a limit is even more vulnerable. This definition is simply not sufficiently precise to be used in proofs. It is hardly clear how one “makes” $f(x)$ close to l (whatever “close” means) by “requiring” x to be sufficiently close to a (however close “sufficiently” close is supposed to be). Despite the criticisms of our definition you may feel (I certainly hope you do) that our arguments were nevertheless quite convincing. In order to present any sort of argument at all, we have been practically forced to invent the real definition. It is possible to arrive at this definition in several steps, each one clarifying some obscure phrase which still remains. Let us begin, once again, with the provisional definition:

The function f approaches the limit l near a , if we can make $f(x)$ as close as we like to l by requiring that x be sufficiently close to, but unequal to, a .

The very first change which we made in this definition was to note that making $f(x)$ close to l meant making $|f(x) - l|$ small, and similarly for x and a :

The function f approaches the limit l near a , if we can make $|f(x) - l|$ as small as we like by requiring that $|x - a|$ be sufficiently small, and $x \neq a$.

The second, more crucial, change was to note that making $|f(x) - l|$ “as small as we like” means making $|f(x) - l| < \varepsilon$ for any $\varepsilon > 0$ that happens to be given us:

The function f approaches the limit l near a , if for every number $\varepsilon > 0$ we can make $|f(x) - l| < \varepsilon$ by requiring that $|x - a|$ be sufficiently small, and $x \neq a$.

There is a common pattern to all the demonstrations about limits which we have given. For each number $\varepsilon > 0$ we found some other positive number, δ say, with the property that if $x \neq a$ and $|x - a| < \delta$, then $|f(x) - l| < \varepsilon$. For the function $f(x) = x \sin 1/x$ (with $a = 0$, $l = 0$), the number δ was just the number ε ; for $f(x) = \sqrt{|x|} \sin 1/x$, it was ε^2 ; for $f(x) = x^2$ it was the minimum of 1 and $\varepsilon/(2|a| + 1)$. In general, it may not be at all clear how to find the number δ , given ε , but it is the condition $|x - a| < \delta$ which expresses how small “sufficiently” small must be:

The function f approaches the limit l near a , if for every $\varepsilon > 0$ there is some $\delta > 0$ such that, for all x , if $|x - a| < \delta$ and $x \neq a$, then $|f(x) - l| < \varepsilon$.

This is practically the definition we will adopt. We will make only one trivial change, noting that “ $|x - a| < \delta$ and $x \neq a$ ” can just as well be expressed “ $0 < |x - a| < \delta$.”

DEFINITION

The function **f approaches the limit l near a** means: for every $\varepsilon > 0$ there is some $\delta > 0$ such that, for all x , if $0 < |x - a| < \delta$, then $|f(x) - l| < \varepsilon$.

This definition is so important (*everything* we do from now on depends on it) that proceeding any further without knowing it is hopeless. If necessary memorize it, like a poem! That, at least, is better than stating it incorrectly; if you do this you are doomed to give incorrect proofs. A good exercise in giving correct proofs is to review every fact already demonstrated about functions approaching limits, giving formal proofs of each. In most cases, this will merely involve a bit of rewording to make the arguments conform to our formal definition—all the algebraic work has been done already. When proving that f does *not* approach l at a , be sure to negate the definition correctly:

If it is *not* true that

for every $\varepsilon > 0$ there is some $\delta > 0$ such that, for all x , if $0 < |x - a| < \delta$, then $|f(x) - l| < \varepsilon$,

then

there is *some* $\varepsilon > 0$ such that for *every* $\delta > 0$ there is *some* x which satisfies $0 < |x - a| < \delta$ but not $|f(x) - l| < \varepsilon$.

Thus, to show that the function $f(x) = \sin 1/x$ does not approach 0 near 0, we consider $\varepsilon = \frac{1}{2}$ and note that for every $\delta > 0$ there is some x with $0 < |x - 0| < \delta$ but not $|\sin 1/x - 0| < \frac{1}{2}$ —namely, an x of the form $1/(\pi/2 + 2n\pi)$, where n is so large that $1/(\pi/2 + 2n\pi) < \delta$.

As a final illustration of the use of the definition of a function approaching a limit, we have reserved the function shown in Figure 15, a standard example, but one of the most complicated:

$$f(x) = \begin{cases} 0, & x \text{ irrational, } 0 < x < 1 \\ 1/q, & x = p/q \text{ in lowest terms, } 0 < x < 1. \end{cases}$$

(Recall that p/q is in lowest terms if p and q are integers with no common factor and $q > 0$.)

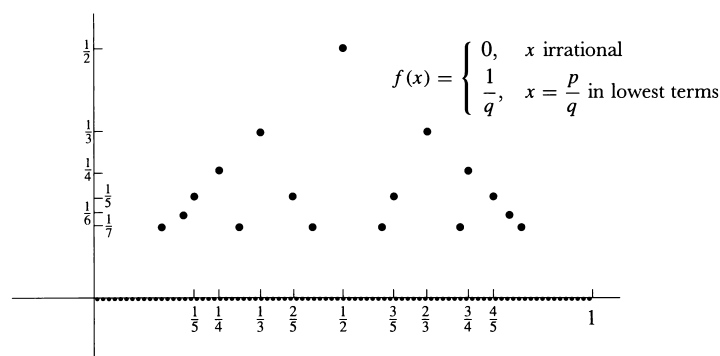


FIGURE 15

For any number a , with $0 < a < 1$, the function f approaches 0 at a . To prove this, consider any number $\varepsilon > 0$. Let n be a natural number so large that $1/n \leq \varepsilon$. Notice that the only numbers x for which $|f(x) - 0| < \varepsilon$ could be false are:

$$\frac{1}{2}, \frac{1}{3}, \frac{2}{3}, \frac{1}{4}, \frac{3}{4}, \frac{1}{5}, \frac{2}{5}, \frac{3}{5}, \frac{4}{5}; \dots; \frac{1}{n}, \dots, \frac{n-1}{n}.$$

(If a is rational, then a might be one of these numbers.) However many of these numbers there may be, there are, at any rate, only finitely many. Therefore, of all these numbers, one is closest to a ; that is, $|p/q - a|$ is smallest for one p/q among these numbers. (If a happens to be one of these numbers, then consider only the values $|p/q - a|$ for $p/q \neq a$.) This closest distance may be chosen as the δ . For if $0 < |x - a| < \delta$, then x is *not* one of

$$\frac{1}{2}, \dots, \frac{n-1}{n}$$

and therefore $|f(x) - 0| < \varepsilon$ is true. This completes the proof. Note that our description of the δ which works for a given ε is completely adequate—there is no reason why we must give a formula for δ in terms of ε .

Armed with our definition, we are now prepared to prove our first theorem; you have probably assumed the result all along, which is a very reasonable thing to do. This theorem is really a test case for our definition: if the theorem could not be proved, our definition would be useless.

THEOREM 1 A function cannot approach two different limits near a . In other words, if f approaches l near a , and f approaches m near a , then $l = m$.

PROOF Since this is our first theorem about limits it will certainly be necessary to translate the hypotheses according to the definition.

Since f approaches l near a , we know that for any $\varepsilon > 0$ there is some number $\delta_1 > 0$ such that, for all x ,

$$\text{if } 0 < |x - a| < \delta_1, \text{ then } |f(x) - l| < \varepsilon.$$

We also know, since f approaches m near a , that there is some $\delta_2 > 0$ such that, for all x ,

$$\text{if } 0 < |x - a| < \delta_2, \text{ then } |f(x) - m| < \varepsilon.$$

We have had to use two numbers, δ_1 and δ_2 , since there is no guarantee that the δ which works in one definition will work in the other. But, in fact, it is now easy to conclude that for any $\varepsilon > 0$ there is some $\delta > 0$ such that, for all x ,

$$\text{if } 0 < |x - a| < \delta, \text{ then } |f(x) - l| < \varepsilon \text{ and } |f(x) - m| < \varepsilon;$$

we simply choose $\delta = \min(\delta_1, \delta_2)$.

To complete the proof we just have to pick a particular $\varepsilon > 0$ for which the two conditions

$$|f(x) - l| < \varepsilon \quad \text{and} \quad |f(x) - m| < \varepsilon$$

cannot both hold, if $l \neq m$. The proper choice is suggested by Figure 16. If $l \neq m$, so that $|l - m| > 0$, we can choose $|l - m|/2$ as our ε . It follows that there is a $\delta > 0$ such that, for all x ,

$$\begin{aligned} \text{if } 0 < |x - a| < \delta, \text{ then } |f(x) - l| &< \frac{|l - m|}{2} \\ \text{and } |f(x) - m| &< \frac{|l - m|}{2}. \end{aligned}$$

This implies that for $0 < |x - a| < \delta$ we have

$$\begin{aligned} |l - m| &= |l - f(x) + f(x) - m| \leq |l - f(x)| + |f(x) - m| \\ &< \frac{|l - m|}{2} + \frac{|l - m|}{2} \\ &= |l - m|, \end{aligned}$$

a contradiction. ■

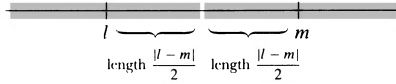


FIGURE 16

The number l which f approaches near a is denoted by $\lim_{x \rightarrow a} f(x)$ (read: the limit of $f(x)$ as x approaches a). This definition is possible only because of Theorem 1, which ensures that $\lim_{x \rightarrow a} f(x)$ never has to stand for two different numbers. The equation

$$\lim_{x \rightarrow a} f(x) = l$$

has exactly the same meaning as the phrase

f approaches l near a .

The possibility still remains that f does not approach l near a , for any l , so that $\lim_{x \rightarrow a} f(x) = l$ is false for every number l . This is usually expressed by saying that “ $\lim_{x \rightarrow a} f(x)$ does not exist.”

Notice that our new notation introduces an extra, utterly irrelevant letter x , which could be replaced by t , y , or any other letter which does not already appear—the symbols

$$\lim_{x \rightarrow a} f(x), \quad \lim_{t \rightarrow a} f(t), \quad \lim_{y \rightarrow a} f(y),$$

all denote precisely the same number, which depends on f and a , and has nothing to do with x , t , or y (these letters, in fact, do not denote anything at all). A more logical symbol would be something like $\lim_a f$, but this notation, despite its brevity, is so infuriatingly rigid that almost no one has seriously tried to use it. The notation $\lim_{x \rightarrow a} f(x)$ is much more useful because a function f often has no simple name, even though it might be possible to express $f(x)$ by a simple formula involving x . Thus, the short symbol

$$\lim_{x \rightarrow a} (x^2 + \sin x)$$

could be paraphrased only by the awkward expression

$$\lim_a f, \text{ where } f(x) = x^2 + \sin x.$$

Another advantage of the standard symbolism is illustrated by the expressions

$$\begin{aligned} \lim_{x \rightarrow a} x + t^3, \\ \lim_{t \rightarrow a} x + t^3. \end{aligned}$$

The first means the number which f approaches near a when

$$f(x) = x + t^3, \quad \text{for all } x;$$

the second means the number which f approaches near a when

$$f(t) = x + t^3, \quad \text{for all } t.$$

You should have little difficulty (especially if you consult Theorem 2) proving that

$$\begin{aligned} \lim_{x \rightarrow a} x + t^3 &= a + t^3, \\ \lim_{t \rightarrow a} x + t^3 &= x + a^3. \end{aligned}$$

These examples illustrate the main advantage of our notation, which is its flexibility. In fact, the notation $\lim_{x \rightarrow a} f(x)$ is so flexible that there is some danger of forgetting what it really means. Here is a simple exercise in the use of this notation, which will be important later: first interpret precisely, and then prove the equality of the expressions

$$\lim_{x \rightarrow a} f(x) \quad \text{and} \quad \lim_{h \rightarrow 0} f(a + h).$$

An important part of this chapter is the proof of a theorem which will make it easy to find many limits, as we promised long ago. The proof depends upon certain properties of inequalities and absolute values, hardly surprising when one considers the definition of limit. Although these facts have already been stated in Problems 1-20, 1-21, and 1-22, because of their importance they will be presented once again, in the form of a lemma (a lemma is an auxiliary theorem, a result that justifies its existence only by virtue of its prominent role in the proof of another theorem). The lemma says, roughly, that if x is close to x_0 , and y is close to y_0 , then $x + y$ will be close to $x_0 + y_0$, and xy will be close to $x_0 y_0$, and $1/y$ will be close to $1/y_0$. This intuitive statement is much easier to remember than the precise estimates of the lemma, and it is not unreasonable to read the proof of Theorem 2 first, in order to see just how these estimates are used.

LEMMA (1) If

$$|x - x_0| < \frac{\varepsilon}{2} \quad \text{and} \quad |y - y_0| < \frac{\varepsilon}{2},$$

then

$$|(x + y) - (x_0 + y_0)| < \varepsilon.$$

(2) If

$$|x - x_0| < \min\left(1, \frac{\varepsilon}{2(|y_0| + 1)}\right) \quad \text{and} \quad |y - y_0| < \frac{\varepsilon}{2(|x_0| + 1)},$$

then

$$|xy - x_0 y_0| < \varepsilon.$$

(3) If $y_0 \neq 0$ and

$$|y - y_0| < \min\left(\frac{|y_0|}{2}, \frac{\varepsilon|y_0|^2}{2}\right),$$

then $y \neq 0$ and

$$\left|\frac{1}{y} - \frac{1}{y_0}\right| < \varepsilon.$$

PROOF

$$\begin{aligned} (1) \quad |(x + y) - (x_0 + y_0)| &= |(x - x_0) + (y - y_0)| \\ &\leq |x - x_0| + |y - y_0| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \end{aligned}$$

(2) Since $|x - x_0| < 1$ we have

$$|x| - |x_0| \leq |x - x_0| < 1,$$

so that

$$|x| < 1 + |x_0|.$$

Thus

$$\begin{aligned} |xy - x_0y_0| &= |x(y - y_0) + y_0(x - x_0)| \\ &\leq |x| \cdot |y - y_0| + |y_0| \cdot |x - x_0| \\ &< (1 + |x_0|) \cdot \frac{\varepsilon}{2(|x_0| + 1)} + |y_0| \cdot \frac{\varepsilon}{2(|y_0| + 1)} \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \end{aligned}$$

(3) We have

$$|y_0| - |y| \leq |y - y_0| < \frac{|y_0|}{2},$$

so $|y| > |y_0|/2$. In particular, $y \neq 0$, and

$$\frac{1}{|y|} < \frac{2}{|y_0|}.$$

Thus

$$\left| \frac{1}{y} - \frac{1}{y_0} \right| = \frac{|y_0 - y|}{|y| \cdot |y_0|} < \frac{2}{|y_0|} \cdot \frac{1}{|y_0|} \cdot \frac{\varepsilon |y_0|^2}{2} = \varepsilon. \blacksquare$$

THEOREM 2 If $\lim_{x \rightarrow a} f(x) = l$ and $\lim_{x \rightarrow a} g(x) = m$, then

$$(1) \quad \lim_{x \rightarrow a} (f + g)(x) = l + m;$$

$$(2) \quad \lim_{x \rightarrow a} (f \cdot g)(x) = l \cdot m.$$

Moreover, if $m \neq 0$, then

$$(3) \quad \lim_{x \rightarrow a} \left(\frac{1}{g} \right)(x) = \frac{1}{m}.$$

PROOF The hypothesis means that for every $\varepsilon > 0$ there are $\delta_1, \delta_2 > 0$ such that, for all x ,

$$\text{if } 0 < |x - a| < \delta_1, \text{ then } |f(x) - l| < \varepsilon,$$

$$\text{and if } 0 < |x - a| < \delta_2, \text{ then } |g(x) - m| < \varepsilon.$$

This means (since, after all, $\varepsilon/2$ is also a positive number) that there are $\delta_1, \delta_2 > 0$ such that, for all x ,

$$\text{if } 0 < |x - a| < \delta_1, \text{ then } |f(x) - l| < \frac{\varepsilon}{2},$$

$$\text{and if } 0 < |x - a| < \delta_2, \text{ then } |g(x) - m| < \frac{\varepsilon}{2}.$$

Now let $\delta = \min(\delta_1, \delta_2)$. If $0 < |x - a| < \delta$, then $0 < |x - a| < \delta_1$ and $0 < |x - a| < \delta_2$ are both true, so both

$$|f(x) - l| < \frac{\varepsilon}{2} \quad \text{and} \quad |g(x) - m| < \frac{\varepsilon}{2}$$

Consulting the proof of Theorem 2(1), we see that we must first find δ_1 and $\delta_2 > 0$ such that, for all x ,

$$\begin{aligned} &\text{if } 0 < |x - a| < \delta_1, \text{ then } |x^2 - a^2| < \frac{\varepsilon}{2} \\ &\text{and if } 0 < |x - a| < \delta_2, \text{ then } |x - a| < \frac{\varepsilon}{2}. \end{aligned}$$

Since we have already given proofs that $\lim_{x \rightarrow a} x^2 = a^2$ and $\lim_{x \rightarrow a} x = a$, we know how to do this:

$$\begin{aligned} \delta_1 &= \min \left(1, \frac{\frac{\varepsilon}{2}}{2|a| + 1} \right), \\ \delta_2 &= \frac{\varepsilon}{2}. \end{aligned}$$

Thus we can take

$$\delta = \min(\delta_1, \delta_2) = \min \left(\min \left(1, \frac{\frac{\varepsilon}{2}}{2|a| + 1} \right), \frac{\varepsilon}{2} \right).$$

If $a \neq 0$, the same method can be used to find a $\delta > 0$ such that, for all x ,

$$\text{if } 0 < |x - a| < \delta, \text{ then } \left| \frac{1}{x^2} - \frac{1}{a^2} \right| < \varepsilon.$$

The proof of Theorem 2(3) shows that the second condition will follow if we find a $\delta > 0$ such that, for all x ,

$$\text{if } 0 < |x - a| < \delta, \text{ then } |x^2 - a^2| < \min \left(\frac{|a|^2}{2}, \frac{\varepsilon|a|^4}{2} \right).$$

Thus we can take

$$\delta = \min \left(1, \frac{\min \left(\frac{|a|^2}{2}, \frac{\varepsilon|a|^4}{2} \right)}{2|a| + 1} \right).$$

Naturally, these complicated expressions for δ can be simplified considerably, after they have been derived.

One technical detail in the proof of Theorem 2 deserves some discussion. In order for $\lim_{x \rightarrow a} f(x)$ to be defined it is, as we know, not necessary for f to be defined at a , nor is it necessary for f to be defined at all points $x \neq a$. However, there must be some $\delta > 0$ such that $f(x)$ is defined for x satisfying $0 < |x - a| < \delta$; otherwise the clause

$$\text{“if } 0 < |x - a| < \delta, \text{ then } |f(x) - l| < \varepsilon\text{”}$$

would make no sense at all, since the symbol $f(x)$ would make no sense for some x 's. If f and g are two functions for which the definition makes sense, it is easy to see that the same is true for $f + g$ and $f \cdot g$. But this is not so clear for $1/g$, since $1/g$ is undefined for x with $g(x) = 0$. However, this fact gets established in the proof of Theorem 2(3).

are true. But by part (1) of the lemma this implies that $|(f + g)(x) - (l + m)| < \varepsilon$. This proves (1).

To prove (2) we proceed similarly, after consulting part (2) of the lemma. If $\varepsilon > 0$ there are $\delta_1, \delta_2 > 0$ such that, for all x ,

$$\text{if } 0 < |x - a| < \delta_1, \text{ then } |f(x) - l| < \min\left(1, \frac{\varepsilon}{2(|m| + 1)}\right),$$

$$\text{and if } 0 < |x - a| < \delta_2, \text{ then } |g(x) - m| < \frac{\varepsilon}{2(|l| + 1)}.$$

Again let $\delta = \min(\delta_1, \delta_2)$. If $0 < |x - a| < \delta$, then

$$|f(x) - l| < \min\left(1, \frac{\varepsilon}{2(|m| + 1)}\right) \quad \text{and} \quad |g(x) - m| < \frac{\varepsilon}{2(|l| + 1)}.$$

So, by the lemma, $|(f \cdot g)(x) - l \cdot m| < \varepsilon$, and this proves (2).

Finally, if $\varepsilon > 0$ there is a $\delta > 0$ such that, for all x ,

$$\text{if } 0 < |x - a| < \delta, \text{ then } |g(x) - m| < \min\left(\frac{|m|}{2}, \frac{\varepsilon|m|^2}{2}\right).$$

But according to part (3) of the lemma this means, first, that $g(x) \neq 0$, so $(1/g)(x)$ makes sense, and second that

$$\left| \left(\frac{1}{g}\right)(x) - \frac{1}{m} \right| < \varepsilon.$$

This proves (3). ■

Using Theorem 2 we can prove, trivially, such facts as

$$\lim_{x \rightarrow a} \frac{x^3 + 7x^5}{x^2 + 1} = \frac{a^3 + 7a^5}{a^2 + 1},$$

without going through the laborious process of finding a δ , given an ε . We must begin with

$$\begin{aligned} \lim_{x \rightarrow a} 7 &= 7, \\ \lim_{x \rightarrow a} 1 &= 1, \\ \lim_{x \rightarrow a} x &= a, \end{aligned}$$

but these are easy to prove directly. If we *want* to find the δ , however, the proof of Theorem 2 amounts to a prescription for doing this. Suppose, to take a simpler example, that we want to find a δ such that, for all x ,

$$\text{if } 0 < |x - a| < \delta, \text{ then } |x^2 + x - (a^2 + a)| < \varepsilon.$$

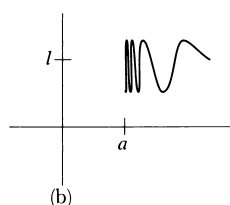
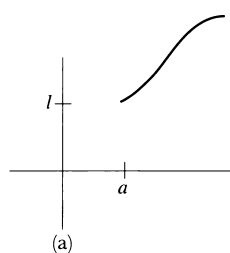


FIGURE 17

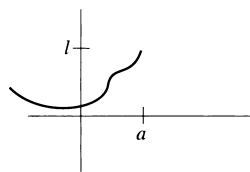


FIGURE 18

There are times when we would like to speak of the limit which f approaches at a , even though there is no $\delta > 0$ such that $f(x)$ is defined for x satisfying $0 < |x - a| < \delta$. For example, we want to distinguish the behavior of the two functions shown in Figure 17, even though they are not defined for numbers less than a . For the function of Figure 17(a) we write

$$\lim_{x \rightarrow a^+} f(x) = l \quad \text{or} \quad \lim_{x \downarrow a} f(x) = l.$$

(The symbols on the left are read: the limit of $f(x)$ as x approaches a from above.) These “limits from above” are obviously closely related to ordinary limits, and the definition is very similar: $\lim_{x \rightarrow a^+} f(x) = l$ means that for every $\varepsilon > 0$ there is a $\delta > 0$ such that, for all x ,

$$\text{if } 0 < x - a < \delta, \text{ then } |f(x) - l| < \varepsilon.$$

(The condition “ $0 < x - a < \delta$ ” is equivalent to “ $0 < |x - a| < \delta$ and $x > a$.”)

“Limits from below” (Figure 18) are defined similarly: $\lim_{x \rightarrow a^-} f(x) = l$ (or $\lim_{x \uparrow a} f(x) = l$) means that for every $\varepsilon > 0$ there is a $\delta > 0$ such that, for all x ,

$$\text{if } 0 < a - x < \delta, \text{ then } |f(x) - l| < \varepsilon.$$

It is quite possible to consider limits from above and below even if f is defined for numbers both greater and less than a . Thus, for the function f of Figure 14, we have

$$\lim_{x \rightarrow 0^+} f(x) = 1 \quad \text{and} \quad \lim_{x \rightarrow 0^-} f(x) = -1.$$

It is an easy exercise (Problem 29) to show that $\lim_{x \rightarrow a} f(x)$ exists if and only if $\lim_{x \rightarrow a^+} f(x)$ and $\lim_{x \rightarrow a^-} f(x)$ both exist and are equal.

Like the definitions of limits from above and below, which have been smuggled into the text informally, there are other modifications of the limit concept which will be found useful. In Chapter 4 it was claimed that if x is large, then $\sin 1/x$ is close to 0. This assertion is usually written

$$\lim_{x \rightarrow \infty} \sin 1/x = 0.$$

The symbol $\lim_{x \rightarrow \infty} f(x)$ is read “the limit of $f(x)$ as x approaches ∞ ,” or “as x becomes infinite,” and a limit of the form $\lim_{x \rightarrow \infty} f(x)$ is often called a limit at infinity.

Figure 19 illustrates a general situation where $\lim_{x \rightarrow \infty} f(x) = l$. Formally, $\lim_{x \rightarrow \infty} f(x) = l$ means that for every $\varepsilon > 0$ there is a number N such that, for all x ,

$$\text{if } x > N, \text{ then } |f(x) - l| < \varepsilon.$$

The analogy with the definition of ordinary limits should be clear: whereas the condition " $0 < |x - a| < \delta$ " expresses the fact that x is close to a , the condition " $x > N$ " expresses the fact that x is large.

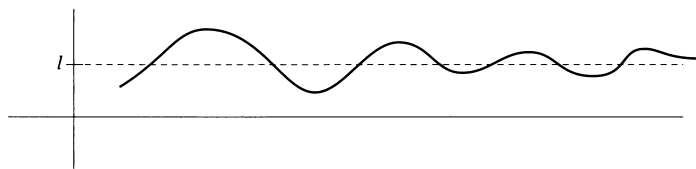


FIGURE 19

We have spent so little time on limits from above and below, and at infinity, because the general philosophy behind the definitions should be clear if you understand the definition of ordinary limits (which are by far the most important). Many exercises on these definitions are provided in the Problems, which also contain several other types of limits which are occasionally useful.

PROBLEMS

1. Find the following limits. (These limits all follow, after some algebraic manipulations, from the various parts of Theorem 2; be sure you know which ones are used in each case, but don't bother listing them.)

- (i) $\lim_{x \rightarrow 1} \frac{x^2 - 1}{x + 1}.$

- (ii) $\lim_{x \rightarrow 2} \frac{x^3 - 8}{x - 2}.$

- (iii) $\lim_{x \rightarrow 3} \frac{x^3 - 8}{x - 2}.$

- (iv) $\lim_{x \rightarrow y} \frac{x^n - y^n}{x - y}.$

- (v) $\lim_{y \rightarrow x} \frac{x^n - y^n}{x - y}.$

- (vi) $\lim_{h \rightarrow 0} \frac{\sqrt{a+h} - \sqrt{a}}{h}.$

2. Find the following limits.

- (i) $\lim_{x \rightarrow 1} \frac{1 - \sqrt{x}}{1 - x}.$

$$(ii) \quad \lim_{x \rightarrow 0} \frac{1 - \sqrt{1 - x^2}}{x}.$$

$$(iii) \quad \lim_{x \rightarrow 0} \frac{1 - \sqrt{1 - x^2}}{x^2}.$$

3. In each of the following cases, determine the limit l for the given a , and prove that it is the limit by showing how to find a δ such that $|f(x) - l| < \varepsilon$ for all x satisfying $0 < |x - a| < \delta$.

$$(i) \quad f(x) = x[3 - \cos(x^2)], \quad a = 0.$$

$$(ii) \quad f(x) = x^2 + 5x - 2, \quad a = 2.$$

$$(iii) \quad f(x) = \frac{100}{x}, \quad a = 1.$$

$$(iv) \quad f(x) = x^4, \quad \text{arbitrary } a.$$

$$(v) \quad f(x) = x^4 + \frac{1}{x}, \quad a = 1.$$

$$(vi) \quad f(x) = \frac{x}{2 - \sin^2 x}, \quad a = 0.$$

$$(vii) \quad f(x) = \sqrt{|x|}, \quad a = 0.$$

$$(viii) \quad f(x) = \sqrt{x}, \quad a = 1.$$

4. For each of the functions in Problem 4-17, decide for which numbers a the limit $\lim_{x \rightarrow a} f(x)$ exists.

- *5. (a) Do the same for each of the functions in Problem 4-19.
 (b) Same problem, if we use infinite decimals ending in a string of 0's instead of those ending in a string of 9's.

6. Suppose the functions f and g have the following property: for all $\varepsilon > 0$ and all x ,

$$\text{if } 0 < |x - 2| < \sin^2\left(\frac{\varepsilon^2}{9}\right) + \varepsilon, \text{ then } |f(x) - 2| < \varepsilon,$$

$$\text{if } 0 < |x - 2| < \varepsilon^2, \text{ then } |g(x) - 4| < \varepsilon.$$

For each $\varepsilon > 0$ find a $\delta > 0$ such that, for all x ,

$$(i) \quad \text{if } 0 < |x - 2| < \delta, \text{ then } |f(x) + g(x) - 6| < \varepsilon.$$

$$(ii) \quad \text{if } 0 < |x - 2| < \delta, \text{ then } |f(x)g(x) - 8| < \varepsilon.$$

$$(iii) \quad \text{if } 0 < |x - 2| < \delta, \text{ then } \left| \frac{1}{g(x)} - \frac{1}{4} \right| < \varepsilon.$$

$$(iv) \quad \text{if } 0 < |x - 2| < \delta, \text{ then } \left| \frac{f(x)}{g(x)} - \frac{1}{2} \right| < \varepsilon.$$

7. Give an example of a function f for which the following assertion is *false*:
 If $|f(x) - l| < \varepsilon$ when $0 < |x - a| < \delta$, then $|f(x) - l| < \varepsilon/2$ when $0 < |x - a| < \delta/2$.

8. (a) If $\lim_{x \rightarrow a} f(x)$ and $\lim_{x \rightarrow a} g(x)$ do not exist, can $\lim_{x \rightarrow a} [f(x) + g(x)]$ exist? Can $\lim_{x \rightarrow a} f(x)g(x)$ exist?
- (b) If $\lim_{x \rightarrow a} f(x)$ exists and $\lim_{x \rightarrow a} [f(x) + g(x)]$ exists, must $\lim_{x \rightarrow a} g(x)$ exist?
- (c) If $\lim_{x \rightarrow a} f(x)$ exists and $\lim_{x \rightarrow a} g(x)$ does not exist, can $\lim_{x \rightarrow a} [f(x) + g(x)]$ exist?
- (d) If $\lim_{x \rightarrow a} f(x)$ exists and $\lim_{x \rightarrow a} f(x)g(x)$ exists, does it follow that $\lim_{x \rightarrow a} g(x)$ exists?
9. Prove that $\lim_{x \rightarrow a} f(x) = \lim_{h \rightarrow 0} f(a + h)$. (This is mainly an exercise in understanding what the terms mean.)
10. (a) Prove that $\lim_{x \rightarrow a} f(x) = l$ if and only if $\lim_{x \rightarrow a} [f(x) - l] = 0$. (First see why the assertion is obvious; then provide a rigorous proof. In this chapter most problems which ask for proofs should be treated in the same way.)
- (b) Prove that $\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow a} f(x - a)$.
- (c) Prove that $\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} f(x^3)$.
- (d) Give an example where $\lim_{x \rightarrow 0} f(x^2)$ exists, but $\lim_{x \rightarrow 0} f(x)$ does not.
11. Suppose there is a $\delta > 0$ such that $f(x) = g(x)$ when $0 < |x - a| < \delta$. Prove that $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} g(x)$. In other words, $\lim_{x \rightarrow a} f(x)$ depends only on the values of $f(x)$ for x near a —this fact is often expressed by saying that limits are a “local property.” (It will clearly help to use δ' , or some other letter, instead of δ , in the definition of limits.)
12. (a) Suppose that $f(x) \leq g(x)$ for all x . Prove that $\lim_{x \rightarrow a} f(x) \leq \lim_{x \rightarrow a} g(x)$, provided that these limits exist.
- (b) How can the hypotheses be weakened?
- (c) If $f(x) < g(x)$ for all x , does it necessarily follow that $\lim_{x \rightarrow a} f(x) < \lim_{x \rightarrow a} g(x)$?
13. Suppose that $f(x) \leq g(x) \leq h(x)$ and that $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} h(x)$. Prove that $\lim_{x \rightarrow a} g(x)$ exists, and that $\lim_{x \rightarrow a} g(x) = \lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} h(x)$. (Draw a picture!)
- *14. (a) Prove that if $\lim_{x \rightarrow 0} f(x)/x = l$ and $b \neq 0$, then $\lim_{x \rightarrow 0} f(bx)/x = bl$. Hint: Write $f(bx)/x = b[f(bx)/bx]$.
- (b) What happens if $b = 0$?
- (c) Part (a) enables us to find $\lim_{x \rightarrow 0} (\sin 2x)/x$ in terms of $\lim_{x \rightarrow 0} (\sin x)/x$. Find this limit in another way.
15. Evaluate the following limits in terms of the number $\alpha = \lim_{x \rightarrow 0} (\sin x)/x$.
- (i) $\lim_{x \rightarrow 0} \frac{\sin 3x}{x}$.
- (ii) $\lim_{x \rightarrow 0} \frac{\sin ax}{\sin bx}$.

- (iii) $\lim_{x \rightarrow 0} \frac{\sin^2 2x}{x}.$
- (iv) $\lim_{x \rightarrow 0} \frac{\sin^2 2x}{x^2}.$
- (v) $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2}.$
- (vi) $\lim_{x \rightarrow 0} \frac{\tan^2 x + 2x}{x + x^2}.$
- (vii) $\lim_{x \rightarrow 0} \frac{x \sin x}{1 - \cos x}.$
- (viii) $\lim_{h \rightarrow 0} \frac{\sin(x+h) - \sin x}{h}.$
- (ix) $\lim_{x \rightarrow 1} \frac{\sin(x^2 - 1)}{x - 1}.$
- (x) $\lim_{x \rightarrow 0} \frac{x^2(3 + \sin x)}{(x + \sin x)^2}.$
- (xi) $\lim_{x \rightarrow 1} (x^2 - 1)^3 \sin \left(\frac{1}{x - 1} \right)^3.$

- 16. (a) Prove that if $\lim_{x \rightarrow a} f(x) = l$, then $\lim_{x \rightarrow a} |f|(x) = |l|$.
 (b) Prove that if $\lim_{x \rightarrow a} f(x) = l$ and $\lim_{x \rightarrow a} g(x) = m$, then $\lim_{x \rightarrow a} \max(f, g)(x) = \max(l, m)$ and similarly for min.
- 17. (a) Prove that $\lim_{x \rightarrow 0} 1/x$ does not exist, i.e., show that $\lim_{x \rightarrow 0} 1/x = l$ is false for every number l .
 (b) Prove that $\lim_{x \rightarrow 1} 1/(x - 1)$ does not exist.
- 18. Prove that if $\lim_{x \rightarrow a} f(x) = l$, then there is a number $\delta > 0$ and a number M such that $|f(x)| < M$ if $0 < |x - a| < \delta$. (What does this mean pictorially?)
 Hint: Why does it suffice to prove that $l - 1 < f(x) < l + 1$ for $0 < |x - a| < \delta$?
- 19. Prove that if $f(x) = 0$ for irrational x and $f(x) = 1$ for rational x , then $\lim_{x \rightarrow a} f(x)$ does not exist for any a .
- *20. Prove that if $f(x) = x$ for rational x , and $f(x) = -x$ for irrational x , then $\lim_{x \rightarrow a} f(x)$ does not exist if $a \neq 0$.
- 21. (a) Prove that if $\lim_{x \rightarrow 0} g(x) = 0$, then $\lim_{x \rightarrow 0} g(x) \sin 1/x = 0$.
 (b) Generalize this fact as follows: If $\lim_{x \rightarrow 0} g(x) = 0$ and $|h(x)| \leq M$ for all x , then $\lim_{x \rightarrow 0} g(x)h(x) = 0$. (Naturally it is unnecessary to do part (a) if you succeed in doing part (b); actually the statement of part (b) may make it easier than (a)—that's one of the values of generalization.)

22. Consider a function f with the following property: if g is any function for which $\lim_{x \rightarrow 0} g(x)$ does not exist, then $\lim_{x \rightarrow 0} [f(x) + g(x)]$ also does not exist. Prove that this happens if and only if $\lim_{x \rightarrow 0} f(x)$ *does* exist. Hint: This is actually very easy: the assumption that $\lim_{x \rightarrow 0} f(x)$ does not exist leads to an immediate contradiction if you consider the right g .
- *23.** This problem is the analogue of Problem 22 when $f + g$ is replaced by $f \cdot g$. In this case the situation is considerably more complex, and the analysis requires several steps (those in search of an especially challenging problem can attempt an independent solution).
- Suppose that $\lim_{x \rightarrow 0} f(x)$ exists and is $\neq 0$. Prove that if $\lim_{x \rightarrow 0} g(x)$ does not exist, then $\lim_{x \rightarrow 0} f(x)g(x)$ also does not exist.
 - Prove the same result if $\lim_{x \rightarrow 0} |f(x)| = \infty$. (The precise definition of this sort of limit is given in Problem 37.)
 - Prove that if neither of these two conditions holds, then there is a function g such that $\lim_{x \rightarrow 0} g(x)$ does not exist, but $\lim_{x \rightarrow 0} f(x)g(x)$ does exist.
Hint: Consider separately the following two cases: (1) for some $\varepsilon > 0$ we have $|f(x)| > \varepsilon$ for all sufficiently small x . (2) For every $\varepsilon > 0$, there are arbitrarily small x with $|f(x)| < \varepsilon$. In the second case, begin by choosing points x_n with $|x_n| < 1/n$ and $|f(x_n)| < 1/n$.
- *24.** Suppose that A_n is, for each natural number n , some *finite* set of numbers in $[0, 1]$, and that A_n and A_m have no members in common if $m \neq n$. Define f as follows:

$$f(x) = \begin{cases} 1/n, & x \text{ in } A_n \\ 0, & x \text{ not in } A_n \text{ for any } n. \end{cases}$$

Prove that $\lim_{x \rightarrow a} f(x) = 0$ for all a in $[0, 1]$.

25. Explain why the following definitions of $\lim_{x \rightarrow a} f(x) = l$ are all correct:
For every $\delta > 0$ there is an $\varepsilon > 0$ such that, for all x ,
- if $0 < |x - a| < \varepsilon$, then $|f(x) - l| < \delta$.
 - if $0 < |x - a| < \varepsilon$, then $|f(x) - l| \leq \delta$.
 - if $0 < |x - a| < \varepsilon$, then $|f(x) - l| < 5\delta$.
 - if $0 < |x - a| < \varepsilon/10$, then $|f(x) - l| < \delta$.
- *26.** Give examples to show that the following definitions of $\lim_{x \rightarrow a} f(x) = l$ are *not* correct.
- For all $\delta > 0$ there is an $\varepsilon > 0$ such that if $0 < |x - a| < \delta$, then $|f(x) - l| < \varepsilon$.
 - For all $\varepsilon > 0$ there is a $\delta > 0$ such that if $|f(x) - l| < \varepsilon$, then $0 < |x - a| < \delta$.

27. For each of the functions in Problem 4-17 indicate for which numbers a the one-sided limits $\lim_{x \rightarrow a^+} f(x)$ and $\lim_{x \rightarrow a^-} f(x)$ exist.
- *28. (a) Do the same for each of the functions in Problem 4-19.
 (b) Also consider what happens if decimals ending in 0's are used instead of decimals ending in 9's.
29. Prove that $\lim_{x \rightarrow a} f(x)$ exists if $\lim_{x \rightarrow a^+} f(x) = \lim_{x \rightarrow a^-} f(x)$.
30. Prove that
- (i) $\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^-} f(-x)$.
 - (ii) $\lim_{x \rightarrow 0} f(|x|) = \lim_{x \rightarrow 0^+} f(x)$.
 - (iii) $\lim_{x \rightarrow 0} f(x^2) = \lim_{x \rightarrow 0^+} f(x)$.
- (These equations, and others like them, are open to several interpretations. They might mean only that the two limits are equal if they both exist; or that if a certain one of the limits exists, the other also exists and is equal to it; or that if either limit exists, then the other exists and is equal to it. Decide for yourself which interpretations are suitable.)
31. Suppose that $\lim_{x \rightarrow a^-} f(x) < \lim_{x \rightarrow a^+} f(x)$. (Draw a picture to illustrate this assertion.) Prove that there is some $\delta > 0$ such that $f(x) < f(y)$ whenever $x < a < y$ and $|x - a| < \delta$ and $|y - a| < \delta$. Is the converse true?
32. Prove that $\lim_{x \rightarrow \infty} (a_n x^n + \cdots + a_0) / (b_m x^m + \cdots + b_0)$ (with $a_n \neq 0$ and $b_m \neq 0$) exists if and only if $m \geq n$. What is the limit when $m = n$? When $m > n$? Hint: the one easy limit is $\lim_{x \rightarrow \infty} 1/x^k = 0$; do some algebra so that this is the only information you need.
33. Find the following limits.
- (i) $\lim_{x \rightarrow \infty} \frac{x + \sin^3 x}{5x + 6}$.
 - (ii) $\lim_{x \rightarrow \infty} \frac{x \sin x}{x^2 + 5}$.
 - (iii) $\lim_{x \rightarrow \infty} \sqrt{x^2 + x} - x$.
 - (iv) $\lim_{x \rightarrow \infty} \frac{x^2(1 + \sin^2 x)}{(x + \sin x)^2}$.
34. Prove that $\lim_{x \rightarrow 0^+} f(1/x) = \lim_{x \rightarrow \infty} f(x)$.
35. Find the following limits in terms of the number $\alpha = \lim_{x \rightarrow 0} (\sin x)/x$.
- (i) $\lim_{x \rightarrow \infty} \frac{\sin x}{x}$.
 - (ii) $\lim_{x \rightarrow \infty} x \sin \frac{1}{x}$.

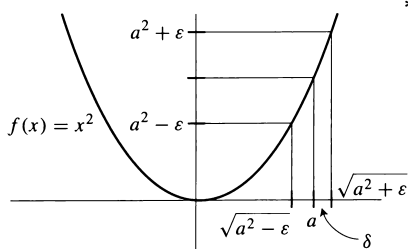


FIGURE 20

41. (a) For $c > 1$, show that $c^{1/n} = \sqrt[n]{c}$ approaches 1 as n becomes very large.
Hint: Show that for any $\varepsilon > 0$ we cannot have $c^{1/n} > 1 + \varepsilon$ for large n .
(b) More generally, if $c > 0$, then $c^{1/n}$ approaches 1 as n becomes very large.
- *42. After sending the manuscript for the first edition of this book off to the printer, I thought of a much simpler way to prove that $\lim_{x \rightarrow a} x^2 = a^2$ and $\lim_{x \rightarrow a} x^3 = a^3$, without going through all the factoring tricks on page 92. Suppose, for example, that we want to prove that $\lim_{x \rightarrow a} x^2 = a^2$, where $a > 0$. Given $\varepsilon > 0$, we simply let δ be the minimum of $\sqrt{a^2 + \varepsilon} - a$ and $a - \sqrt{a^2 - \varepsilon}$ (see Figure 19); then $|x - a| < \delta$ implies that $\sqrt{a^2 - \varepsilon} < x < \sqrt{a^2 + \varepsilon}$, so $a^2 - \varepsilon < x^2 < a^2 + \varepsilon$, or $|x^2 - a^2| < \varepsilon$. It is fortunate that these pages had already been set, so that I couldn't make these changes, because this "proof" is completely fallacious. Wherein lies the fallacy?

36. Define “ $\lim_{x \rightarrow -\infty} f(x) = l$.”

- (a) Find $\lim_{x \rightarrow -\infty} (a_n x^n + \cdots + a_0) / (b_m x^m + \cdots + b_0)$.
- (b) Prove that $\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow -\infty} f(-x)$.
- (c) Prove that $\lim_{x \rightarrow 0^-} f(1/x) = \lim_{x \rightarrow -\infty} f(x)$.

37. We define $\lim_{x \rightarrow a} f(x) = \infty$ to mean that for all N there is a $\delta > 0$ such that, for all x , if $0 < |x - a| < \delta$, then $f(x) > N$. (Draw an appropriate picture!) (Of course, we may still say that $\lim_{x \rightarrow a} f(x)$ “does not exist” in the usual sense.)

- (a) Show that $\lim_{x \rightarrow 3} 1/(x - 3)^2 = \infty$.
- (b) Prove that if $f(x) > \varepsilon > 0$ for all x , and $\lim_{x \rightarrow a} g(x) = 0$, then

$$\lim_{x \rightarrow a} f(x)/|g(x)| = \infty.$$

38. (a) Define $\lim_{x \rightarrow a^+} f(x) = \infty$ and $\lim_{x \rightarrow a^-} f(x) = \infty$. (Or at least convince yourself that you could write down the definitions if you had the energy. How many other such symbols can you define?)

- (b) Prove that $\lim_{x \rightarrow 0^+} 1/x = \infty$.
- (c) Prove that $\lim_{x \rightarrow 0^+} f(x) = \infty$ if and only if $\lim_{x \rightarrow \infty} f(1/x) = \infty$.

39. Find the following limits, when they exist.

- (i) $\lim_{x \rightarrow \infty} \frac{x^3 + 4x - 7}{7x^2 - x + 1}$
- (ii) $\lim_{x \rightarrow \infty} x(1 + \sin^2 x)$.
- (iii) $\lim_{x \rightarrow \infty} x \sin^2 x$.
- (iv) $\lim_{x \rightarrow \infty} x^2 \sin \frac{1}{x}$.
- (v) $\lim_{x \rightarrow \infty} \sqrt{x^2 + 2x} - x$.
- (vi) $\lim_{x \rightarrow \infty} x(\sqrt{x + 2} - \sqrt{x})$.
- (vii) $\lim_{x \rightarrow \infty} \frac{\sqrt{|x|}}{x}$.

40. (a) Find the perimeter of a regular n -gon inscribed in a circle of radius r . [Answer: $2rn \sin(\pi/n)$.]

- (b) What value does this perimeter approach as n becomes very large?
- (c) What limit can you guess from this?

CHAPTER 6 CONTINUOUS FUNCTIONS

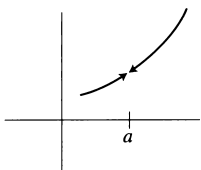


FIGURE 1

If f is an arbitrary function, it is not necessarily true that

$$\lim_{x \rightarrow a} f(x) = f(a).$$

In fact, there are many ways this can fail to be true. For example, f might not even be defined at a , in which case the equation makes no sense (Figure 1).

Again, $\lim_{x \rightarrow a} f(x)$ might not exist (Figure 2). Finally, as illustrated in Figure 3, even if f is defined at a and $\lim_{x \rightarrow a} f(x)$ exists, the limit might not equal $f(a)$.

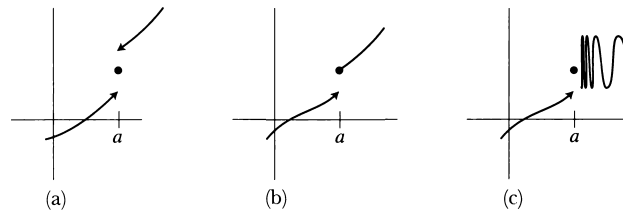


FIGURE 2

We would like to regard all behavior of this type as abnormal and honor, with some complimentary designation, functions which do not exhibit such peculiarities. The term which has been adopted is “continuous.” Intuitively, a function f is continuous if the graph contains no breaks, jumps, or wild oscillations. Although this description will usually enable you to decide whether a function is continuous simply by looking at its graph (a skill well worth cultivating) it is easy to be fooled, and the precise definition is *very* important.

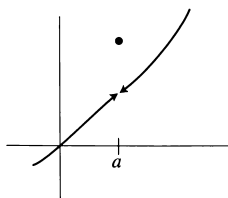


FIGURE 3

DEFINITION

The function f is **continuous at a** if

$$\lim_{x \rightarrow a} f(x) = f(a).$$

We will have no difficulty finding many examples of functions which are, or are not, continuous at some number a —every example involving limits provides an example about continuity, and Chapter 5 certainly provides enough of these.

The function $f(x) = \sin 1/x$ is not continuous at 0, because it is not even defined at 0, and the same is true of the function $g(x) = x \sin 1/x$. On the other hand, if we are willing to extend the second of these functions, that is, if we wish to define

a new function G by

$$G(x) = \begin{cases} x \sin 1/x, & x \neq 0 \\ a, & x = 0, \end{cases}$$

then the choice of $a = G(0)$ can be made in such a way that G will be continuous at 0—to do this we can (if fact, we must) define $G(0) = 0$ (Figure 4). This sort of extension is not possible for f ; if we define

$$F(x) = \begin{cases} \sin 1/x, & x \neq 0 \\ a, & x = 0, \end{cases}$$

then F will not be continuous at 0, no matter what a is, because $\lim_{x \rightarrow 0} f(x)$ does not exist.

The function

$$f(x) = \begin{cases} x, & x \text{ rational} \\ 0, & x \text{ irrational} \end{cases}$$

is not continuous at a , if $a \neq 0$, since $\lim_{x \rightarrow a} f(x)$ does not exist. However, $\lim_{x \rightarrow 0} f(x) = 0 = f(0)$, so f is continuous at precisely one point, 0.

The functions $f(x) = c$, $g(x) = x$, and $h(x) = x^2$ are continuous at all numbers a , since

$$\begin{aligned} \lim_{x \rightarrow a} f(x) &= \lim_{x \rightarrow a} c = c = f(a), \\ \lim_{x \rightarrow a} g(x) &= \lim_{x \rightarrow a} x = a = g(a), \\ \lim_{x \rightarrow a} h(x) &= \lim_{x \rightarrow a} x^2 = a^2 = h(a). \end{aligned}$$

Finally, consider the function

$$f(x) = \begin{cases} 0, & x \text{ irrational} \\ 1/q, & x = p/q \text{ in lowest terms.} \end{cases}$$

In Chapter 5 we showed that $\lim_{x \rightarrow a} f(x) = 0$ for all a (actually, only for $0 < a < 1$, but you can easily see that this is true for all a). Since $0 = f(a)$ only when a is irrational, this function is continuous at a if a is irrational, but not if a is rational.

It is even easier to give examples of continuity if we prove two simple theorems.

THEOREM 1 If f and g are continuous at a , then

- (1) $f + g$ is continuous at a ,
- (2) $f \cdot g$ is continuous at a .

Moreover, if $g(a) \neq 0$, then

- (3) $1/g$ is continuous at a .

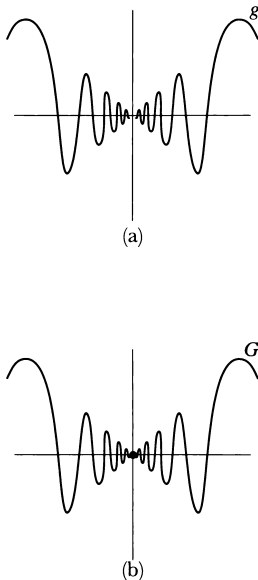


FIGURE 4

PROOF Since f and g are continuous at a ,

$$\lim_{x \rightarrow a} f(x) = f(a) \quad \text{and} \quad \lim_{x \rightarrow a} g(x) = g(a).$$

By Theorem 2(1) of Chapter 5 this implies that

$$\lim_{x \rightarrow a} (f + g)(x) = f(a) + g(a) = (f + g)(a),$$

which is just the assertion that $f + g$ is continuous at a . The proofs of parts (2) and (3) are left to you. ■

Starting with the functions $f(x) = c$ and $f(x) = x$, which are continuous at a , for every a , we can use Theorem 1 to conclude that a function

$$f(x) = \frac{b_n x^n + b_{n-1} x^{n-1} + \cdots + b_0}{c_m x^m + c_{m-1} x^{m-1} + \cdots + c_0}$$

is continuous at every point in its domain. But it is harder to get much further than that. When we discuss the sine function in detail it will be easy to prove that $\sin$ is continuous at a for all a ; let us assume this fact meanwhile. A function like

$$f(x) = \frac{\sin^2 x + x^2 + x^4 \sin x}{\sin^{27} x + 4x^2 \sin^2 x}$$

can now be proved continuous at every point in its domain. But we are still unable to prove the continuity of a function like $f(x) = \sin(x^2)$; we obviously need a theorem about the composition of continuous functions. Before stating this theorem, the following point about the definition of continuity is worth noting. If we translate the equation $\lim_{x \rightarrow a} f(x) = f(a)$ according to the definition of limits, we obtain

$$\begin{aligned} &\text{for every } \varepsilon > 0 \text{ there is } \delta > 0 \text{ such that, for all } x, \\ &\text{if } 0 < |x - a| < \delta, \text{ then } |f(x) - f(a)| < \varepsilon. \end{aligned}$$

But in this case, where the limit is $f(a)$, the phrase

$$0 < |x - a| < \delta$$

may be changed to the simpler condition

$$|x - a| < \delta,$$

since if $x = a$ it is certainly true that $|f(x) - f(a)| < \varepsilon$.

THEOREM 2 If g is continuous at a , and f is continuous at $g(a)$, then $f \circ g$ is continuous at a . (Notice that f is required to be continuous at $g(a)$, not at a .)

PROOF Let $\varepsilon > 0$. We wish to find a $\delta > 0$ such that for all x ,

$$\begin{aligned} &\text{if } |x - a| < \delta, \text{ then } |(f \circ g)(x) - (f \circ g)(a)| < \varepsilon, \\ &\text{i.e., } |f(g(x)) - f(g(a))| < \varepsilon. \end{aligned}$$

shows that this description is a little too optimistic, but it is nevertheless true that there are many important results involving functions which are continuous on an interval. These theorems are generally much harder than the ones in this chapter, but there is a simple theorem which forms a bridge between the two kinds of results. The hypothesis of this theorem requires continuity at only a single point, but the conclusion describes the behavior of the function on some interval containing the point. Although this theorem is really a lemma for later arguments, it is included here as a preview of things to come.

THEOREM 3 Suppose f is continuous at a , and $f(a) > 0$. Then $f(x) > 0$ for all x in some interval containing a ; more precisely, there is a number $\delta > 0$ such that $f(x) > 0$ for all x satisfying $|x - a| < \delta$. Similarly, if $f(a) < 0$, then there is a number $\delta > 0$ such that $f(x) < 0$ for all x satisfying $|x - a| < \delta$.

PROOF Consider the case $f(a) > 0$. Since f is continuous at a , for every $\varepsilon > 0$ there is a $\delta > 0$ such that, for all x ,

$$\begin{aligned} \text{if } |x - a| < \delta, \text{ then } |f(x) - f(a)| < \varepsilon, \\ \text{i.e., } -\varepsilon < f(x) - f(a) < \varepsilon. \end{aligned}$$

In particular, this must hold for $\varepsilon = \frac{1}{2}f(a)$, since $\frac{1}{2}f(a) > 0$ (Figure 5). Thus there is $\delta > 0$ so that for all x ,

$$\text{if } |x - a| < \delta, \text{ then } -\frac{1}{2}f(a) < f(x) - f(a) < \frac{1}{2}f(a),$$

and this implies that $f(x) > \frac{1}{2}f(a) > 0$. (We could even have picked ε to be $f(a)$ itself, leading to a proof that is more elegant, but more confusing to picture.)

A similar proof can be given in the case $f(a) < 0$; take $\varepsilon = -\frac{1}{2}f(a)$. Or one can apply the first case to the function $-f$. ■

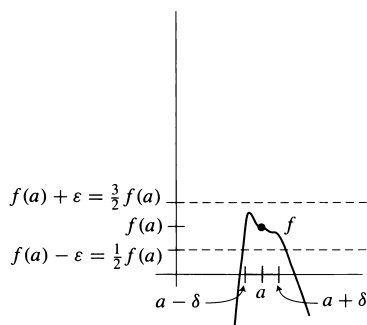


FIGURE 5

PROBLEMS

- For which of the following functions f is there a continuous function F with domain $\mathbf{R}$ such that $F(x) = f(x)$ for all x in the domain of f ?

(i) $f(x) = \frac{x^2 - 4}{x - 2}$.

(ii) $f(x) = \frac{|x|}{x}$.

(iii) $f(x) = 0$, x irrational.

(iv) $f(x) = 1/q$, $x = p/q$ rational in lowest terms.

- At which points are the functions of Problems 4-17 and 4-19 continuous?

We first use continuity of f to estimate how close $g(x)$ must be to $g(a)$ in order for this inequality to hold. Since f is continuous at $g(a)$, there is a $\delta' > 0$ such that for all y ,

$$(1) \quad \text{if } |y - g(a)| < \delta', \text{ then } |f(y) - f(g(a))| < \varepsilon.$$

In particular, this means that

$$(2) \quad \text{if } |g(x) - g(a)| < \delta', \text{ then } |f(g(x)) - f(g(a))| < \varepsilon.$$

We now use continuity of g to estimate how close x must be to a in order for the inequality $|g(x) - g(a)| < \delta'$ to hold. The number δ' is a positive number just like any other positive number; we can therefore take δ' as the ε (!) in the definition of continuity of g at a . We conclude that there is a $\delta > 0$ such that, for all x ,

$$(3) \quad \text{if } |x - a| < \delta, \text{ then } |g(x) - g(a)| < \delta'.$$

Combining (2) and (3) we see that for all x ,

$$\text{if } |x - a| < \delta, \text{ then } |f(g(x)) - f(g(a))| < \varepsilon. \blacksquare$$

We can now reconsider the function

$$f(x) = \begin{cases} x \sin 1/x, & x \neq 0 \\ 0, & x = 0. \end{cases}$$

We have already noted that f is continuous at 0. A few applications of Theorems 1 and 2, together with the continuity of $\sin$, show that f is also continuous at a , for $a \neq 0$. Functions like $f(x) = \sin(x^2 + \sin(x + \sin^2(x^3)))$ should be equally easy for you to analyze.

The few theorems of this chapter have all been related to continuity of functions at a single point, but the concept of continuity doesn't begin to be really interesting until we focus our attention on functions which are continuous at all points of some interval. If f is continuous at x for all x in (a, b) , then f is called **continuous on** (a, b) ; as a "special case", f is **continuous on** $\mathbf{R} = (-\infty, \infty)$ [see page 57] if it is continuous at x for all x in $\mathbf{R}$. Continuity on a closed interval must be defined a little differently; a function f is called **continuous on** $[a, b]$ if

- (1) f is continuous at x for all x in (a, b) ,
- (2) $\lim_{x \rightarrow a^+} f(x) = f(a)$ and $\lim_{x \rightarrow b^-} f(x) = f(b)$.

(We also often simply say that a function is continuous if it is continuous at x for all x in its domain.)

Functions which are continuous on an interval are usually regarded as especially well behaved; indeed continuity might be specified as the first condition which a "reasonable" function ought to satisfy. A continuous function is sometimes described, intuitively, as one whose graph can be drawn without lifting your pencil from the paper. Consideration of the function

$$f(x) = \begin{cases} x \sin 1/x, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

3. (a) Suppose that f is a function satisfying $|f(x)| \leq |x|$ for all x . Show that f is continuous at 0. (Notice that $f(0)$ must equal 0.)
 (b) Give an example of such a function f which is not continuous at any $a \neq 0$.
 (c) Suppose that g is continuous at 0 and $g(0) = 0$, and $|f(x)| \leq |g(x)|$. Prove that f is continuous at 0.
4. Give an example of a function f such that f is continuous nowhere, but $|f|$ is continuous everywhere.
5. For each number a , find a function which is continuous at a , but not at any other points.
6. (a) Find a function f which is discontinuous at $1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots$ but continuous at all other points.
 (b) Find a function f which is discontinuous at $1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots$, and at 0, but continuous at all other points.
7. Suppose that f satisfies $f(x + y) = f(x) + f(y)$, and that f is continuous at 0. Prove that f is continuous at a for all a .
8. Suppose that f is continuous at a and $f(a) = 0$. Prove that if $\alpha \neq 0$, then $f + \alpha$ is nonzero in some open interval containing a .
9. (a) Suppose f is defined at a but is not continuous at a . Prove that for some number $\varepsilon > 0$ there are numbers x arbitrarily close to a with $|f(x) - f(a)| > \varepsilon$. Illustrate graphically.
 (b) Conclude that for some number $\varepsilon > 0$ either there are numbers x arbitrarily close to a with $f(x) < f(a) - \varepsilon$ or there are numbers x arbitrarily close to a with $f(x) > f(a) + \varepsilon$.
10. (a) Prove that if f is continuous at a , then so is $|f|$.
 (b) Prove that every function f continuous on $\mathbf{R}$ can be written $f = E + O$, where E is even and continuous and O is odd and continuous.
 (c) Prove that if f and g are continuous, then so are $\max(f, g)$ and $\min(f, g)$.
 (d) Prove that every continuous f can be written $f = g - h$, where g and h are nonnegative and continuous.
11. Prove Theorem 1(3) by using Theorem 2 and continuity of the function $f(x) = 1/x$.
- *12. (a) Prove that if f is continuous at l and $\lim_{x \rightarrow a} g(x) = l$, then $\lim_{x \rightarrow a} f(g(x)) = f(l)$. (You can go right back to the definitions, but it is easier to consider the function G with $G(x) = g(x)$ for $x \neq a$, and $G(a) = l$.)
 (b) Show that if continuity of f at l is not assumed, then it is not generally true that $\lim_{x \rightarrow a} f(g(x)) = f(\lim_{x \rightarrow a} g(x))$. Hint: Try $f(x) = 0$ for $x \neq l$, and $f(l) = 1$.

13. (a) Prove that if f is continuous on $[a, b]$, then there is a function g which is continuous on $\mathbf{R}$, and which satisfies $g(x) = f(x)$ for all x in $[a, b]$. Hint: Since you obviously have a great deal of choice, try making g constant on $(-\infty, a]$ and $[b, \infty)$.
- (b) Give an example to show that this assertion is false if $[a, b]$ is replaced by (a, b) .
14. (a) Suppose that g and h are continuous at a , and that $g(a) = h(a)$. Define $f(x)$ to be $g(x)$ if $x \geq a$ and $h(x)$ if $x \leq a$. Prove that f is continuous at a .
- (b) Suppose g is continuous on $[a, b]$ and h is continuous on $[b, c]$ and $g(b) = h(b)$. Let $f(x)$ be $g(x)$ for x in $[a, b]$ and $h(x)$ for x in $[b, c]$. Show that f is continuous on $[a, c]$. (Thus, continuous functions can be “pasted together”.)
15. Prove that if f is continuous at a , then for any $\varepsilon > 0$ there is a $\delta > 0$ so that whenever $|x - a| < \delta$ and $|y - a| < \delta$, we have $|f(x) - f(y)| < \varepsilon$.
16. (a) Prove the following version of Theorem 3 for “right-hand continuity”: Suppose that $\lim_{x \rightarrow a^+} f(x) = f(a)$, and $f(a) > 0$. Then there is a number $\delta > 0$ such that $f(x) > 0$ for all x satisfying $0 \leq x - a < \delta$. Similarly, if $f(a) < 0$, then there is a number $\delta > 0$ such that $f(x) < 0$ for all x satisfying $0 \leq x - a < \delta$.
- (b) Prove a version of Theorem 3 when $\lim_{x \rightarrow b^-} f(x) = f(b)$.
17. If $\lim_{x \rightarrow a} f(x)$ exists, but is $\neq f(a)$, then f is said to have a **removable discontinuity** at a .
- (a) If $f(x) = \sin 1/x$ for $x \neq 0$ and $f(0) = 1$, does f have a removable discontinuity at 0? What if $f(x) = x \sin 1/x$ for $x \neq 0$, and $f(0) = 1$?
- (b) Suppose f has a removable discontinuity at a . Let $g(x) = f(x)$ for $x \neq a$, and let $g(a) = \lim_{x \rightarrow a} f(x)$. Prove that g is continuous at a . (Don’t work very hard; this is quite easy.)
- (c) Let $f(x) = 0$ if x is irrational, and let $f(p/q) = 1/q$ if p/q is in lowest terms. What is the function g defined by $g(x) = \lim_{y \rightarrow x} f(y)$?
- *(d) Let f be a function with the property that every point of discontinuity is a removable discontinuity. This means that $\lim_{y \rightarrow x} f(y)$ exists for all x , but f may be discontinuous at some (even infinitely many) numbers x . Define $g(x) = \lim_{y \rightarrow x} f(y)$. Prove that g is continuous. (This is not quite so easy as part (b).)
- ** (e) Is there a function f which is discontinuous at every point, and which has only removable discontinuities? (It is worth thinking about this problem now, but mainly as a test of intuition; even if you suspect the correct answer, you will almost certainly be unable to prove it at the present time. See Problem 22-33.)

CHAPTER 7 THREE HARD THEOREMS

This chapter is devoted to three theorems about continuous functions, and some of their consequences. The proofs of the three theorems themselves will not be given until the next chapter, for reasons which are explained at the end of this chapter.

THEOREM 1 If f is continuous on $[a, b]$ and $f(a) < 0 < f(b)$, then there is some x in $[a, b]$ such that $f(x) = 0$.

(Geometrically, this means that the graph of a continuous function which starts below the horizontal axis and ends above it must cross this axis at some point, as in Figure 1.)

THEOREM 2 If f is continuous on $[a, b]$, then f is bounded above on $[a, b]$, that is, there is some number N such that $f(x) \leq N$ for all x in $[a, b]$.

(Geometrically, this theorem means that the graph of f lies below some line parallel to the horizontal axis, as in Figure 2.)

THEOREM 3 If f is continuous on $[a, b]$, then there is some number y in $[a, b]$ such that $f(y) \geq f(x)$ for all x in $[a, b]$ (Figure 3).

These three theorems differ markedly from the theorems of Chapter 6. The hypotheses of those theorems always involved continuity at a single point, while

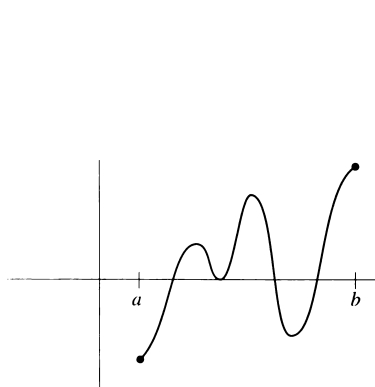


FIGURE 1

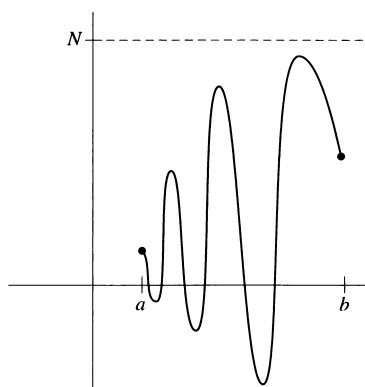


FIGURE 2

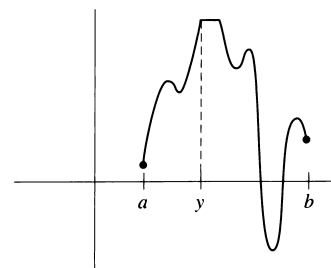


FIGURE 3

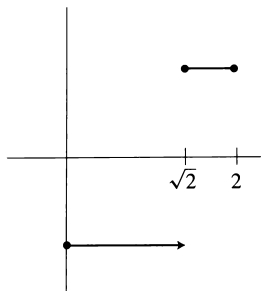


FIGURE 4

the hypotheses of the present theorems require continuity on a whole interval $[a, b]$ —if continuity fails to hold at a single point, the conclusions may fail. For example, let f be the function shown in Figure 4,

$$f(x) = \begin{cases} -1, & 0 \leq x < \sqrt{2} \\ 1, & \sqrt{2} \leq x \leq 2. \end{cases}$$

Then f is continuous at every point of $[0, 2]$ except $\sqrt{2}$, and $f(0) < 0 < f(2)$, but there is no point x in $[0, 2]$ such that $f(x) = 0$; the discontinuity at the single point $\sqrt{2}$ is sufficient to destroy the conclusion of Theorem 1.

Similarly, suppose that f is the function shown in Figure 5,

$$f(x) = \begin{cases} 1/x, & x \neq 0 \\ 0, & x = 0. \end{cases}$$

Then f is continuous at every point of $[0, 1]$ except 0, but f is not bounded above on $[0, 1]$. In fact, for any number $N > 0$ we have $f(1/2N) = 2N > N$.

This example also shows that the closed interval $[a, b]$ in Theorem 2 cannot be replaced by the open interval (a, b) , for the function f is continuous on $(0, 1)$, but is not bounded there.

Finally, consider the function shown in Figure 6,

$$f(x) = \begin{cases} x^2, & x < 1 \\ 0, & x \geq 1. \end{cases}$$

On the interval $[0, 1]$ the function f is bounded above, so f does satisfy the conclusion of Theorem 2, even though f is not continuous on $[0, 1]$. But f does not satisfy the conclusion of Theorem 3—there is no y in $[0, 1]$ such that $f(y) \geq f(x)$ for all x in $[0, 1]$; in fact, it is certainly not true that $f(1) \geq f(x)$ for all x in $[0, 1]$ so we cannot choose $y = 1$, nor can we choose $0 \leq y < 1$ because $f(y) < f(x)$ if x is any number with $y < x < 1$.

This example shows that Theorem 3 is considerably stronger than Theorem 2. Theorem 3 is often paraphrased by saying that a continuous function on a closed interval “takes on its maximum value” on that interval.

As a compensation for the stringency of the hypotheses of our three theorems, the conclusions are of a totally different order than those of previous theorems. They describe the behavior of a function, not just near a point, but on a whole interval; such “global” properties of a function are always significantly more difficult to prove than “local” properties, and are correspondingly of much greater power. To illustrate the usefulness of Theorems 1, 2, and 3, we will soon deduce some important consequences, but it will help to first mention some simple generalizations of these theorems.

THEOREM 4 If f is continuous on $[a, b]$ and $f(a) < c < f(b)$, then there is some x in $[a, b]$ such that $f(x) = c$.

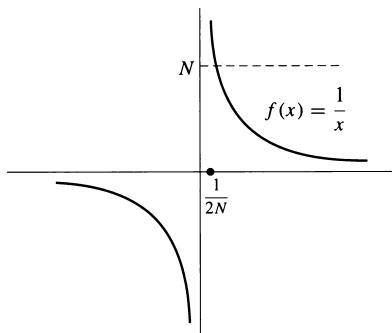


FIGURE 5

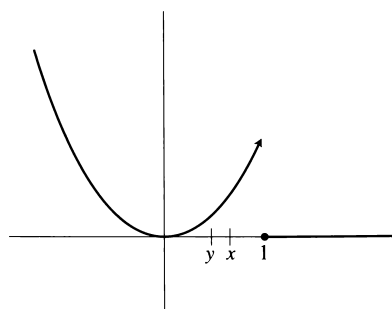


FIGURE 6

PROOF Let $g = f - c$. Then g is continuous, and $g(a) < 0 < g(b)$. By Theorem 1, there is some x in $[a, b]$ such that $g(x) = 0$. But this means that $f(x) = c$. ■

THEOREM 5 If f is continuous on $[a, b]$ and $f(a) > c > f(b)$, then there is some x in $[a, b]$ such that $f(x) = c$.

PROOF The function $-f$ is continuous on $[a, b]$ and $-f(a) < -c < -f(b)$. By Theorem 4 there is some x in $[a, b]$ such that $-f(x) = -c$, which means that $f(x) = c$. ■

Theorems 4 and 5 together show that f takes on any value between $f(a)$ and $f(b)$. We can do even better than this: if c and d are in $[a, b]$, then f takes on any value between $f(c)$ and $f(d)$. The proof is simple: if, for example, $c < d$, then just apply Theorems 4 and 5 to the interval $[c, d]$. Summarizing, if a continuous function on an interval takes on two values, it takes on every value in between; this slight generalization of Theorem 1 is often called the Intermediate Value Theorem.

THEOREM 6 If f is continuous on $[a, b]$, then f is bounded below on $[a, b]$, that is, there is some number N such that $f(x) \geq N$ for all x in $[a, b]$.

PROOF The function $-f$ is continuous on $[a, b]$, so by Theorem 2 there is a number M such that $-f(x) \leq M$ for all x in $[a, b]$. But this means that $f(x) \geq -M$ for all x in $[a, b]$, so we can let $N = -M$. ■

Theorems 2 and 6 together show that a continuous function f on $[a, b]$ is bounded on $[a, b]$, that is, there is a number N such that $|f(x)| \leq N$ for all x in $[a, b]$. In fact, since Theorem 2 ensures the existence of a number N_1 such that $f(x) \leq N_1$ for all x in $[a, b]$, and Theorem 6 ensures the existence of a number N_2 such that $f(x) \geq N_2$ for all x in $[a, b]$, we can take $N = \max(|N_1|, |N_2|)$.

THEOREM 7 If f is continuous on $[a, b]$, then there is some y in $[a, b]$ such that $f(y) \leq f(x)$ for all x in $[a, b]$.

(A continuous function on a closed interval takes on its minimum value on that interval.)

PROOF The function $-f$ is continuous on $[a, b]$; by Theorem 3 there is some y in $[a, b]$ such that $-f(y) \geq -f(x)$ for all x in $[a, b]$, which means that $f(y) \leq f(x)$ for all x in $[a, b]$. ■

Now that we have derived the trivial consequences of Theorems 1, 2, and 3, we can begin proving a few interesting things.

THEOREM 8 Every positive number has a square root. In other words, if $\alpha > 0$, then there is some number x such that $x^2 = \alpha$.

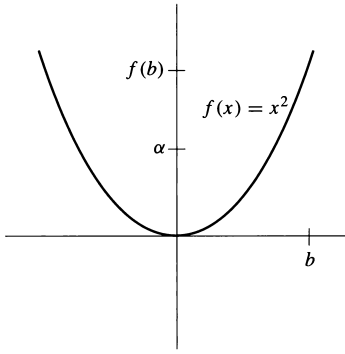


FIGURE 7

PROOF Consider the function $f(x) = x^2$, which is certainly continuous. Notice that the statement of the theorem can be expressed in terms of f : “the number α has a square root” means that f takes on the value α . The proof of this fact about f will be an easy consequence of Theorem 4.

There is obviously a number $b > 0$ such that $f(b) > \alpha$ (as illustrated in Figure 7); in fact, if $\alpha > 1$ we can take $b = \alpha$, while if $\alpha < 1$ we can take $b = 1$. Since $f(0) < \alpha < f(b)$, Theorem 4 applied to $[0, b]$ implies that for some x (in $[0, b]$), we have $f(x) = \alpha$, i.e., $x^2 = \alpha$. ■

Precisely the same argument can be used to prove that a positive number has an n th root, for any natural number n . If n happens to be odd, one can do better: *every* number has an n th root. To prove this we just note that if the positive number α has the n th root x , i.e., if $x^n = \alpha$, then $(-x)^n = -\alpha$ (since n is odd), so $-\alpha$ has the n th root $-x$. The assertion, that for odd n any number α has an n th root, is equivalent to the statement that the equation

$$x^n - \alpha = 0$$

has a root if n is odd. Expressed in this way the result is susceptible of great generalization.

THEOREM 9 If n is odd, then any equation

$$x^n + a_{n-1}x^{n-1} + \cdots + a_0 = 0$$

has a root.

PROOF We obviously want to consider the function

$$f(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_0;$$

we would like to prove that f is sometimes positive and sometimes negative. The intuitive idea is that for large $|x|$, the function is very much like $g(x) = x^n$ and, since n is odd, this function is positive for large positive x and negative for large negative x . A little algebra is all we need to make this intuitive idea work.

The proper analysis of the function f depends on writing

$$f(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_0 = x^n \left(1 + \frac{a_{n-1}}{x} + \cdots + \frac{a_0}{x^n} \right).$$

Note that

$$\left| \frac{a_{n-1}}{x} + \frac{a_{n-2}}{x^2} + \cdots + \frac{a_0}{x^n} \right| \leq \frac{|a_{n-1}|}{|x|} + \cdots + \frac{|a_0|}{|x|^n}.$$

Consequently, if we choose x satisfying

$$(*) \quad |x| > 1, 2n|a_{n-1}|, \dots, 2n|a_0|,$$

then $|x^k| > |x|$ and

$$\frac{|a_{n-k}|}{|x^k|} < \frac{|a_{n-k}|}{|x|} < \frac{|a_{n-k}|}{2n|a_{n-k}|} = \frac{1}{2n},$$

so

$$\left| \frac{a_{n-1}}{x} + \frac{a_{n-2}}{x^2} + \cdots + \frac{a_0}{x^n} \right| \leq \underbrace{\frac{1}{2n} + \cdots + \frac{1}{2n}}_{n \text{ terms}} = \frac{1}{2}.$$

In other words,

$$-\frac{1}{2} \leq \frac{a_{n-1}}{x} + \cdots + \frac{a_0}{x^n} \leq \frac{1}{2},$$

which implies that

$$\frac{1}{2} \leq 1 + \frac{a_{n-1}}{x} + \cdots + \frac{a_0}{x^n}.$$

Therefore, if we choose an $x_1 > 0$ which satisfies (*), then

$$\frac{(x_1)^n}{2} \leq (x_1)^n \left(1 + \frac{a_{n-1}}{x_1} + \cdots + \frac{a_0}{(x_1)^n} \right) = f(x_1),$$

so that $f(x_1) > 0$. On the other hand, if $x_2 < 0$ satisfies (*), then $(x_2)^n < 0$ and

$$\frac{(x_2)^n}{2} \geq (x_2)^n \left(1 + \frac{a_{n-1}}{x_2} + \cdots + \frac{a_0}{(x_2)^n} \right) = f(x_2),$$

so that $f(x_2) < 0$.

Now applying Theorem 1 to the interval $[x_2, x_1]$ we conclude that there is an x in $[x_2, x_1]$ such that $f(x) = 0$. ■

Theorem 9 disposes of the problem of odd degree equations so happily that it would be frustrating to leave the problem of even degree equations completely undiscussed. At first sight, however, the problem seems insuperable. Some equations, like $x^2 - 1 = 0$, have a solution, and some, like $x^2 + 1 = 0$, do not—what more is there to say? If we are willing to consider a more general question, however, something interesting *can* be said. Instead of trying to solve the equation

$$x^n + a_{n-1}x^{n-1} + \cdots + a_0 = 0,$$

let us ask about the possibility of solving the equations

$$x^n + a_{n-1}x^{n-1} + \cdots + a_0 = c$$

for all possible numbers c . This amounts to allowing the constant term a_0 to vary. The information which can be given concerning the solution of these equations depends on a fact which is illustrated in Figure 8.

The graph of the function $f(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_0$, with n even, contains, at least the way we have drawn it, a lowest point. In other words, there is a number y such that $f(y) \leq f(x)$ for all numbers x —the function f takes on a minimum value, not just on each closed interval, but on the whole line. (Notice that this is false if n is odd.) The proof depends on Theorem 7, but a tricky application will be required. We can apply Theorem 7 to any interval $[a, b]$, and obtain a point y_0 such that $f(y_0)$ is the minimum value of f on $[a, b]$; but if $[a, b]$ happens to be the interval shown in Figure 8, for example, then the point y_0 will not be the place where f has its minimum value for the whole line. In the next

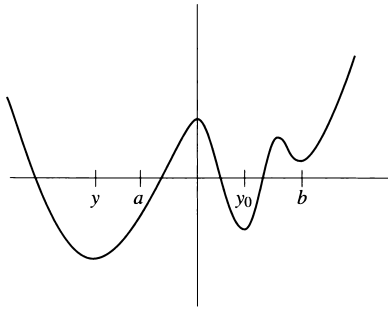


FIGURE 8

and suppose n is even. Then there is a number m such that $(*)$ has a solution for $c \geq m$ and has no solution for $c < m$.

PROOF Let $f(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_0$ (Figure 10).

According to Theorem 10 there is a number y such that $f(y) \leq f(x)$ for all x . Let $m = f(y)$. If $c < m$, then the equation $(*)$ obviously has no solution, since the left side always has a value $\geq m$. If $c = m$, then $(*)$ has y as a solution. Finally, suppose $c > m$. Let b be a number such that $b > y$ and $f(b) > c$. Then $f(y) = m < c < f(b)$. Consequently, by Theorem 4, there is some number x in $[y, b]$ such that $f(x) = c$, so x is a solution of $(*)$. ■

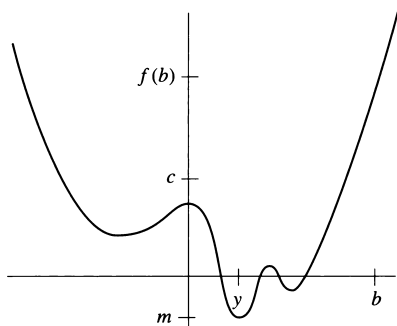


FIGURE 10

These consequences of Theorems 1, 2, and 3 are the only ones we will derive now (these theorems will play a fundamental role in everything we do later, however). Only one task remains—to prove Theorems 1, 2, and 3. Unfortunately, we cannot hope to do this—on the basis of our present knowledge about the real numbers (namely, P1–P12) a proof is *impossible*. There are several ways of convincing ourselves that this gloomy conclusion is actually the case. For example, the proof of Theorem 8 relies only on the proof of Theorem 1; if we could prove Theorem 1, then the proof of Theorem 8 would be complete, and we would have a proof that every positive number has a square root. As pointed out in Part I, it is impossible to prove this on the basis of P1–P12. Again, suppose we consider the function

$$f(x) = \frac{1}{x^2 - 2}$$

If there were no number x with $x^2 = 2$, then f would be continuous, since the denominator would never be 0. But f is not bounded on $[0, 2]$. So Theorem 2 depends essentially on the existence of numbers other than rational numbers, and therefore on some property of the real numbers other than P1–P12.

Despite our inability to prove Theorems 1, 2, and 3, they are certainly results which we want to be true. If the pictures we have been drawing have any connection with the mathematics we are doing, if our notion of continuous function corresponds to any degree with our intuitive notion, Theorems 1, 2, and 3 have got to be true. Since a proof of any of these theorems must require some new property of $\mathbf{R}$ which has so far been overlooked, our present difficulties suggest a way to discover that property: let us try to construct a proof of Theorem 1, for example, and see what goes wrong.

One idea which seems promising is to locate the first point where $f(x) = 0$, that is, the smallest x in $[a, b]$ such that $f(x) = 0$. To find this point, first consider the set A which contains all numbers x in $[a, b]$ such that f is negative on $[a, x]$. In Figure 11, x is such a point, while x' is not. The set A itself is indicated by a heavy line. Since f is negative at a , and positive at b , the set A contains some points greater than a , while all points sufficiently close to b are not in A . (We are here using the continuity of f on $[a, b]$, as well as Problem 6-16.)

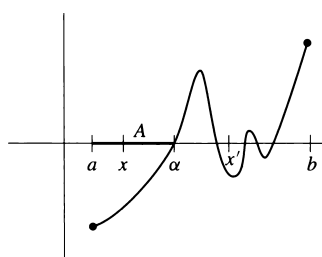


FIGURE 11

theorem the entire point of the proof is to choose an interval $[a, b]$ in such a way that this cannot happen.

THEOREM 10 If n is even and $f(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_0$, then there is a number y such that $f(y) \leq f(x)$ for all x .

PROOF As in the proof of Theorem 9, if

$$M = \max(1, 2n|a_{n-1}|, \dots, 2n|a_0|),$$

then for all x with $|x| \geq M$, we have

$$\frac{1}{2} \leq 1 + \frac{a_{n-1}}{x} + \cdots + \frac{a_0}{x^n}.$$

Since n is even, $x^n \geq 0$ for all x , so

$$\frac{x^n}{2} \leq x^n \left(1 + \frac{a_{n-1}}{x} + \cdots + \frac{a_0}{x^n} \right) = f(x),$$

provided that $|x| \geq M$. Now consider the number $f(0)$. Let $b > 0$ be a number such that $b^n \geq 2f(0)$ and also $b > M$. Then, if $x \geq b$, we have (Figure 9)

$$f(x) \geq \frac{x^n}{2} \geq \frac{b^n}{2} \geq f(0).$$

Similarly, if $x \leq -b$, then

$$f(x) \geq \frac{x^n}{2} \geq \frac{(-b)^n}{2} = \frac{b^n}{2} \geq f(0).$$

Summarizing:

$$\text{if } x \geq b \text{ or } x \leq -b, \text{ then } f(x) \geq f(0).$$

Now apply Theorem 7 to the function f on the interval $[-b, b]$. We conclude that there is a number y such that

$$(1) \quad \text{if } -b \leq x \leq b, \text{ then } f(y) \leq f(x).$$

In particular, $f(y) \leq f(0)$. Thus

$$(2) \quad \text{if } x \leq -b \text{ or } x \geq b, \text{ then } f(x) \geq f(0) \geq f(y).$$

Combining (1) and (2) we see that $f(y) \leq f(x)$ for all x . ■

Theorem 10 now allows us to prove the following result.

THEOREM 11 Consider the equation

$$(*) \quad x^n + a_{n-1}x^{n-1} + \cdots + a_0 = c,$$

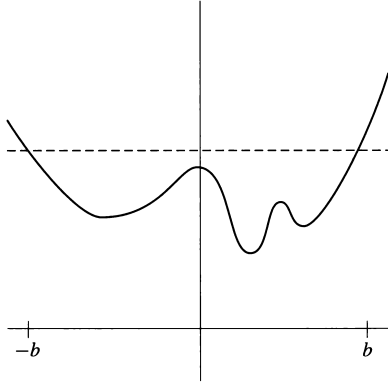


FIGURE 9

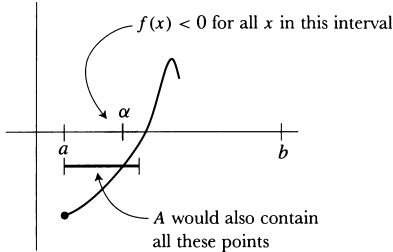


FIGURE 12

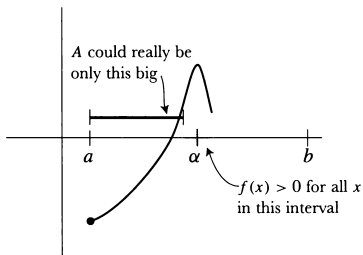


FIGURE 13

Now suppose α is the smallest number which is greater than all members of A ; clearly $a < \alpha < b$. We claim that $f(\alpha) = 0$, and to prove this we only have to eliminate the possibilities $f(\alpha) < 0$ and $f(\alpha) > 0$.

Suppose first that $f(\alpha) < 0$. Then, by Theorem 6-3, $f(x)$ would be less than 0 for all x in a small interval containing α , in particular for some numbers bigger than α (Figure 12); but this contradicts the fact that α is bigger than every member of A , since the larger numbers would also be in A . Consequently, $f(\alpha) < 0$ is false.

On the other hand, suppose $f(\alpha) > 0$. Again applying Theorem 6-3, we see that $f(x)$ would be positive for all x in a small interval containing α , in particular for some numbers smaller than α (Figure 13). This means that these smaller numbers are all *not* in A . Consequently, one could have chosen an even smaller α which would be greater than all members of A . Once again we have a contradiction; $f(\alpha) > 0$ is also false. Hence $f(\alpha) = 0$ and, we are tempted to say, Q.E.D.

We know, however, that something must be wrong, since no new properties of $\mathbf{R}$ were ever used, and it does not require much scrutiny to find the dubious point. It is clear that we can choose a number α which is greater than all members of A (for example, we can choose $\alpha = b$), but it is not so clear that we can choose a *smallest* one. In fact, suppose A consists of all numbers $x \geq 0$ such that $x^2 < 2$. If the number $\sqrt{2}$ did not exist, there would not be a least number greater than all the members of A ; for any $y > \sqrt{2}$ we chose, we could always choose a still smaller one.

Now that we have discovered the fallacy, it is almost obvious what additional property of the real numbers we need. All we must do is say it properly and use it. That is the business of the next chapter.

PROBLEMS

1. For each of the following functions, decide which are bounded above or below on the indicated interval, and which take on their maximum or minimum value. (Notice that f *might* have these properties even if f is not continuous, and even if the interval is not a closed interval.)
 - (i) $f(x) = x^2$ on $(-1, 1)$.
 - (ii) $f(x) = x^3$ on $(-1, 1)$.
 - (iii) $f(x) = x^2$ on $\mathbf{R}$.
 - (iv) $f(x) = x^2$ on $[0, \infty)$.
 - (v) $f(x) = \begin{cases} x^2, & x \leq a \\ a+2, & x > a \end{cases}$ on $(-a-1, a+1)$. (We assume $a > -1$, so that $-a-1 < a+1$; it will be necessary to consider several possibilities for a .)
 - (vi) $f(x) = \begin{cases} x^2, & x < a \\ a+2, & x \geq a \end{cases}$ on $[-a-1, a+1]$. (Again assume $a > -1$.)
 - (vii) $f(x) = \begin{cases} 0, & x \text{ irrational} \\ 1/q, & x = p/q \text{ in lowest terms} \end{cases}$ on $[0, 1]$.

- (viii) $f(x) = \begin{cases} 1, & x \text{ irrational} \\ 1/q, & x = p/q \text{ in lowest terms} \end{cases}$ on $[0, 1]$.
- (ix) $f(x) = \begin{cases} 1, & x \text{ irrational} \\ -1/q, & x = p/q \text{ in lowest terms} \end{cases}$ on $[0, 1]$.
- (x) $f(x) = \begin{cases} x, & x \text{ rational} \\ 0, & x \text{ irrational} \end{cases}$ on $[0, a]$.
- (xi) $f(x) = \sin^2(\cos x + \sqrt{a + a^2})$ on $[0, a^3]$.
- (xii) $f(x) = [x]$ on $[0, a]$.
2. For each of the following polynomial functions f , find an integer n such that $f(x) = 0$ for some x between n and $n + 1$.
- (i) $f(x) = x^3 - x + 3$.
- (ii) $f(x) = x^5 + 5x^4 + 2x + 1$.
- (iii) $f(x) = x^5 + x + 1$.
- (iv) $f(x) = 4x^2 - 4x + 1$.
3. Prove that there is some number x such that
- (i) $x^{179} + \frac{163}{1 + x^2 + \sin^2 x} = 119$.
- (ii) $\sin x = x - 1$.
4. This problem is a continuation of Problem 3-7.
- (a) If $n - k$ is even, and ≥ 0 , find a polynomial function of degree n with exactly k roots.
- (b) A root a of the polynomial function f is said to have **multiplicity** m if $f(x) = (x - a)^m g(x)$, where g is a polynomial function that does *not* have a as a root. Let f be a polynomial function of degree n . Suppose that f has k roots, counting multiplicities, i.e., suppose that k is the sum of the multiplicities of all the roots. Show that $n - k$ is even.
5. Suppose that f is continuous on $[a, b]$ and that $f(x)$ is always rational. What can be said about f ?
6. Suppose that f is a *continuous* function on $[-1, 1]$ such that $x^2 + (f(x))^2 = 1$ for all x . (This means that $(x, f(x))$ always lies on the unit circle.) Show that either $f(x) = \sqrt{1 - x^2}$ for all x , or else $f(x) = -\sqrt{1 - x^2}$ for all x .
7. How many continuous functions f are there which satisfy $(f(x))^2 = x^2$ for all x ?
8. Suppose that f and g are continuous, that $f^2 = g^2$, and that $f(x) \neq 0$ for all x . Prove that either $f(x) = g(x)$ for all x , or else $f(x) = -g(x)$ for all x .
9. (a) Suppose that f is continuous, that $f(x) = 0$ only for $x = a$, and that $f(x) > 0$ for some $x > a$ as well as for some $x < a$. What can be said about $f(x)$ for all $x \neq a$?

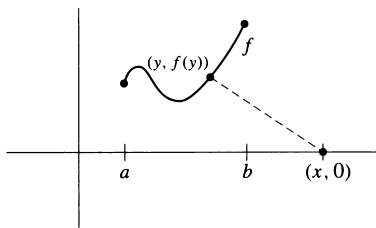


FIGURE 15

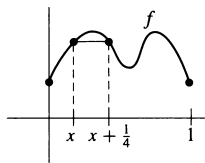


FIGURE 16

- *18.** Suppose that f is a continuous function with $f(x) > 0$ for all x , and $\lim_{x \rightarrow \infty} f(x) = 0 = \lim_{x \rightarrow -\infty} f(x)$. (Draw a picture.) Prove that there is some number y such that $f(y) \geq f(x)$ for all x .
- *19.** (a) Suppose that f is continuous on $[a, b]$, and let x be any number. Prove that there is a point on the graph of f which is closest to $(x, 0)$; in other words there is some y in $[a, b]$ such that the distance from $(x, 0)$ to $(y, f(y))$ is $\leq$ distance from $(x, 0)$ to $(z, f(z))$ for all z in $[a, b]$. (See Figure 15.)
- (b) Show that this same assertion is not necessarily true if $[a, b]$ is replaced by (a, b) throughout.
- (c) Show that the assertion is true if $[a, b]$ is replaced by $\mathbf{R}$ throughout.
- (d) In cases (a) and (c), let $g(x)$ be the minimum distance from $(x, 0)$ to a point on the graph of f . Prove that $g(y) \leq g(x) + |x - y|$, and conclude that g is continuous.
- (e) Prove that there are numbers x_0 and x_1 in $[a, b]$ such that the distance from $(x_0, 0)$ to $(x_1, f(x_1))$ is $\leq$ the distance from $(x_0', 0)$ to $(x_1', f(x_1'))$ for any x_0', x_1' in $[a, b]$.
- 20.** (a) Suppose that f is continuous on $[0, 1]$ and $f(0) = f(1)$. Let n be any natural number. Prove that there is some number x such that $f(x) = f(x + 1/n)$, as shown in Figure 16 for $n = 4$. Hint: Consider the function $g(x) = f(x) - f(x + 1/n)$; what would be true if $g(x) \neq 0$ for all x ?
- *16.** (b) Suppose $0 < a < 1$, but that a is not equal to $1/n$ for any natural number n . Find a function f which is continuous on $[0, 1]$ and which satisfies $f(0) = f(1)$, but which does not satisfy $f(x) = f(x + a)$ for any x .
- *21.** (a) Prove that there does not exist a continuous function f defined on $\mathbf{R}$ which takes on every value exactly twice. Hint: If $f(a) = f(b)$ for $a < b$, then either $f(x) > f(a)$ for all x in (a, b) or $f(x) < f(a)$ for all x in (a, b) . Why? In the first case all values close to $f(a)$, but slightly larger than $f(a)$, are taken on somewhere in (a, b) ; this implies that $f(x) < f(a)$ for $x < a$ and $x > b$.
- (b) Refine part (a) by proving that there is no continuous function f which takes on each value either 0 times or 2 times, i.e., which takes on exactly twice each value that it does take on. Hint: The previous hint implies that f has either a maximum or a minimum value (which must be taken on twice). What can be said about values close to the maximum value?
- (c) Find a continuous function f which takes on every value exactly 3 times. More generally, find one which takes on every value exactly n times, if n is odd.
- (d) Prove that if n is even, then there is no continuous f which takes on every value exactly n times. Hint: To treat the case $n = 4$, for example, let $f(x_1) = f(x_2) = f(x_3) = f(x_4)$. Then either $f(x) > 0$ for all x in two of the three intervals (x_1, x_2) , (x_2, x_3) , (x_3, x_4) , or else $f(x) < 0$ for all x in two of these three intervals.

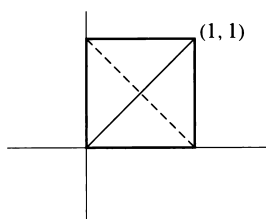


FIGURE 14

- (b) Again assume that f is continuous and that $f(x) = 0$ only for $x = a$, but suppose, instead, that $f(x) > 0$ for some $x > a$ and $f(x) < 0$ for some $x < a$. Now what can be said about $f(x)$ for $x \neq a$?
- *(c) Discuss the sign of $x^3 + x^2y + xy^2 + y^3$ when x and y are not both 0.
10. Suppose f and g are continuous on $[a, b]$ and that $f(a) < g(a)$, but $f(b) > g(b)$. Prove that $f(x) = g(x)$ for some x in $[a, b]$. (If your proof isn't very short, it's not the right one.)
11. Suppose that f is a continuous function on $[0, 1]$ and that $f(x)$ is in $[0, 1]$ for each x (draw a picture). Prove that $f(x) = x$ for some number x .
12. (a) Problem 11 shows that f intersects the diagonal of the square in Figure 14 (solid line). Show that f must also intersect the other (dashed) diagonal.
- (b) Prove the following more general fact: If g is continuous on $[0, 1]$ and $g(0) = 0, g(1) = 1$ or $g(0) = 1, g(1) = 0$, then $f(x) = g(x)$ for some x .
13. (a) Let $f(x) = \sin 1/x$ for $x \neq 0$ and let $f(0) = 0$. Is f continuous on $[-1, 1]$? Show that f satisfies the conclusion of the Intermediate Value Theorem on $[-1, 1]$; in other words, if f takes on two values somewhere on $[-1, 1]$, it also takes on every value in between.
- *(b) Suppose that f satisfies the conclusion of the Intermediate Value Theorem, and that f takes on each value *only once*. Prove that f is continuous.
- *(c) Generalize to the case where f takes on each value only finitely many times.
14. If f is a continuous function on $[0, 1]$, let $\|f\|$ be the maximum value of $|f|$ on $[0, 1]$.
- (a) Prove that for any number c we have $\|cf\| = |c| \cdot \|f\|$.
- *(b) Prove that $\|f + g\| \leq \|f\| + \|g\|$. Give an example where $\|f + g\| \neq \|f\| + \|g\|$.
- (c) Prove that $\|h - f\| \leq \|h - g\| + \|g - f\|$.
- *15. Suppose that ϕ is continuous and $\lim_{x \rightarrow \infty} \phi(x)/x^n = 0 = \lim_{x \rightarrow -\infty} \phi(x)/x^n$.
- (a) Prove that if n is odd, then there is a number x such that $x^n + \phi(x) = 0$.
- (b) Prove that if n is even, then there is a number y such that $y^n + \phi(y) \leq x^n + \phi(x)$ for all x .
- Hint: Of which proofs does this problem test your understanding?
- *16. (a) Suppose that f is continuous on (a, b) and $\lim_{x \rightarrow a^+} f(x) = \lim_{x \rightarrow b^-} f(x) = \infty$. Prove that f has a minimum on all of (a, b) .
- (b) Prove the corresponding result when $a = -\infty$ and/or $b = \infty$.
- *17. Let f be any polynomial function. Prove that there is some number y such that $|f(y)| \leq |f(x)|$ for all x .

This chapter reveals the most important property of the real numbers. Nevertheless, it is merely a sequel to Chapter 7; the path which must be followed has already been indicated, and further discussion would be useless delay.

DEFINITION

A set A of real numbers is **bounded above** if there is a number x such that

$$x \geq a \quad \text{for every } a \text{ in } A.$$

Such a number x is called an **upper bound** for A .

Obviously A is bounded above if and only if there is a number x which is an upper bound for A (and in this case there will be lots of upper bounds for A); we often say, as a concession to idiomatic English, that “ A has an upper bound” when we mean that there is a number which is an upper bound for A .

Notice that the term “bounded above” has now been used in two ways—first, in Chapter 7, in reference to functions, and now in reference to sets. This dual usage should cause no confusion, since it will always be clear whether we are talking about a set of numbers or a function. Moreover, the two definitions are closely connected: if A is the set $\{f(x) : a \leq x \leq b\}$, then the function f is bounded above on $[a, b]$ if and only if the set A is bounded above.

The entire collection $\mathbf{R}$ of real numbers, and the natural numbers $\mathbf{N}$, are both examples of sets which are *not* bounded above. An example of a set which *is* bounded above is

$$A = \{x : 0 \leq x < 1\}.$$

To show that A is bounded above we need only name some upper bound for A , which is easy enough; for example, 138 is an upper bound for A , and so are 2, $1\frac{1}{2}$, $1\frac{1}{4}$, and 1. Clearly, 1 is the least upper bound of A ; although the phrase just introduced is self-explanatory, in order to avoid any possible confusion (in particular, to ensure that we all know what the superlative of “less” means), we define this explicitly.

DEFINITION

A number x is a **least upper bound** of A if

- (1) x is an upper bound of A ,
- and (2) if y is an upper bound of A , then $x \leq y$.

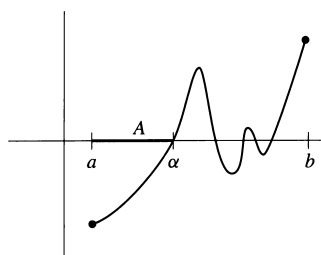


FIGURE 1

(P13) (The least upper bound property) If A is a set of real numbers, $A \neq \emptyset$, and A is bounded above, then A has a least upper bound.

Property P13 may strike you as anticlimactic, but that is actually one of its virtues. To complete our list of basic properties for the real numbers we require no particularly abstruse proposition, but only a property so simple that we might feel foolish for having overlooked it. Of course, the least upper bound property is not really so innocent as all that; after all, it does *not* hold for the rational numbers $\mathbb{Q}$. For example, if A is the set of all rational numbers x satisfying $x^2 < 2$, then there is no *rational* number y which is an upper bound for A and which is less than or equal to every other *rational* number which is an upper bound for A . It will become clear only gradually how significant P13 is, but we are already in a position to demonstrate its power, by supplying the proofs which were omitted in Chapter 7.

THEOREM 7-1

If f is continuous on $[a, b]$ and $f(a) < 0 < f(b)$, then there is some number x in $[a, b]$ such that $f(x) = 0$.

PROOF

Our proof is merely a rigorous version of the outline developed at the end of Chapter 7—we will locate the smallest number x in $[a, b]$ with $f(x) = 0$.

Define the set A , shown in Figure 1, as follows:

$$A = \{x : a \leq x \leq b, \text{ and } f \text{ is negative on the interval } [a, x]\}.$$

Clearly $A \neq \emptyset$, since a is in A ; in fact, there is some $\delta > 0$ such that A contains all points x satisfying $a \leq x < a + \delta$; this follows from Problem 6-16, since f is continuous on $[a, b]$ and $f(a) < 0$. Similarly, b is an upper bound for A and, in fact, there is a $\delta > 0$ such that all points x satisfying $b - \delta < x \leq b$ are upper bounds for A ; this also follows from Problem 6-16, since $f(b) > 0$.

From these remarks it follows that A has a least upper bound α and that $a < \alpha < b$. We now wish to show that $f(\alpha) = 0$, by eliminating the possibilities $f(\alpha) < 0$ and $f(\alpha) > 0$.

Suppose first that $f(\alpha) < 0$. By Theorem 6-3, there is a $\delta > 0$ such that $f(x) < 0$ for $\alpha - \delta < x < \alpha + \delta$ (Figure 2). Now there is some number x_0 in A which satisfies $\alpha - \delta < x_0 < \alpha$ (because otherwise α would not be the *least* upper bound of A). This means that f is negative on the whole interval $[a, x_0]$. But if x_1 is a number between α and $\alpha + \delta$, then f is also negative on the whole interval $[x_0, x_1]$. Therefore f is negative on the interval $[a, x_1]$, so x_1 is in A . But this contradicts the fact that α is an upper bound for A ; our original assumption that $f(\alpha) < 0$ must be false.

Suppose, on the other hand, that $f(\alpha) > 0$. Then there is a number $\delta > 0$ such that $f(x) > 0$ for $\alpha - \delta < x < \alpha + \delta$ (Figure 3). Once again we know that there is an x_0 in A satisfying $\alpha - \delta < x_0 < \alpha$; but this means that f is negative on $[a, x_0]$.

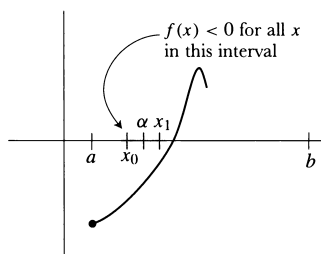


FIGURE 2

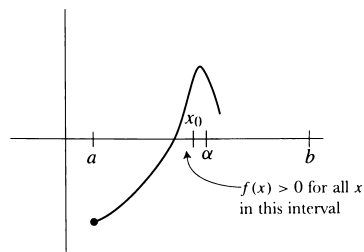


FIGURE 3

which is impossible, since $f(x_0) > 0$. Thus the assumption $f(a) > 0$ also leads to a contradiction, leaving $f(a) = 0$ as the only possible alternative. ■

The proofs of Theorems 2 and 3 of Chapter 7 require a simple preliminary result, which will play much the same role as Theorem 6-3 played in the previous proof.

THEOREM 1 If f is continuous at a , then there is a number $\delta > 0$ such that f is bounded above on the interval $(a - \delta, a + \delta)$ (see Figure 4).

PROOF Since $\lim_{x \rightarrow a} f(x) = f(a)$, there is, for every $\varepsilon > 0$, a $\delta > 0$ such that, for all x ,

$$\text{if } |x - a| < \delta, \text{ then } |f(x) - f(a)| < \varepsilon.$$

It is only necessary to apply this statement to some particular ε (any one will do), for example, $\varepsilon = 1$. We conclude that there is a $\delta > 0$ such that, for all x ,

$$\text{if } |x - a| < \delta, \text{ then } |f(x) - f(a)| < 1.$$

It follows, in particular, that if $|x - a| < \delta$, then $f(x) - f(a) < 1$. This completes the proof: on the interval $(a - \delta, a + \delta)$ the function f is bounded above by $f(a) + 1$. ■

It should hardly be necessary to add that we can now also prove that f is bounded below on some interval $(a - \delta, a + \delta)$, and, finally, that f is bounded on some open interval containing a .

A more significant point is the observation that if $\lim_{x \rightarrow a^+} f(x) = f(a)$, then there

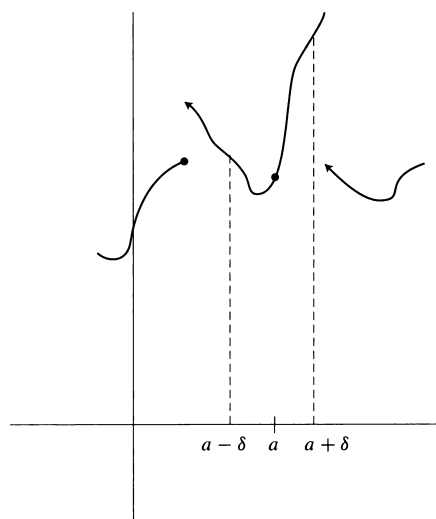


FIGURE 4

The use of the indefinite article “a” in this definition was merely a concession to temporary ignorance. Now that we have made a precise definition, it is easily seen that if x and y are both least upper bounds of A , then $x = y$. Indeed, in this case

$$\begin{aligned} x &\leq y, \quad \text{since } y \text{ is an upper bound, and } x \text{ is a least upper bound,} \\ \text{and } y &\leq x, \quad \text{since } x \text{ is an upper bound, and } y \text{ is a least upper bound;} \end{aligned}$$

it follows that $x = y$. For this reason we speak of *the* least upper bound of A . The term **supremum** of A is synonymous and has one advantage. It abbreviates quite nicely to

$$\sup A \quad (\text{pronounced “soup } A\text{”})$$

and saves us from the abbreviation

$$\text{lub } A$$

(which is nevertheless used by some authors).

There is a series of important definitions, analogous to those just given, which can now be treated more briefly. A set A of real numbers is **bounded below** if there is a number x such that

$$x \leq a \quad \text{for every } a \text{ in } A.$$

Such a number x is called a **lower bound** for A . A number x is the **greatest lower bound** of A if

$$\begin{aligned} (1) \quad &x \text{ is a lower bound of } A, \\ \text{and } (2) \quad &\text{if } y \text{ is a lower bound of } A, \text{ then } x \geq y. \end{aligned}$$

The greatest lower bound of A is also called the **infimum** of A , abbreviated

$$\inf A;$$

some authors use the abbreviation

$$\text{glb } A.$$

One detail has been omitted from our discussion so far—the question of which sets have at least one, and hence exactly one, least upper bound or greatest lower bound. We will consider only least upper bounds, since the question for greatest lower bounds can then be answered easily (Problem 2).

If A is not bounded above, then A has no upper bound at all, so A certainly cannot be expected to have a least upper bound. It is tempting to say that A does have a least upper bound if it has *some* upper bound, but, like the principle of mathematical induction, this assertion can fail to be true in a rather special way. If $A = \emptyset$, then A is bounded above. Indeed, any number x is an upper bound for $\emptyset$:

$$x \geq y \quad \text{for every } y \text{ in } \emptyset$$

simply because there is no y in $\emptyset$. Since *every* number is an upper bound for $\emptyset$, there is surely no least upper bound for $\emptyset$. With this trivial exception however,

is a $\delta > 0$ such that f is bounded on the set $\{x : a \leq x < a + \delta\}$, and a similar observation holds if $\lim_{x \rightarrow b^-} f(x) = f(b)$. Having made these observations (and assuming that you will supply the proofs), we tackle our second major theorem.

THEOREM 7-2 If f is continuous on $[a, b]$, then f is bounded above on $[a, b]$.

PROOF Let

$$A = \{x : a \leq x \leq b \text{ and } f \text{ is bounded above on } [a, x]\}.$$

Clearly $A \neq \emptyset$ (since a is in A), and A is bounded above (by b), so A has a least upper bound α . Notice that we are here applying the term “bounded above” both to the set A , which can be visualized as lying on the horizontal axis, and to f , i.e., to the sets $\{f(y) : a \leq y \leq x\}$, which can be visualized as lying on the vertical axis (Figure 5).

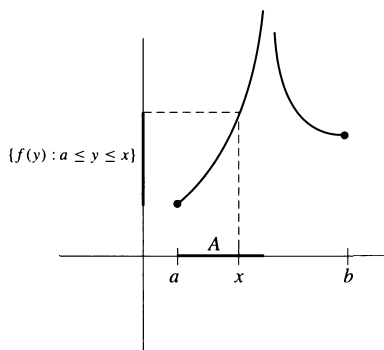


FIGURE 5

Our first step is to prove that we actually have $\alpha = b$. Suppose, instead, that $\alpha < b$. By Theorem 1 there is $\delta > 0$ such that f is bounded on $(\alpha - \delta, \alpha + \delta)$. Since α is the least upper bound of A there is some x_0 in A satisfying $\alpha - \delta < x_0 < \alpha$. This means that f is bounded on $[a, x_0]$. But if x_1 is any number with $\alpha < x_1 < \alpha + \delta$, then f is also bounded on $[x_0, x_1]$. Therefore f is bounded on $[a, x_1]$, so x_1 is in A , contradicting the fact that α is an upper bound for A . This contradiction shows that $\alpha = b$. One detail should be mentioned: this demonstration implicitly assumed that $a < \alpha$ [so that f would be defined on some interval $(\alpha - \delta, \alpha + \delta)$]; the possibility $a = \alpha$ can be ruled out similarly, using the existence of a $\delta > 0$ such that f is bounded on $\{x : a \leq x < a + \delta\}$.

The proof is not quite complete—we only know that f is bounded on $[a, x]$ for every $x < b$, not necessarily that f is bounded on $[a, b]$. However, only one small argument needs to be added.

There is a $\delta > 0$ such that f is bounded on $\{x : b - \delta < x \leq b\}$. There is x_0 in A such that $b - \delta < x_0 < b$. Thus f is bounded on $[a, x_0]$ and also on $[x_0, b]$, so f is bounded on $[a, b]$. ■

To prove the third important theorem we resort to a trick.

THEOREM 7-3 If f is continuous on $[a, b]$, then there is a number y in $[a, b]$ such that $f(y) \geq f(x)$ for all x in $[a, b]$.

PROOF We already know that f is bounded on $[a, b]$, which means that the set

$$\{f(x) : x \text{ in } [a, b]\}$$

is bounded. This set is obviously not $\emptyset$, so it has a least upper bound α . Since $\alpha \geq f(x)$ for x in $[a, b]$ it suffices to show that $\alpha = f(y)$ for some y in $[a, b]$.

Suppose instead that $\alpha \neq f(y)$ for all y in $[a, b]$. Then the function g defined by

$$g(x) = \frac{1}{\alpha - f(x)}, \quad x \text{ in } [a, b]$$

is continuous on $[a, b]$, since the denominator of the right side is never 0. On the other hand, α is the least upper bound of $\{f(x) : x \text{ in } [a, b]\}$; this means that

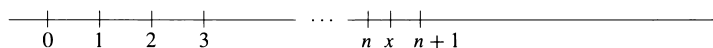
$$\text{for every } \varepsilon > 0 \text{ there is } x \text{ in } [a, b] \text{ with } \alpha - f(x) < \varepsilon.$$

This, in turn, means that

$$\text{for every } \varepsilon > 0 \text{ there is } x \text{ in } [a, b] \text{ with } g(x) > 1/\varepsilon.$$

But *this* means that g is not bounded on $[a, b]$, contradicting the previous theorem. ■

At the beginning of this chapter the set of natural numbers $\mathbf{N}$ was given as an example of an unbounded set. We are now going to *prove* that $\mathbf{N}$ is unbounded. After the difficult theorems proved in this chapter you may be startled to find such an “obvious” theorem winding up our proceedings. If so, you are, perhaps, allowing the geometrical picture of $\mathbf{R}$ to influence you too strongly. “Look,” you may say, “the real numbers look like



so every number x is between two integers $n, n+1$ (unless x is itself an integer).” Basing the argument on a geometric picture is not a proof, however, and even the geometric picture contains an assumption: that if you place unit segments end-to-end you will eventually get a segment larger than any given segment. This axiom, often omitted from a first introduction to geometry, is usually attributed (not quite justly) to Archimedes, and the corresponding property for numbers, that $\mathbf{N}$ is not bounded, is called the *Archimedean property* of the real numbers. This property is *not* a consequence of P1–P12 (see reference [14] of the Suggested Reading), although it does hold for $\mathbf{Q}$, of course. Once we have P13 however, there are no longer any problems.

THEOREM 2 $\mathbf{N}$ is not bounded above.

PROOF Suppose $\mathbf{N}$ were bounded above. Since $\mathbf{N} \neq \emptyset$, there would be a least upper bound α for $\mathbf{N}$. Then

$$\alpha \geq n \quad \text{for all } n \text{ in } \mathbf{N}.$$

Consequently,

$$\alpha \geq n+1 \quad \text{for all } n \text{ in } \mathbf{N},$$

since $n+1$ is in $\mathbf{N}$ if n is in $\mathbf{N}$. But this means that

$$\alpha - 1 \geq n \quad \text{for all } n \text{ in } \mathbf{N},$$

and *this* means that $\alpha - 1$ is also an upper bound for $\mathbf{N}$, contradicting the fact that α is the least upper bound. ■

There is a consequence of Theorem 2 (actually an equivalent formulation) which we have very often assumed implicitly.

THEOREM 3 For any $\varepsilon > 0$ there is a natural number n with $1/n < \varepsilon$.

PROOF Suppose not; then $1/n \geq \varepsilon$ for all n in $\mathbf{N}$. Thus $n \leq 1/\varepsilon$ for all n in $\mathbf{N}$. But this means that $1/\varepsilon$ is an upper bound for $\mathbf{N}$, contradicting Theorem 2. ■

A brief glance through Chapter 6 will show you that the result of Theorem 3 was used in the discussion of many examples. Of course, Theorem 3 was not available at the time, but the examples were so important that in order to give them some cheating was tolerated. As partial justification for this dishonesty we can claim that this result was never used in the proof of a *theorem*, but if your faith has been shaken, a review of all the proofs given so far is in order. Fortunately, such deception will not be necessary again. We have now stated every property of the real numbers that we will ever need. Henceforth, no more lies.

PROBLEMS

1. Find the least upper bound and the greatest lower bound (if they exist) of the following sets. Also decide which sets have greatest and least elements (i.e., decide when the least upper bound and greatest lower bound happens to belong to the set).
 - (i) $\left\{ \frac{1}{n} : n \text{ in } \mathbf{N} \right\}$.
 - (ii) $\left\{ \frac{1}{n} : n \text{ in } \mathbf{Z} \text{ and } n \neq 0 \right\}$.
 - (iii) $\{x : x = 0 \text{ or } x = 1/n \text{ for some } n \text{ in } \mathbf{N}\}$.
 - (iv) $\{x : 0 \leq x \leq \sqrt{2} \text{ and } x \text{ is rational}\}$.
 - (v) $\{x : x^2 + x + 1 \geq 0\}$.
 - (vi) $\{x : x^2 + x - 1 < 0\}$.
 - (vii) $\{x : x < 0 \text{ and } x^2 + x - 1 < 0\}$.
 - (viii) $\left\{ \frac{1}{n} + (-1)^n : n \text{ in } \mathbf{N} \right\}$.
2. (a) Suppose $A \neq \emptyset$ is bounded below. Let $-A$ denote the set of all $-x$ for x in A . Prove that $-A \neq \emptyset$, that $-A$ is bounded above, and that $-\sup(-A)$ is the greatest lower bound of A .
 (b) If $A \neq \emptyset$ is bounded below, let B be the set of all lower bounds of A . Show that $B \neq \emptyset$, that B is bounded above, and that $\sup B$ is the greatest lower bound of A .
3. Let f be a continuous function on $[a, b]$ with $f(a) < 0 < f(b)$.
 (a) The proof of Theorem 7-1 showed that there is a smallest x in $[a, b]$ with $f(x) = 0$. If there is more than one x in $[a, b]$ with $f(x) = 0$, is there necessarily a second smallest? Show that there is a largest x in

- $[a, b]$ with $f(x) = 0$. (Try to give an easy proof by considering a new function closely related to f .)
- (b) The proof of Theorem 7-1 depended upon considering $A = \{x : a \leq x \leq b \text{ and } f \text{ is negative on } [a, x]\}$. Give another proof of Theorem 7-1, which depends upon consideration of $B = \{x : a \leq x \leq b \text{ and } f(x) < 0\}$. Which point x in $[a, b]$ with $f(x) = 0$ will this proof locate? Give an example where the sets A and B are not the same.
- *4. (a) Suppose that f is continuous on $[a, b]$ and that $f(a) = f(b) = 0$. Suppose also that $f(x_0) > 0$ for some x_0 in $[a, b]$. Prove that there are numbers c and d with $a \leq c < x_0 < d \leq b$ such that $f(c) = f(d) = 0$, but $f(x) > 0$ for all x in (c, d) . Hint: The previous problem can be used to good advantage.
- (b) Suppose that f is continuous on $[a, b]$ and that $f(a) < f(b)$. Prove that there are numbers c and d with $a \leq c < d \leq b$ such that $f(c) = f(a)$ and $f(d) = f(b)$ and $f(a) < f(x) < f(d)$ for all x in (c, d) .
5. (a) Suppose that $y - x > 1$. Prove that there is an integer k such that $x < k < y$. Hint: Let l be the largest integer satisfying $l \leq x$, and consider $l + 1$.
- (b) Suppose $x < y$. Prove that there is a rational number r such that $x < r < y$. Hint: If $1/n < y - x$, then $ny - nx > 1$. (Query: Why have parts (a) and (b) been postponed until this problem set?)
- (c) Suppose that $r < s$ are rational numbers. Prove that there is an irrational number between r and s . Hint: As a start, you know that there is an irrational number between 0 and 1.
- (d) Suppose that $x < y$. Prove that there is an irrational number between x and y . Hint: It is unnecessary to do any more work; this follows from (b) and (c).
- *6. A set A of real numbers is said to be **dense** if every open interval contains a point of A . For example, Problem 5 shows that the set of rational numbers and the set of irrational numbers are each dense.
- (a) Prove that if f is continuous and $f(x) = 0$ for all numbers x in a dense set A , then $f(x) = 0$ for all x .
- (b) Prove that if f and g are continuous and $f(x) = g(x)$ for all x in a dense set A , then $f(x) = g(x)$ for all x .
- (c) If we assume instead that $f(x) \geq g(x)$ for all x in A , show that $f(x) \geq g(x)$ for all x . Can $\geq$ be replaced by $>$ throughout?
7. Prove that if f is continuous and $f(x + y) = f(x) + f(y)$ for all x and y , then there is a number c such that $f(x) = cx$ for all x . (This conclusion can be demonstrated simply by combining the results of two previous problems.) Point of information: There *do* exist *noncontinuous* functions f satisfying $f(x + y) = f(x) + f(y)$ for all x and y , but we cannot prove this now; in fact, this simple question involves ideas that are usually never mentioned in undergraduate courses (see reference [7] in the Suggested Reading).

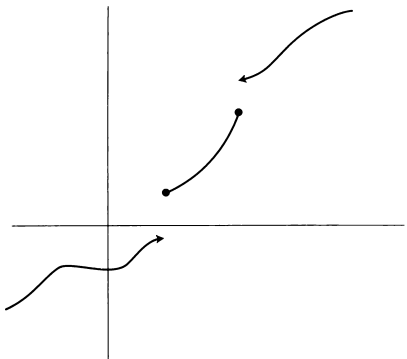


FIGURE 6

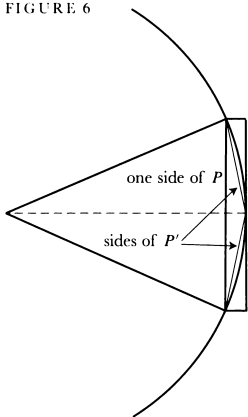


FIGURE 7

*8. Suppose that f is a function such that $f(a) \leq f(b)$ whenever $a < b$ (Figure 6).

- Prove that $\lim_{x \rightarrow a^-} f(x)$ and $\lim_{x \rightarrow a^+} f(x)$ both exist. Hint: Why is this problem in this chapter?
- Prove that f never has a removable discontinuity (this terminology comes from Problem 6-17).
- Prove that if f satisfies the conclusions of the Intermediate Value Theorem, then f is continuous.

*9. If f is a bounded function on $[0, 1]$, let $\|f\| = \sup\{|f(x)| : x \text{ in } [0, 1]\}$. Prove analogues of the properties of $\| \cdot \|$ in Problem 7-14.

10. Suppose $\alpha > 0$. Prove that every number x can be written uniquely in the form $x = k\alpha + x'$, where k is an integer, and $0 \leq x' < \alpha$.

- Suppose that $a_1, a_2, a_3, \dots$ is a sequence of positive numbers with $a_{n+1} \leq a_n/2$. Prove that for any $\varepsilon > 0$ there is some n with $a_n < \varepsilon$.
 - Suppose P is a regular polygon inscribed inside a circle. If P' is the inscribed regular polygon with twice as many sides, show that the difference between the area of the circle and the area of P' is less than half the difference between the area of the circle and the area of P (use Figure 7).
 - Prove that there is a regular polygon P inscribed in a circle with area as close as desired to the area of the circle. In order to do part (c) you will need part (a). This was clear to the Greeks, who used part (a) as the basis for their entire treatment of proportion and area. By calculating the areas of polygons, this method ("the method of exhaustion") allows computations of π to any desired accuracy; Archimedes used it to show that $\frac{223}{71} < \pi < \frac{22}{7}$. But it has far greater theoretical importance:

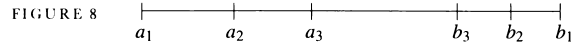
*(d) Using the fact that the areas of two regular polygons with the same number of sides have the same ratio as the square of their sides, prove that the areas of two circles have the same ratios as the square of their radii. Hint: Deduce a contradiction from the assumption that the ratio of the areas is greater, or less, than the ratio of the square of the radii by inscribing appropriate polygons.

12. Suppose that A and B are two nonempty sets of numbers such that $x \leq y$ for all x in A and all y in B .

- Prove that $\sup A \leq y$ for all y in B .
- Prove that $\sup A \leq \inf B$.

13. Let A and B be two nonempty sets of numbers which are bounded above, and let $A+B$ denote the set of all numbers $x+y$ with x in A and y in B . Prove that $\sup(A+B) = \sup A + \sup B$. Hint: The inequality $\sup(A+B) \leq \sup A + \sup B$ is easy. Why? To prove that $\sup A + \sup B \leq \sup(A+B)$ it suffices to prove that $\sup A + \sup B \leq \sup(A+B) + \varepsilon$ for all $\varepsilon > 0$; begin by choosing x in A

and y in B with $\sup A - x < \varepsilon/2$ and $\sup B - y < \varepsilon/2$.



14. (a) Consider a sequence of closed intervals $I_1 = [a_1, b_1]$, $I_2 = [a_2, b_2]$, $\dots$. Suppose that $a_n \leq a_{n+1}$ and $b_{n+1} \leq b_n$ for all n (Figure 8). Prove that there is a point x which is in every I_n .
- (b) Show that this conclusion is false if we consider open intervals instead of closed intervals.

The simple result of Problem 14(a) is called the “Nested Interval Theorem.” It may be used to give alternative proofs of Theorems 1 and 2. The appropriate reasoning, outlined in the next two problems, illustrates a general method, called a “bisection argument.”

- *15. Suppose f is continuous on $[a, b]$ and $f(a) < 0 < f(b)$. Then either $f((a+b)/2) = 0$, or f has different signs at the end points of the interval $[a, (a+b)/2]$, or f has different signs at the end points of $[(a+b)/2, b]$. Why? If $f((a+b)/2) \neq 0$, let I_1 be the interval on which f changes sign. Now bisect I_1 . Either f is 0 at the midpoint, or f changes sign on one of the two intervals. Let I_2 be that interval. Continue in this way, to define I_n for each n (unless f is 0 at some midpoint). Use the Nested Interval Theorem to find a point x where $f(x) = 0$.
- *16. Suppose f were continuous on $[a, b]$, but not bounded on $[a, b]$. Then f would be unbounded on either $[a, (a+b)/2]$ or $[(a+b)/2, b]$. Why? Let I_1 be one of these intervals on which f is unbounded. Proceed as in Problem 15 to obtain a contradiction.
17. (a) Let $A = \{x : x < \alpha\}$. Prove the following (they are all easy):
- (i) If x is in A and $y < x$, then y is in A .
 - (ii) $A \neq \emptyset$.
 - (iii) $A \neq \mathbf{R}$.
 - (iv) If x is in A , then there is some number x' in A such that $x < x'$.
- (b) Suppose, conversely, that A satisfies (i)–(iv). Prove that $A = \{x : x < \sup A\}$.
- *18. A number x is called an **almost upper bound** for A if there are only finitely many numbers y in A with $y \geq x$. An **almost lower bound** is defined similarly.
- (a) Find all almost upper bounds and almost lower bounds of the sets in Problem 1.
 - (b) Suppose that A is a bounded infinite set. Prove that the set B of all almost upper bounds of A is nonempty, and bounded below.

- (c) It follows from part (b) that $\inf B$ exists; this number is called the **limit superior** of A , and denoted by $\overline{\lim} A$ or $\limsup A$. Find $\overline{\lim} A$ for each set A in Problem 1.
- (d) Define $\underline{\lim} A$, and find it for all A in Problem 1.

***19.** If A is a bounded infinite set prove

- (a) $\underline{\lim} A \leq \overline{\lim} A$.
- (b) $\overline{\lim} A \leq \sup A$.
- (c) If $\overline{\lim} A < \sup A$, then A contains a largest element.
- (d) The analogues of parts (b) and (c) for $\underline{\lim}$.

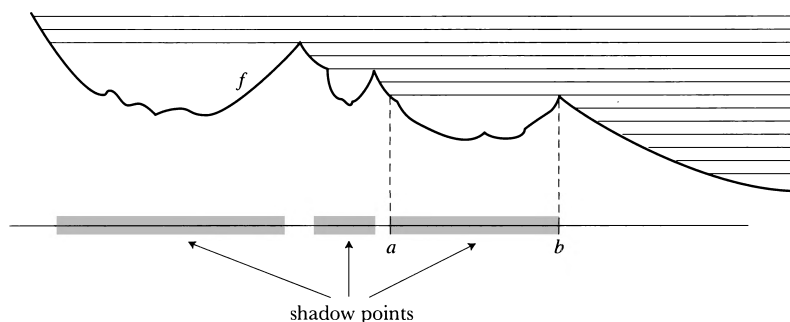


FIGURE 9

- 20.** Let f be a continuous function on $\mathbf{R}$. A point x is called a **shadow point** of f if there is a number $y > x$ with $f(y) > f(x)$. The rationale for this terminology is indicated in Figure 9; the parallel lines are the rays of the sun rising in the east (you are facing north). Suppose that all points of (a, b) are shadow points, but that a and b are not shadow points. Clearly, $f(a) \geq f(b)$.
- (a) Suppose that $f(a) > f(b)$. Show that the point where f takes on its maximum value on $[a, b]$ must be a .
- (b) Then show that this leads to a contradiction, so that in fact we must have $f(a) = f(b)$.

This little result, known as the Rising Sun Lemma, is instrumental in proving several beautiful theorems that do not appear in this book; see page 450.

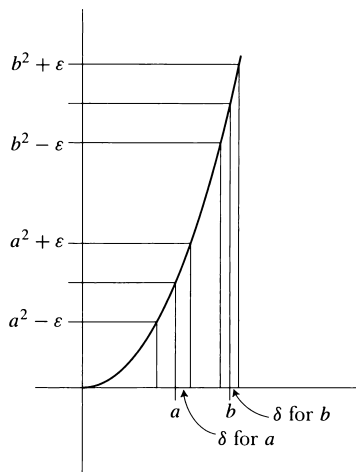


FIGURE 1

APPENDIX. UNIFORM CONTINUITY

Now that we've come to the end of the "foundations," it might be appropriate to slip in one further fundamental concept. This notion is not used crucially in the rest of the book, but it can help clarify many points later on.

We know that the function $f(x) = x^2$ is continuous at a for all a . In other words,

if a is any number, then for every $\varepsilon > 0$ there is some $\delta > 0$
such that, for all x , if $|x - a| < \delta$, then $|x^2 - a^2| < \varepsilon$.

Of course, δ depends on ε . But δ *also depends* on a —the δ that works at a might not work at b (Figure 1). Indeed, it's clear that given $\varepsilon > 0$ there is no one $\delta > 0$ that works for all a , or even for all positive a . In fact, the number $a + \delta/2$ will certainly satisfy $|x - a| < \delta$, but if $a > 0$, then

$$\left| \left(a + \frac{\delta}{2} \right)^2 - a^2 \right| = \left| a\delta + \frac{\delta^2}{4} \right| \geq a\delta,$$

and this won't be $< \varepsilon$ once $a > \varepsilon/\delta$. (This is just an admittedly confusing computational way of saying that f is growing faster and faster!)

On the other hand, for any $\varepsilon > 0$ there *will* be one $\delta > 0$ that works for all a in any interval $[-N, N]$. In fact, the δ which works at N or $-N$ will also work everywhere else in the interval.

As a final example, consider the function $f(x) = \sin 1/x$, or the function whose graph appears in Figure 18 on page 62. It is easy to see that, so long as $\varepsilon < 1$, there will not be one $\delta > 0$ that works for these functions at all points a in the open interval $(0, 1)$.

These examples illustrate important distinctions between the behavior of various continuous functions on certain intervals, and there is a special term to signal this distinction.

DEFINITION

The function f is **uniformly continuous on an interval** A if for every $\varepsilon > 0$ there is some $\delta > 0$ such that, for all x and y in A ,

$$\text{if } |x - y| < \delta, \text{ then } |f(x) - f(y)| < \varepsilon.$$

We've seen that a function can be continuous on the whole line, or on an open interval, without being uniformly continuous there. On the other hand, the function $f(x) = x^2$ did turn out to be uniformly continuous on any closed interval. This shouldn't be too surprising—it's the same sort of thing that occurs when we ask whether a function is bounded on an interval—and we would be led to suspect that any continuous function on a closed interval is also uniformly continuous on that interval. In order to prove this, we'll need to deal first with one subtle point.

Suppose that we have two intervals $[a, b]$ and $[b, c]$ with the common endpoint b , and a function f that is continuous on $[a, c]$. Let $\varepsilon > 0$ and suppose that

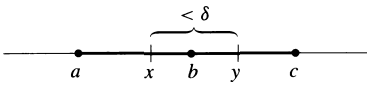


FIGURE 2

the following two statements hold:

- (i) if x and y are in $[a, b]$ and $|x - y| < \delta_1$, then $|f(x) - f(y)| < \varepsilon$,
- (ii) if x and y are in $[b, c]$ and $|x - y| < \delta_2$, then $|f(x) - f(y)| < \varepsilon$.

We'd like to know if there is some $\delta > 0$ such that $|f(x) - f(y)| < \varepsilon$ whenever x and y are points in $[a, c]$ with $|x - y| < \delta$. Our first inclination might be to choose δ as the minimum of δ_1 and δ_2 . But it is easy to see what goes wrong (Figure 2): we might have x in $[a, b]$ and y in $[b, c]$, and then neither (i) nor (ii) tells us anything about $|f(x) - f(y)|$. So we have to be a little more cagey, and also use continuity of f at b .

LEMMA Let $a < b < c$ and let f be continuous on the interval $[a, c]$. Let $\varepsilon > 0$, and suppose that statements (i) and (ii) hold. Then there is a $\delta > 0$ such that,

if x and y are in $[a, c]$ and $|x - y| < \delta$, then $|f(x) - f(y)| < \varepsilon$.

PROOF Since f is continuous at b , there is a $\delta_3 > 0$ such that,

$$\text{if } |x - b| < \delta_3, \text{ then } |f(x) - f(b)| < \frac{\varepsilon}{2}.$$

It follows that

$$\text{(iii) if } |x - b| < \delta_3 \text{ and } |y - b| < \delta_3, \text{ then } |f(x) - f(y)| < \varepsilon.$$

Choose δ to be the minimum of δ_1 , δ_2 , and δ_3 . We claim that this δ works. In fact, suppose that x and y are any two points in $[a, c]$ with $|x - y| < \delta$. If x and y are both in $[a, b]$, then $|f(x) - f(y)| < \varepsilon$ by (i); and if x and y are both in $[b, c]$, then $|f(x) - f(y)| < \varepsilon$ by (ii). The only other possibility is that

$$x < b < y \quad \text{or} \quad y < b < x.$$

In either case, since $|x - y| < \delta$, we also have $|x - b| < \delta$ and $|y - b| < \delta$. So $|f(x) - f(y)| < \varepsilon$ by (iii). ■

THEOREM 1 If f is continuous on $[a, b]$, then f is uniformly continuous on $[a, b]$.

PROOF It's the usual trick, but we've got to be a little bit careful about the mechanism of the proof. For $\varepsilon > 0$ let's say that f is ε -good on $[a, b]$ if there is some $\delta > 0$ such that, for all y and z in $[a, b]$,

$$\text{if } |y - z| < \delta, \text{ then } |f(y) - f(z)| < \varepsilon.$$

Then we're trying to prove that f is ε -good on $[a, b]$ for all $\varepsilon > 0$.

Consider any particular $\varepsilon > 0$. Let

$$A = \{x : a \leq x \leq b \text{ and } f \text{ is } \varepsilon\text{-good on } [a, x]\}.$$

Then $A \neq \emptyset$ (since a is in A), and A is bounded above (by b), so A has a least upper bound α . We really should write α_ε , since A and α might depend on ε . But we won't since we intend to prove that $\alpha = b$, no matter what ε is.

Suppose that we had $\alpha < b$. Since f is continuous at α , there is some $\delta_0 > 0$ such that, if $|y - \alpha| < \delta_0$, then $|f(y) - f(\alpha)| < \varepsilon/2$. Consequently, if $|y - \alpha| < \delta_0$ and $|z - \alpha| < \delta_0$, then $|f(y) - f(z)| < \varepsilon$. So f is surely ε -good on the interval $[\alpha - \delta_0, \alpha + \delta_0]$. On the other hand, since α is the least upper bound of A , it is also clear that f is ε -good on $[a, \alpha - \delta_0]$. Then the Lemma implies that f is ε -good on $[a, \alpha + \delta_0]$, so $\alpha + \delta_0$ is in A , contradicting the fact that α is an upper bound.

To complete the proof we just have to show that $\alpha = b$ is actually in A . The argument for this is practically the same: Since f is continuous at b , there is some $\delta_0 > 0$ such that, if $b - \delta_0 < y < b$, then $|f(y) - f(b)| < \varepsilon/2$. So f is ε -good on $[b - \delta_0, b]$. But f is also ε -good on $[a, b - \delta_0]$, so the Lemma implies that f is ε -good on $[a, b]$. ■

PROBLEMS

1. (a) For which of the following values of α is the function $f(x) = x^\alpha$ uniformly continuous on $[0, \infty)$: $\alpha = 1/3, 1/2, 2, 3$?
 (b) Find a function f that is continuous and bounded on $(0, 1]$, but not uniformly continuous on $(0, 1]$.
 (c) Find a function f that is continuous and bounded on $[0, \infty)$ but which is not uniformly continuous on $[0, \infty)$.
2. (a) Prove that if f and g are uniformly continuous on A , then so is $f + g$.
 (b) Prove that if f and g are uniformly continuous and bounded on A , then fg is uniformly continuous on A .
 (c) Show that this conclusion does not hold if one of them isn't bounded.
 (d) Suppose that f is uniformly continuous on A , that g is uniformly continuous on B , and that $f(x)$ is in B for all x in A . Prove that $g \circ f$ is uniformly continuous on A .
3. Use a "bisection argument" (page 142) to give another proof of Theorem 1.
4. Derive Theorem 7-2 as a consequence of Theorem 1.

